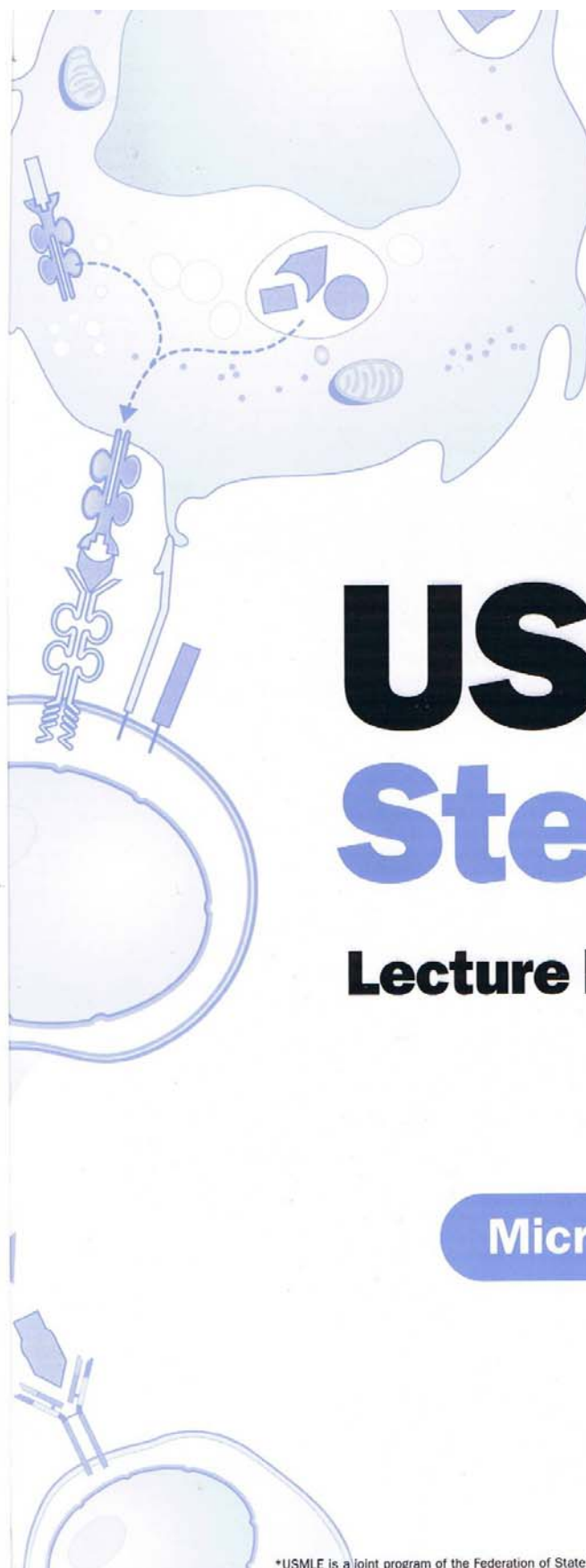




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USMLE **Step 1**

Lecture Notes

Microbiology/Immunology

*USMLE is a joint program of the Federation of State Medical Boards of the United States, Inc. and the National Board of Medical Examiners.

MICROBIOLOGY

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SECTION I

Microbiology

General Microbiology

1

What the USMLE Requires You to Know

- Differences among viruses, fungi, bacteria, and parasites
- Differences between eukaryotic and prokaryotic cells
- Important normal flora
- Major mechanisms of pathogenicity

MAJOR MICROBIAL GROUPS

Table I-1-1. Comparison of Medically Important Microbial Groups

Characteristic	Viruses*	Bacteria	Fungi	Parasites
Diameter**	Minute (0.02–0.3 μ)	Small (0.3–2 μ)	3–10 μ	15–25 μ (trophozoites)
Cell type	Acellular (not cell) No nucleus	Prokaryotic cells Nucleoid region: no nuclear membrane	Eukaryotic cells Nucleus with nuclear membrane	Eukaryotic cells
	DNA <u>or</u> RNA 1 nucleocapsid except in segmented or diploid viruses	DNA and RNA 1 chromosome No histones	DNA and RNA More than 1 chromosome	
	Replicates in host cells	DNA replicates continuously	G and S phases	
		Exons, no introns	Introns and exons	
	Some have poly- cistronic mRNA*** and post translational cleavage	Mono- and poly- cistronic mRNA	Monocistronic RNA	
	Uses host organelles; obligate intracellular parasites	No membrane bound organelles	Mitochondria and other membrane-bound organelles	
	No ribosomes	70S ribosomes (30s+50s)	80S ribosomes (40S+60S)	
Cellular membrane	Some are enveloped: but no membrane function	Membranes have no sterols except Mycoplasmas , which have cholesterol	Membrane ergosterol is major sterol.	Sterols such as cholesterol
Cell wall	No cell wall	Peptidoglycan	Complex carbo- hydrate cell wall : chitin , glucans, or mannans	No cell wall
Replication	Make and assemble viral components	Binary fission (asexual)	Cytokinesis with mitosis / meiosis	Cytokinesis with mitosis / meiosis

*Besides viruses, **two other acellular forms exist**:

- Viroids: obligate intracellular but acellular parasites of plants; naked RNA; no human diseases.
- Prions: acellular particles associated with Kuru, etc.; insensitive to nucleases.

Abnormal prion proteins (PrP) modify folding of normal prion-like proteins found in the body (coded for by human genes).

If the **diameter of a cell described in a clinical case is $>2 \mu$, then it is **probably a eukaryotic cell**.

***Polycistronic mRNA carries the genetic code for several proteins. (It has multiple Shine-Dalgarno sites.)

Epidemiology

Normal Flora

- Is found on body surfaces contiguous with the outside environment
- Is semi-permanent, varying with major life changes
- Can cause infection
 - if misplaced, e.g., fecal flora to urinary tract or abdominal cavity, or skin flora to catheter
 - or, if person becomes compromised, normal flora may overgrow (oral thrush)
- Contributes to health
 - protective host defense by maintaining conditions such as pH so other organisms may not grow
 - serves nutritional function by synthesizing: K and B vitamins

In A Nutshell

Definitions

Carrier: person colonized by a potential pathogen without overt disease.

Bacteremia: bacteria in bloodstream without overt clinical signs.

Septicemia: bacteria in bloodstream (multiplying) with clinical symptoms.

Table I-1-2. Important Normal Flora

Site	Common or Medically Important Organisms	Less Common but Notable Organisms
Blood, internal organs	None, generally sterile	
Cutaneous surfaces including urethra and outer ear	<i>Staphylococcus epidermidis</i>	<i>Staphylococcus aureus</i> , Corynebacteria (diphtheroids), streptococci, anaerobes, e.g., peptostreptococci, yeasts (<i>Candida</i> spp.)
Nose	<i>Staphylococcus aureus</i>	<i>S. epidermidis</i> , diphtheroids, assorted streptococci
Oropharynx	Viridans streptococci including <i>Strep. mutans</i> ¹	Assorted streptococci, nonpathogenic <i>Neisseria</i>, nontypeable² <i>Haemophilus influenzae</i>
Gingival crevices	Anaerobes: <i>Bacteroides</i> , <i>Prevotella</i> , <i>Fusobacterium</i> , <i>Streptococcus</i> , <i>Actinomyces</i>	
Stomach	None	
Colon (microaerophilic/anaerobic)	Babies; breast-fed only: <i>Bifidobacterium</i> Adult: <i>Bacteroides/Prevotella</i> (Predominant organism) <i>Escherichia</i> <i>Bifidobacterium</i>	<i>Lactobacillus</i> , streptococci <i>Eubacterium</i> , <i>Fusobacterium</i> , <i>Lactobacillus</i> , assorted Gram-negative anaerobic rods, <i>Enterococcus faecalis</i> and other streptococci
Vagina	<i>Lactobacillus</i> ³	Assorted streptococci, Gram-negative rods, diphtheroids, yeasts

¹*S. mutans* secretes a biofilm that glues it and other oral flora to teeth, producing **dental plaque**.

²(Nontypeable for *Haemophilus* means no capsule.)

³Group B streptococci colonize vagina of 15–20% of women and may infect the infant during labor or delivery, causing septicemia and/or meningitis (as may *E. coli* from fecal flora).

PATHOGENICITY (INFECTIVITY AND TOXICITY) MAJOR MECHANISMS

Colonization

(Important unless organism is traumatically implanted.)

Adherence to cell surfaces involves

- **Pili/fimbriae:** primary mechanism in most gram-negative cells.
- **Teichoic acids:** primary mechanism of gram-positive cells.
- **Adhesins:** colonizing factor adhesins, pertussis toxin, and hemagglutinins.
- **IgA proteases:** cleaved Fc portion may coat bacteria and bind them to cellular Fc receptors.

Partial Adherence to inert materials, **biofilms:** *Staph. epidermidis*, *Streptococcus mutans*

Avoiding immediate destruction by host defense system:

- **Anti-phagocytic surface components** (inhibit phagocytic uptake):
 - **Capsules/slime layers:**
 - Streptococcus pyogenes* M protein
 - Neisseria gonorrhoeae* pili
 - Staphylococcus aureus* A protein
- **IgA proteases,** destruction of mucosal IgA: *Neisseria*, *Haemophilus*, *S. pneumoniae*

“Hunting and gathering” needed nutrients:

- **Siderophores** steal (chelate) and import iron.

Ability to Survive Intracellularly

- **Evading intracellular killing by professional phagocytic cells** allows intracellular growth:
 - *M. tuberculosis* survives by inhibiting phagosome-lysosome fusion.
 - *Listeria* quickly escapes the phagosome into the cytoplasm **before** phagosome-lysosome fusion.
- **Invasins:** surface proteins that allow an organism to bind to and invade normally non-phagocytic human cells, escaping the immune system. Best studied invasin is on *Yersinia pseudotuberculosis* (an organism causing diarrhea).
- Damage from viruses is largely from intracellular replication, which either kills cells, transforms them or, in the case of latent viruses, may do no noticeable damage.

Inflammation or Immune-Mediated Damage

Examples

- **Cross-reaction of bacteria-induced antibodies with tissue antigens** causes disease. Rheumatic fever is one example.
- **Delayed hypersensitivity and the granulomatous response** stimulated by the presence of intracellular bacteria is responsible for neurological damage in leprosy, cavitation in tuberculosis, and fallopian tube blockage resulting in infertility from *Chlamydia* PID (pelvic inflammatory disease).

Note

Mnemonic

Streptococcus pneumoniae
Klebsiella pneumoniae
Haemophilus influenzae
Pseudomonas aeruginosa
Neisseria meningitidis
Cryptococcus neoformans

(Some Killers Have Pretty Nice Capsules)

Note

Intracellular organisms

- Elicit different immune responses
- Different pathology
- Different antibiotics
- Different cultural techniques

- **Immune complexes** damage the kidney in post streptococcal acute glomerulonephritis.
- **Peptidoglycan-teichoic acid** (large fragments) of Gram-positive cells:
Serves as a structural toxin released when cells die.
Chemotactic for neutrophils.

Physical Damage

Swelling from infection in a fixed space damages tissues; examples: meningitis and cysticercosis.
Large physical size of organism may cause problems; example: *Ascaris lumbricoides* blocking bile duct.

Aggressive tissue invasion from *Entamoeba histolytica* causes intestinal ulceration and releases intestinal bacteria, compounding problems.

TOXINS

Toxins may aid in invasiveness, damage cells, inhibit cellular processes, or trigger immune response and damage.

Structural Toxins

- **Endotoxin** (Lipopolysaccharide = LPS)
 - LPS is part of the **Gram-negative outer membrane**.
 - **Toxic portion is lipid A**: generally not released (and toxic) until death of cell.
Exception: *N. meningitidis*, which over-produces outer membrane fragments.
 - LPS is **heat stable** and not strongly immunogenic so it **cannot be converted to a toxoid**.
 - Mechanism
 - LPS **activates macrophages**, leading to release of TNF-alpha, IL-1, and IL-6.
 - IL-1 is a major mediator of fever.
 - Macrophage activation and products lead to tissue damage.
 - Damage to the endothelium from **bradykinin-induced vasodilation leads to shock**.
 - Coagulation (DIC) is mediated through the activation of Hageman factor.
- **Peptidoglycan, Teichoic Acids**
- **Exotoxins**
 - are **protein toxins**, generally quite **toxic** and **secreted by bacterial cells** (some Gram +, some Gram -)
 - **can be modified** by chemicals or heat to produce a **toxoid** that still is **immunogenic, but no longer toxic** so can be used as a vaccine
 - **A-B (or "two") component protein toxins**
 - B component **binds** to specific cell receptors to facilitate the internalization of A.
 - A component is the **active (toxic) component** (often an enzyme such as an ADP ribosyl transferase).
 - Exotoxins may be subclassed as enterotoxins, neurotoxins, or cytotoxins.
 - **Cytolysins**: lyse cells from outside by damaging membrane.
 - *C. perfringens* **alpha toxin** is a **lecithinase**.
 - *Staphylococcus aureus* **alpha toxin** inserts itself to form **pores** in the membrane.

Table I-1-3. Major Exotoxins

	Organism (Gram)	Toxin	Mode of Action	Role in Disease
Protein Inhibitors	<i>Corynebacterium diphtheriae</i> (+)	Diphtheria toxin	ADP ribosyl transferase; inactivates EF-2; 1' targets: heart/nerves/epithelium	Inhibits eukaryotic cell protein synthesis
	<i>Pseudomonas aeruginosa</i> (-)	Exotoxin A	ADP ribosyl transferase; inactivates EF-2; 1' target: liver.	Inhibits eukaryotic cell protein synthesis
	<i>Shigella dysenteriae</i> (-)	Shiga toxin	Interferes with 60S ribosomal subunit	Inhibits protein synthesis in eukaryotic cells. Enterotoxic, cytotoxic, and neurotoxic
	Enterohemorrhagic <i>E. coli</i> (EHEC)	Verotoxin (a shiga-like toxin)	Interferes with 60S ribosomal subunit	Inhibits protein synthesis in eukaryotic cells
Neurotoxins	<i>Clostridium tetani</i> (+)	Tetanus toxin	Blocks release of the inhibitory transmitters glycine and GABA	Inhibits neurotransmission in inhibitory synapses
	<i>Clostridium botulinum</i> (+)	Botulinum toxin	Blocks release of acetylcholine	Inhibits cholinergic synapses
Endotoxin Enhancers	<i>Staphylococcus aureus</i> (+)	TSST-1	Pyrogenic, decreases liver clearance of LPS, superantigen	Fever, increased susceptibility to LPS, rash, shock, capillary leakage
	<i>Streptococcus pyogenes</i> (+)	Exotoxin A, a.k.a.: erythrogenic or pyrogenic toxin	Similar to TSST-1	Fever, increased susceptibility to LPS, rash, shock, capillary leakage, cardiotoxicity
cAMP Inducers	Enterotoxigenic <i>Escherichia coli</i> (-)	Heat labile toxin (LT)	LT stimulates an adenylate cyclase by ADP ribosylation of GTP binding protein	Both LT and ST promote secretion of fluid and electrolytes from intestinal epithelium
	<i>Vibrio cholerae</i> (-)	Cholera toxin	Similar to <i>E. coli</i> LT	Profuse, watery diarrhea
	<i>Bacillus anthracis</i> (+)	Anthrax toxin (3 proteins make 2 toxins)	EF = edema factor = adenylate cyclase LF = lethal factor PA = protective antigen (B component for both)	Decreases phagocytosis; causes edema, kills cells
	<i>Bordetella pertussis</i> (-)	Pertussis toxin	ADP ribosylates G _i , the negative regulator of adenylate cyclase → increased cAMP	Histamine-sensitizing Lymphocytosis promotion Islet activation
Cytolysins	<i>Clostridium perfringens</i> (+)	Alpha toxin	Lecithinase	Damages cell membranes; myonecrosis
	<i>Staphylococcus aureus</i> (+)	Alpha toxin	Toxin intercalates forming pores	Cell membrane becomes leaky

Chapter Summary

The size, cell type, cellular membrane, cell walls, and modes of replication of viruses, bacteria, fungi, and protozoan parasites are compared, as are the differences between prokaryotes and eukaryotes. There is also a brief consideration of the properties and pathogenicity of prions and plant viroids.

The important normal microflora typically associated with various body sites are described.

For the successful induction of pathogenicity, microorganisms must be infective and toxic. Infectivity requires colonization, which in turn requires adhesion, the avoidance of destruction by the host, and the invasion of cells. Modes of damage include damage caused by immune-mediated inflammation, physical-damage-associated tissue invasion, and/or production of endo- or exotoxins.

Review Questions

1. Your laboratory isolates an entirely new and unknown pathogen from one of your patients, which has all the characteristics of an aerobic filamentous fungus except that the ribosomes are prokaryotic. Unfortunately, your patient with this pathogen is very ill. Which agent would most likely be successful in treating your patient?
 - A. Third generation of cephalosporins
 - B. Isoniazid
 - C. Metronidazole
 - D. Careful limited usage of Shiga toxin
 - E. Tetracycline
2. Mitochondria are missing in
 - A. Filamentous fungi
 - B. Protozoan parasites
 - C. Viruses
 - D. Yeasts
 - E. Cestodes

Answers

1. **Answer: E.** The cephalosporin that inhibits prokaryotic cell peptidoglycan cross linkage will not likely be effective against the complex carbohydrate cell wall. Isoniazid, which appears to inhibit mycolic acid synthesis, also would not likely work. Metronidazole would not work on an aerobic organism. Shiga toxin is only effective against eukaryotic ribosomes. Tetracycline (the correct answer) would have the greatest chance of success. However, it may not be taken up by the cell, or the cell could have an effective pump mechanism to get rid of it quickly.
2. **Answer: C.** Mitochondria are found only in eukaryotic organisms so both viruses and bacteria lack them.

Medically Important Bacteria

2

What the USMLE Requires You to Know

The type of disease (major diseases) from presenting symptoms

- You must know the common etiologic agents of the disease and be able to determine the causative agent of the particular case from case clues.
- No distinguishing clues given? Know most common agent(s).
- Epidemiologic clues, symptomatic clues, or organism information given? Know the specific agent.
- Be able to answer basic science questions about disease or organism, predisposing conditions, epidemiology, mechanism of pathogenicity, and major tests used in identification.

The basic science used as clues or tested directly

Morphology

- Gram reaction, basic morphology, motility (*Listeria*), spore formation (*Bacillus* and *Clostridium*)

Physiology

- Obligate aerobes/anaerobes
- A few specific fermentations
- A few specific enzymes (oxidase, urease, catalase, coagulase, superoxide dismutase, hemolysins)
- How bacterial cells grow, divide, and die

Bacterial structures

- Their composition, function, and role in disease

Determinants of pathogenicity

- Toxins
- Factors aiding in invasiveness, pathogenicity, or immune system evasion
- Intracellular parasites (obligate and facultative)

Epidemiology/transmission

- Know how each major disease is acquired
- Particularly important for organisms with animal or arthropod vectors

(Continued)

Note

Nomenclature

Latin bacterial **family** names have “-aceae,” e.g., Enterobacteriaceae.

Genus and species names are **italicized and abbreviated**, e.g., *Enterobacter aerogenes* = *E. aerogenes*.

What the USMLE Requires You to Know (continued)

Laboratory diagnosis

- Serologic/skin tests: specific serology for syphilis
- Stains: acid fast and Gram
- Unusual growth requirements
- Specific media
- Steps in Gram stain and acid fast stain

Diseases

- Common presenting symptoms
- Stages of multistage diseases
- Common complications

Treatment (drug of choice – pharmacology)/prevention

(vaccination, public health, and prophylaxis, where regularly used)

(Basically the same for all other pathogens, too!)

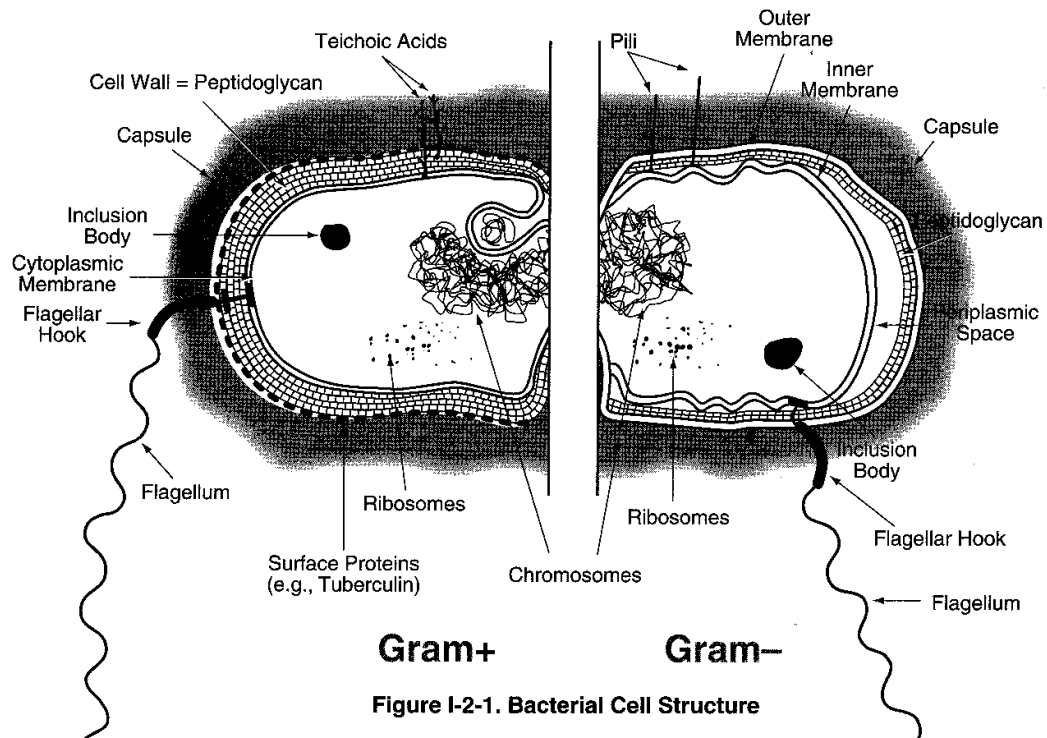


Figure I-2-1. Bacterial Cell Structure

Table I-2-1. Bacterial Envelope (All the Concentric Surface Layers of the Bacterial Cell)

Envelope Structure	Gram + or -	Chemical Composition	Function
Capsule (Non-essential) = Slime = Glycocalyx	Both Gram + Gram -	Polysaccharide gel*	Pathogenicity factor protecting against phagocytosis until opsonized; immunogenic**
Outer membrane	Gram-negative only	Phospholipid/proteins: Lipopolysaccharide Lipid A Polysaccharide	Hydrophobic membrane: LPS = endotoxin Lipid A = toxic moiety PS = immunogenic portion
		Outer membrane proteins	Attachment, virulence, etc.
		Protein porins	Passive transport
Cell wall = peptidoglycan	Gram + (thick) Gram - (thin)	Peptidoglycan-open 3-D net of: N-acetyl-glucosamine N-acetyl-muramic acid amino acids (DAP)	Rigid support, cell shape, and protection from osmotic damage Synthesis inhibited by penicillins and cephalosporins Confers Gram reaction
	Gram-positive only	Teichoic acids***	Immunogenic induces TNF-alpha, IL-1 Attachment
	Acid-fast only	Mycolic acids	Acid-fastness Resistance to drying and chemicals
Periplasmic space	Gram-negative only	"Storage space" between the inner and outer membranes	Enzymes to break down large molecules, (β -lactamases) Aids regulation of osmolarity
Cytoplasmic membrane = inner membrane = cell membrane = plasma membrane	Gram + Gram -	Phospholipid bilayer with many embedded proteins	Hydrophobic cell "sack" Selective permeability and active transport Carrier for enzymes for: Oxidative metabolism Phosphorylation Phospholipid synthesis DNA replication Peptidoglycan cross linkage Penicillin Binding Proteins (PBPs)

* Except *Bacillus anthracis*, which is a polypeptide of poly D-glutamate.

** Except *S. pyogenes* (hyaluronic acid) and type B *N. meningitidis* (sialic acid), which are nonimmunogenic.

*** Teichoic acid: polymers of ribitol or glycerol, bound to cell membrane or peptidoglycan.

Table I-2-2. Other Surface Structures of the Bacterial Cell

Pilus or fimbria 1. Common 2. Sex 3. Virulence	Primarily Gram – *	Glycoprotein (pilin)	Adherence to cell surfaces, including attachment to other bacteria during conjugation
Flagellum	+ and –	Protein (flagellin)	Motility
Axial filaments (internal flagellum)	Spirochetes Gram –	Protein	Motility

*M-protein of group A Strep described as diffuse fimbriate layer or fimbriae.

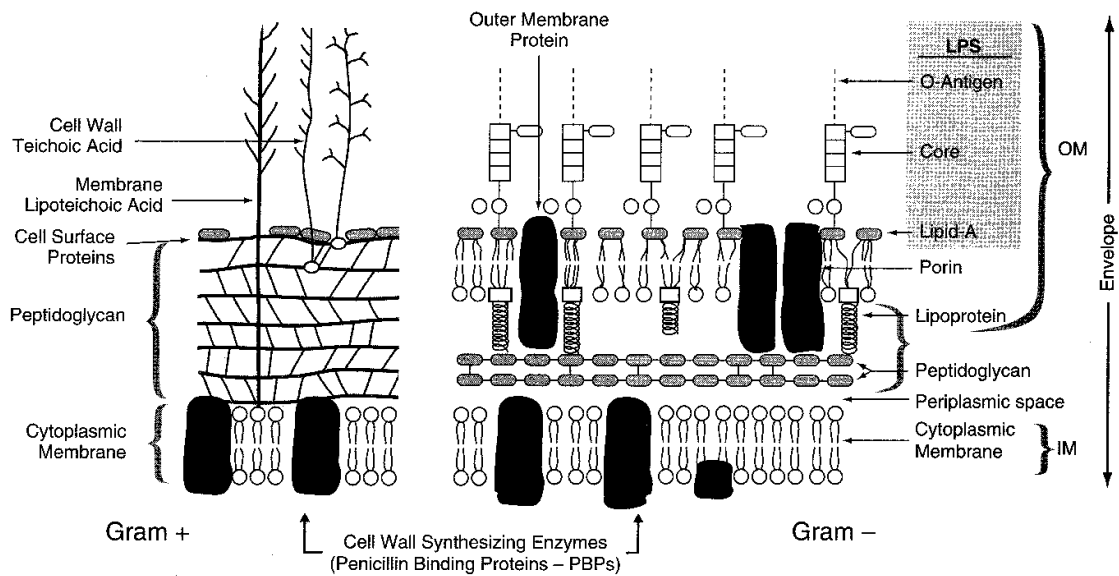


Figure I-2-2. Details of Cell Envelope and Peptidoglycan

STAINS

Table I-2-3. Gram Stain

Reagent	Gram-Positive	Gram-Negative
Crystal Violet (a very intense purple, small dye molecule)	Purple/Blue	Purple/Blue
Gram's Iodine	Purple/Blue (a large dye complex)	Purple/Blue (a large dye complex)
Acetone or Alcohol	Purple/Blue	Colorless
Safranin (a pale dye)	Purple/Blue	Red/Pink

All cocci are Gram-positive except *Neisseria* and *Moraxella*.

All spore formers are Gram-positive.

Background in stain modified for tissues will be pale red.

Table I-2-4. Ziehl-Neelsen Acid Fast Stain (or Kinyoun)

Reagent	Acid Fast	Non-Acid Fast*
Carbol Fuchsin with heat**	Red (Hot Pink)	Red (Hot Pink)
Acid Alcohol	Red	Colorless
Methylene Blue***	Red	Blue

* *Mycobacterium* is acid fast. *Nocardia* is partially acid fast. All other bacteria are non-acid fast except *Legionella micdadei*. Two protozoan parasites (*Cryptosporidium* and *Isospora*) have acid fast oocysts.

** Without the heat, the dye would not go in the mycobacterial cells.

*** Sputa and human cells will be blue.

Processing Sputa

(In labs with fluorescent microscopes)

Screen Sputum Using Auramine-Rhodamine Fluorescent Stain

- Mycobacteria, *Nocardia*, and some other bacteria fluoresce a bright apple green on a black background.
- No antibody is involved.
- Test is sensitive (picks up high percent of AFB) but not very specific (picks up others, too).
- Negatives can be screened in 5 minutes (as opposed to 20 minutes for negative acid fast stains).

All positives are confirmed by acid fast stain.

Table I-2-5. Internal Bacterial Structures*

Structure	Cell Type	Chemical Composition	Function
Nucleoid region No membrane No histones No introns	Gram + and Gram -	DNA RNA Proteins	Genetic material (all essential genes) Primers, mRNA Linker proteins, polymerases
Plasmids	Gram + and Gram -	DNA	Non-essential genetic material Roles in conjugation, drug resistance, toxin production
Ribosomes	Gram + and Gram -	70s (protein/RNA) 30s (16S RNA) 50s (23 and 5s)	Protein synthesis
Granules (various types)	Gram + and Gram -	Glycogen, lipids, polyphosphate, etc.	Storage: polymerization of molecules present in high numbers in cells reduces osmotic pressure. Volutin granules of <i>Corynebacterium diphtheriae</i> are used in clinical identification.
Endospores	Gram + <i>only</i>	Keratin coat, calcium dipicolinate	Resistance to heat, chemicals, and dehydration

*Note that there are no mitochondria or membrane-bound structures like chloroplasts.

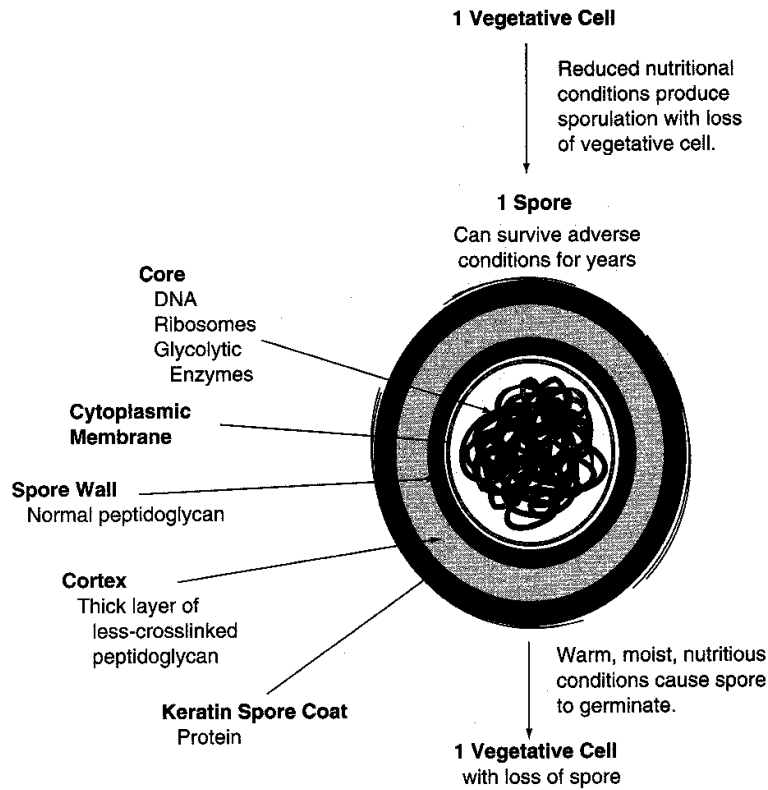


Figure I-2-3. Endospore

Note

Spores of fungi have a reproductive role.

ENDOSPORES

Organisms: *Bacillus* and *Clostridium*

Function

- Survival not reproductive (1 bacterium → 1 spore)
- Resistance to chemicals, dessiccation, radiation, freezing, and heat

Mechanism of Resistance

- New enzymes (i.e., dipicolinic acid synthetase, heat-resistant catalase)
- Increases or decreases in other enzymes
- Dehydration: calcium dipicolinate in core
- Keratin spore coat

BACTERIAL GROWTH AND DEATH

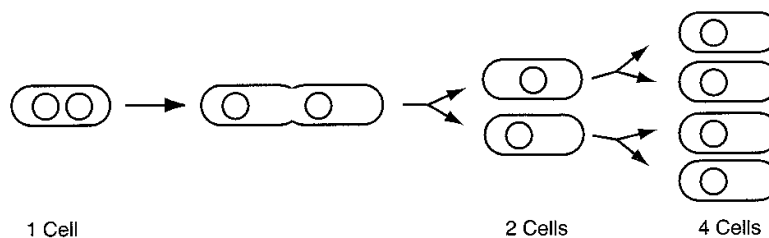


Figure I-2-4. Exponential Growth by Binary Fission

In A Nutshell

Lag Phase

- Initial Phase (only 1 lag phase)
- Detoxifying medium
- Turning on enzymes to utilize medium
- For exam, number of cells at beginning equals number of cells at end of lag phase.

Log Phase

- Rapid exponential growth
- Generation time = time it takes one cell to divide into two. This is determined during log phase.

Stationary Phase

- Nutrients used up
- Toxic products like acids and alkalis begin to accumulate.
- Number of new cells equals the number of dying cells.

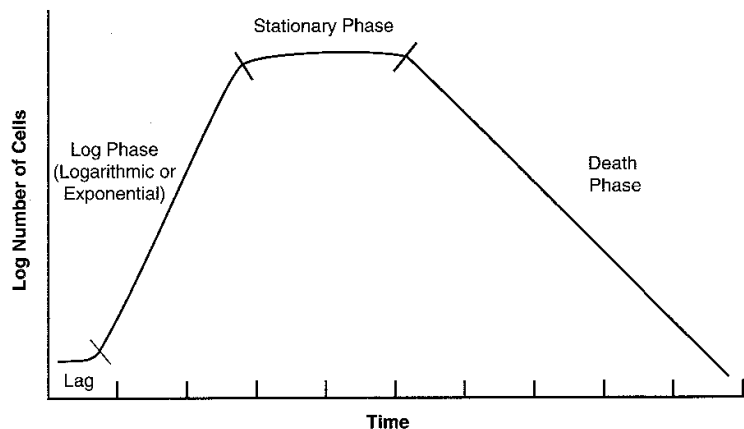


Figure I-2-5. Bacterial Growth Curve

Typical question:

A flask is inoculated to a density of 3×10^3 cells/ml. What is the density of cells in the culture after 50 minutes if the generation time is 20 minutes and the lag time is 10 minutes?

CULTURE OF MICROORGANISMS

- Obligate intracellular pathogens (viruses, rickettsias, chlamydias, etc.): Tissue cultures (cell cultures), eggs, animals, or not at all
- Facultative intracellular or extracellular organisms: Inert lab media (broths and agars)
 - Selective medium (S): A medium that selects for certain bacteria by inclusion of special nutrients and/or antibiotics.
 - Differential medium (D): A medium on which different bacteria can be distinguished by differences in colonial morphology or color.

Table I-2-6. Special Media for Selected Organisms

Organism	Medium
Anaerobes	Thioglycolate
<i>Corynebacterium</i>	Löffler's coagulated serum medium (S) Tellurite agar (D)
Enteric bacteria	Eosin methylene blue (D) MacConkeys (D)
Enteric pathogens	Hektoen enteric agar (D) Xylose-lysine-deoxycholate agar
<i>Vibrio cholerae</i> (likes alkaline growth medium)	TCBS (Thiosulfate Citrate Bile Salts Sucrose agar) (S)
<i>Legionella</i>	Charcoal-yeast extract agar (CYE agar) (S)
<i>Mycobacterium</i>	Löwenstein-Jensen medium (S)
<i>Neisseria</i> from normally sterile sites, <i>Haemophilus</i>	Chocolate agar
<i>Neisseria</i> from sites with normal flora	Thayer-Martin selective medium* (S)

*Thayer-Martin media is a chocolate agar supplemented with vancomycin, nystatin and colistin to inhibit the normal flora, including nonpathogenic *Neisseria*.

Table I-2-7. Miscellaneous Growth Requirements

Cholesterol and purines and pyrimidines	<i>Mycoplasma</i>
Cysteine*	<i>Francisella</i> , <i>Brucella</i> , <i>Legionella</i> , <i>Pasteurella</i>
X (protoporphyrin) and V (NAD)	<i>Haemophilus</i> (<i>influenzae</i> and <i>aegypticus</i> require both)

*The four Sisters Ella and the Cysteine Chapel.

Note

Mnemonic

The four sisters "Ella" worship in the "Cysteine" chapel:

- *Francisella*
- *Brucella*
- *Legionella*
- *Pasteurella*

ANAEROBIC AND AEROBIC

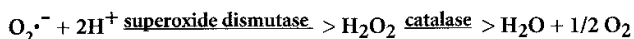


Table I-2-8. Oxygen Requirements and Toxicity

Classification	Characteristics	Important Genera
Obligate aerobes	Require oxygen Have no fermentative pathways Generally produce superoxide dismutase	<i>Mycobacterium</i> <i>Pseudomonas</i> (<i>Bacillus</i>)
Microaerophilic	Requires low but not full oxygen tension	<i>Campylobacter</i> <i>Helicobacter</i>
Facultative anaerobes	Will respire aerobically until oxygen is depleted and then ferment or respire anaerobically	Most bacteria, i.e., <i>Enterobacteriaceae</i>
Obligate anaerobes	1. Lack superoxide dismutase 2. Generally lack catalase 3. Are fermenters 4. Cannot use O ₂ as terminal electron acceptor	<i>Actinomyces</i> * <i>Bacteroides</i> <i>Clostridium</i>

*ABCs of anaerobiosis = *Actinomyces*, *Bacteroides*, and *Clostridium*.

In A Nutshell

Vaccine Efficacy

Neisseria vaccine

50% of cases are serotype B. It is not in the vaccine.

Haemophilus Hib vaccine

95% of cases are invasive disease caused by type B. The vaccine is against type B and is 95% effective

BACTERIAL VACCINES

Childhood Vaccines

DTaP

- Totally acellular
- Components of *B. pertussis*:
 - Pertussis toxoid
 - $\pm$ Filamentous hemagglutinin
 - $\pm$ pertactin (adhesin)
- Diphtheria: diphtheria toxoid
(= inactivated toxin that no longer causes disease but produces immunity)
- Tetanus: tetanus toxoid

DTP

- Same D and T components as in DTaP, except the pertussis component is killed.

HIB (*Haemophilus influenzae* type b)

H. influenzae capsular polysaccharide conjugated to protein (diphtheria toxoid or *Neisseria meningitidis* outer membrane proteins), making it a **T-cell-dependent vaccine** that infants respond to.

Senior Citizens or Asplenic

Streptococcus pneumoniae

Capsular polysaccharides of 23 different *Pneumococcus* strains

Limited Usage

Neisseria meningitidis

- Four capsular polysaccharides: Y, W-135, C, and A
- B serotype about 50% of cases in USA but capsule is sialic acid so not good immunogen.
- Used in outbreaks along with antibiotic prophylaxis
- Routine usage is in military recruits.

Salmonella typhi (ty21)

- Attenuated bacterium for travelers to endemic typhoid areas

Yersinia pestis

- Killed **cellular** vaccine (F-1 antigen)
- Military in endemic areas and *Y. pestis* laboratory workers

Bacillus anthracis

- **Supernatant** of partially purified proteins
- Military or occupational usage

BCG = Bacille Calmette Guérin (BCG)

- **Attenuated** (living) strain of *Mycobacterium bovis*
- Doesn't prevent pulmonary tuberculosis but reduces dissemination

GRAM-STAINING REACTIONS

(†Marked organisms have high numbers of questions in the pool.)

Table I-2-9. Gram-Positive Bacteria

Cocci	<i>Staphylococcus</i> † <i>Streptococcus</i> †
Rods	Aerobic or facultative anaerobic <i>Bacillus</i> <i>Listeria</i> <i>Corynebacterium</i> † <i>Nocardia</i> <i>Mycobacterium</i> † Anaerobic <i>Clostridium</i> † <i>Actinomyces</i> <i>Eubacterium</i> <i>Propionibacterium</i> <i>Lactobacillus</i>

*Spore formers are *Bacillus* and *Clostridium*.

Table I-2-10. Non-Gram-staining Bacteria*

Mycoplasmataceae
<i>Mycoplasma</i> † <i>Ureaplasma</i>

*Note also:

Poorly visible on traditional Gram stain: *Mycobacterium* does not stain well with the Gram stain due to its waxy cell wall. It is considered Gram-positive.

Most **spirochetes**, **chlamydiae**, and **rickettsias** are so thin that the color of the Gram stain cannot be seen. All have Gram-negative cell walls.

Legionella (Gram-negative) also does not stain well with the traditional Gram stain unless counterstain time is increased.

Table I-2-11. Gram-Negative Bacteria

Aerobic	Cocci <i>Neisseria</i> † <i>Moraxella</i>
	Rods <i>Pseudomonas</i> <i>Legionella</i> <i>Brucella</i> <i>Bordetella</i> † <i>Francisella</i> Helical (and microaerophilic) <i>Campylobacter</i> <i>Helicobacter</i>
Facultative anaerobic rods	Enterobacteriaceae † <i>Escherichia</i> † <i>Shigella</i> <i>Salmonella</i> † <i>Citrobacter</i> <i>Klebsiella</i> <i>Enterobacter</i> <i>Serratia</i> <i>Proteus</i> <i>Yersinia</i> †
	Vibrionaceae <i>Vibrio</i>
	Pasteurellaceae <i>Pasteurella</i> <i>Haemophilus</i> †
Anaerobic straight to helical rods	<i>Bacteroides/Prevotella</i> <i>Fusobacterium</i>
Spirochetes	<i>Treponema</i> † <i>Borrelia</i> <i>Leptospira</i>
Rickettsiaceae and relatives	<i>Rickettsia</i> † <i>Bartonella (Rochalimea)</i> <i>Coxiella</i> <i>Ehrlichia</i>
Chlamydiaceae	<i>Chlamydia</i> †

GENUS: *STAPHYLOCOCCUS*

- Gram-positive
- Cocci arranged in clusters
- Catalase positive

Table I-2-12. Medically Important Staphylococci*

Species	Coagulase/ Hemolysis	Additional Virulence Factors and Lab ID	Common Diseases
<i>Staphylococcus aureus</i>	Coagulase positive/ β -hemolysis	Protein A TSST-1 Enterotoxins Exfoliatins Cytolysins	Infective endocarditis (dominant cause in IV drug abusers) Abscesses Toxic shock syndrome Gastroenteritis Suppurative lesions Osteomyelitis
<i>Staphylococcus epidermidis</i>	Coagulase negative/ no hemolysis	Susceptible to novobiocin Normal skin flora Adherence: biofilm	Catheter & prosthetic device and infections Endocarditis in IV drug abusers
<i>Staphylococcus saprophyticus</i>	Coagulase negative/ no hemolysis	Resistant to novobiocin	Urinary tract infections in newly active adolescent women

*All members of the genus *Staphylococcus* are catalase positive.

Staphylococcus aureus

Distinguishing Characteristics

- β -hemolytic, yellow colonies of Gram + cocci (clusters) on blood agar (BA)
- Catalase-positive, coagulase-positive
- Salt tolerant; ferments mannitol on mannitol salt agar

Reservoir

- Normal flora on nasal mucosa and skin.

Transmission

- Spread via the hands and sneezing
- Surgical wounds; lungs of cystic fibrosis patients
- Foods associated with food poisoning:
Ham or canned meats, custard pastries, and potato salad

Predisposing Factors for Infections

- Surgery or any break in skin, surgical packing or sutures or any foreign body (e.g., tampons); ventilators
- Severe neutropenia ($<500/\mu\text{L}$); cystic fibrosis
- IV drug abuse (IV drug abusers in general have more *S. aureus* on skin than *S. epidermidis*)
- Chronic granulomatous disease (CGD) Staph are catalase +

Pathogenesis

- Numerous enzymes and exotoxins
 - Protein A inhibits phagocytosis**; binds Fc portion of antibody.
 - Enterotoxins A-E** (heat stable 60°C, 10 min); fast 2–6 hours
 - TSST-1** causes toxic shock syndrome. **Decreases normal liver clearance of endotoxin**; **superantigen** nonspecifically activates large numbers of T helper cells without processed antigen.
 - Coagulases: convert fibrinogen to a fibrin clot**
 - Cytolytic toxins**, e.g., **Staphylococcal alpha toxin**, a pore-forming toxin that damages human cell membranes
 - Exfoliatins** involved in scalded skin syndrome (SSS) and formation of the bullae seen in staphylococcal impetigo

Table I-2-13. Staphylococcal Diseases

Diseases*	Clinical Symptoms	Pathogenicity Factors
Gastroenteritis (food poisoning) - toxin ingested preformed in food	2–6 hours after ingesting toxin: nausea, abdominal pain, vomiting, followed by diarrhea	Enterotoxins A–E preformed in food
Infective endocarditis	Fever, malaise, leukocytosis, heart murmur (may be absent initially)	Fibrin-platelet mesh, cytolytic toxins
Abscesses and mastitis	Subcutaneous tenderness, redness and swelling; hot	Coagulase, probably the cytolsins
Toxic shock syndrome	Fever, hypotension, scarlatiniform rash which desquamates (particularly on palms and soles), multiorgan failure	TSST-1
Impetigo	Erythematous papules to bullae	Coagulase, exfoliatins
Pneumonia	Productive pneumonia with rapid onset, high rate of necrosis and high fatality; nosocomial, ventilator, post-influenza, IV drug abuse, CF, CGD*, etc.	All
Surgical infections	Fever with cellulitis and/or abscesses	Coagulase, exfoliatins, ± TSSTs
Osteomyelitis	Bone pain, fever, ± tissue swelling, redness; lytic bone lesions on imaging	Cytolsins, coagulase

CF = cystic fibrosis; CGD = chronic granulomatous disease

*All except gastroenteritis may have scalded skin syndrome associated with the infection, especially in babies and the elderly.

Treatment

- Early on *S. aureus* (now known as methicillin-sensitive *S. aureus*, **MSSA**) acquired a multiple drug resistant plasmid with resistance to early beta-lactams (via a beta-lactamase) and most other antibiotics.
- Methicillin (and nafcillin) were developed.
- Methicillin-resistant *S. aureus* (**MRSA**) (due to changes in major penicillin-binding proteins) is **commonly resistant to all antibiotics except vancomycin and fusidic acid**.
- Topical mupirocin used to reduce nasal colonization

Prevention

- Basic hospital infection control

GENUS: STREPTOCOCCUS

- Gram-positive
- Chains or pairs of cocci
- Catalase negative

Hemolysis varies by species: Beta = clear; alpha = partial (green); gamma = non-hemolytic.

Streptococci are **serogrouped** using known antibodies to the cell wall carbohydrates

(Lancefield's Groups A-O)

Streptococci are **serotyped** using known antibodies to the

- Capsules for *Streptococcus pneumoniae*
- M-protein for *Streptococcus pyogenes*

Table I-2-14. Medically Important Streptococci¹

Species	Lancefield Group	Typical Hemolysis	Important Lab Characteristics
<i>S. pyogenes</i>	A	beta	Bacitracin-sensitive PYR test ² positive
<i>S. agalactiae</i>	B	beta	Bacitracin-resistant Hippurate utilized CAMP test positive
<i>Enterococcus faecalis</i> ³	D	alpha, beta or none	Growth in 6.5% NaCl PYR test positive
<i>S. bovis</i> ⁴	D	alpha or none	No growth in 6.5% NaCl
<i>S. pneumoniae</i>	Not groupable ⁵	alpha	Bile-soluble Inhibited by optochin
Viridans group	Not groupable	alpha	Not bile-soluble Not inhibited by optochin

¹All streptococci are catalase negative.

²PYR test demonstrates the presence of pyrrolidonyl arylamidase. *S. pyogenes* is the one beta hemolytic *Streptococcus* that is positive; enterococci are positive.

³*Enterococcus faecalis* = *Streptococcus faecalis*. The enterococci belong to the Streptococcaceae family.

⁴*S. bovis* is a non-enterococcal group D organism.

⁵Not serogrouped because they lack the carbohydrate cell wall antigens.

Streptococcus pyogenes* (Group A Streptococcus)*Distinguishing Characteristics**

- Beta-hemolytic colonies inhibited by bacitracin on BA
- Gram positive cocci in chains, catalase-negative, PYR+

Reservoir

Human throat and skin

Transmission

Spread by respiratory droplets or direct contact

In a Nutshell

S. pyogenes

- group A
- beta hemolytic
- bacitracin sensitive

Pathogenesis

- **Hyaluronic acid capsule** (a polysaccharide) is **non-immunogenic**; inhibits phagocytic uptake.
- **M-protein**: virulence factor, **antiphagocytic**, used to **TYPE** group A Strep; **M12 strains**-associated with **acute glomerulonephritis**

Toxins

- **Streptolysin O**: **immunogenic**, hemolysin/cytolysin
- **Streptolysin S**: not immunogenic, hemolysin/cytolysin

Spreading factors:

- **Streptokinase**: breaks down fibrin clot
- **Streptococcal DNase**: liquefies pus, extension of lesion
- **Hyaluronidase**: hydrolyzes the ground substances of the connective tissues; important to spread in cellulitis

Exotoxins A–C

(Pyrogenic or erythrogenic exotoxins.)

- **Phage-coded** (i.e., the cells are lysogenized by a phage.)
- Cause **fever and the rash** of Scarlet fever
- **Inhibit liver clearance of endotoxin** (from normal flora), creating shock-like conditions
- **Superantigens**: activate many helper T cells by bridging T cell receptors and MHC class II markers without processed antigen

Diseases

- *Streptococcus pyogenes* causes a wide variety of acute infections; some have immunologic sequelae.

Table I-2-15. Acute (Suppurative) *Streptococcus Pyogenes* Infections

Disease	Clinical Symptoms (Sx)
Pharyngitis	Abrupt onset of sore throat, fever, malaise and headache; tonsillar abscesses and tender anterior cervical lymph nodes
Scarlet fever	Above followed by a blanching, "sandpaper" rash; circumoral pallor; palms and soles are generally spared; strawberry tongue; nausea/vomiting
Pyoderma/impetigo	Pus-producing skin infection (honey-crusted lesions)

Also, cellulitis/necrotizing fasciitis (flesh-eating bacteria!), puerperal fever, lymphangitis, pneumonia, a toxic shock-like syndrome, etc.

Table I-2-16. Non-suppurative Sequelae to Group A Streptococcal Infections

Disease	Sequelae to	Mechanism/Symptoms (Sx)
Rheumatic fever	Pharyngitis with Group A strep (not Group C)	Antibodies to heart tissue; mean = 19 d: fever, joint inflammation, carditis, erythema marginatum (chorea later) (Type II)
Acute glomerulonephritis (M12 serotype)	Pharyngitis or cutaneous	Immune complexes bound to glomeruli/pulmonary edema and hypertension, dark urine (Type III)

Lab Notes

- For Strep throat: **rapid antigen test** (misses about 25% of the Strep throats); culture all “negatives.”
- **ASO titer for rheumatic fever (>200 is significant)**

Treatment

- Beta-lactam drugs or erythromycin

Prevention

- Penicillin in rheumatic fever patients to prevent recurrent *S. pyogenes* pharyngitis

***Streptococcus agalactiae* = Group B Streptococci (GBS)**

Distinguishing Characteristics

- **Beta-hemolytic**, bacitracin-resistant on BA
- **Gram-positive cocci in chains**
- **Group B** (determined by antiserum against cell wall carbohydrate in precipitin test)
- Catalase-negative, **hydrolyze hippurate**
- **CAMP test-positive: CAMP factor is a polypeptide** that “compliments” a *Staph aureus* sphingomyelinase to make an area of new complete beta-hemolysis.

Reservoir

- Colonizes human vagina (15–20% of women)

Transmission

- Newborn infected during birth
- Increased risk with **prolonged rupture of membranes**

Pathogenesis

- **Capsule**
- **Beta-hemolysin** and CAMP factor (an incomplete hemolysin)

Diseases

Neonatal septicemia and meningitis

- Group B Strep is the most common causative agent.

Treatment

- Ampicillin with cefotaxime or gentamicin

Prevention

- Treat mother prior to delivery if she has had a previous baby with GBS, has documented GBS colonization, or prolonged rupture of membranes.

In a Nutshell

S. agalactiae

- group B
- beta hemolytic
- bacitracin resistant
- hydrolyzes hippurate



Figure I-2-6.
Streptococcus pneumoniae

In a Nutshell

S. pneumoniae

- alpha hemolytic
- bile soluble
- optochin sensitive

In A Nutshell

Typical Pneumonia

Bacterial pneumonia such as *Streptococcus pneumoniae* elicits neutrophils; arachidonic acid metabolites (acute inflammatory mediators) cause pain and fever.

Pneumococcus produces a lobar pneumonia with a productive cough, grows on blood agar, and usually responds well to penicillin treatment.

Streptococcus pneumoniae (Pneumococcus)

Distinguishing Characteristics

- Alpha-hemolytic colonies **inhibited by optochin** on BA
- Gram-positive, lancet-shaped diplococci (or short chains)
- Lysed by bile

Reservoir

Human upper respiratory tract

Transmission

- **Respiratory droplets**; not considered highly communicable
- Often colonizes without causing disease.

Predisposing Conditions for Pneumonia

- Antecedent influenza or measles infection: damage to mucociliary elevator
- Chronic obstructive pulmonary disorders
- Congestive heart failure
- Alcoholism
- Asplenia predisposes to septicemia

Pathogenesis

- IgA protease: colonization
- Teichoic acids: attachment
- **Polysaccharide capsule: major virulence factor; retards phagocytosis** through inhibition of antibody-independent opsonization via the alternative complement pathway.
- **Quellung reaction** positive (swelling of capsule with type-specific antiserum)
- **Latex particle agglutination** test for capsular antigen in spinal fluid diagnostic for meningitis
- **Antibody to the capsule (>80 serotypes) provides type-specific immunity.**
- **Pneumolysin O:** hemolysin/cytolysin
 - **Damages respiratory epithelium** (hemolysin similar to streptolysin O, which damages eukaryotic cells).
 - (Inhibits leukocyte respiratory burst and inhibits classical complement fixation.)
- **Pneumococcus in alveoli** stimulate release of fluid and red and white cells producing "rusty sputum."
- **Peptidoglycan/teichoic acids highly inflammatory** in CNS.

Diseases

Bacterial Pneumonia

- Most common bacterial cause, especially after 65 years but also in infants
- Sx: "big" shaking chill, high fever, lobar with productive blood-tinged sputum

Adult Meningitis

Most common cause. CSF generally has very high white cell count, low glucose.

Otitis Media and Sinusitis in Children

Most common cause

Treatment

- Penicillin G
- Resistance (both low level and high level) is chromosomal (altered penicillin-binding proteins); major concern in meningitis. (vancomycin $\pm$ rifampin used.)

Prevention

Vaccine 23 serotypes of capsule

Viridans Streptococci (*S. sanguis*, *S. mutans*, etc.)

Distinguishing Characteristics

- Alpha-hemolytic, resistant to optochin
- Gram-positive cocci in chains

Reservoir

Human oropharynx (normal flora)

Diseases

Dental Caries

S. mutans dextran-mediated adherence glues oral flora onto teeth, forming plaque and causing caries.

Infective Endocarditis

- Sx: malaise, fatigue, anorexia, night sweats, weight loss
- Predisposing conditions: damaged (or prosthetic) heart valve and dental work without prophylactic antibiotics or extremely poor oral hygiene

Pathogenesis

Dextran (biofilm)-mediated adherence onto tooth enamel or damaged heart valve and to each other (vegetation). Growth in vegetation protects organism from immune system.

Treatment

Penicillin G with aminoglycoside for endocarditis

Prevention

For individuals with damaged heart valve: prophylactic penicillin prior to dental work

In a Nutshell

Viridans streptococci

- alpha hemolytic
- bile resistant
- optochin resistant

In a Nutshell

E. faecalis

- group D
- bile esculin positive
- grows in 6.5% salt

GENUS: *ENTEROCOCCUS*

- Catalase negative
- PYR+

Enterococcus faecalis

Distinguishing Characteristics

- Group D Gram-positive cocci in chains
- PYR test +
- Catalase-negative, varied hemolysis
- Hydrolyzes esculin in 40% bile and 6.5% NaCl (bile esculin agar turns black)

Reservoir

Human colon, urethra ± and female genital tract

Pathogenesis/ Predisposing Conditions

- Bile/salt tolerance allows survival in bowel and gall bladder.
- During medical procedures on GI or GU tract: *E. faecalis* → bloodstream → previously damaged heart valves → endocarditis

Diseases

Urinary, biliary tract infections

Infective (subacute) endocarditis in persons (often elderly) with damaged heart valves

Treatment

- All strains carry some drug resistance.
- Some **vancomycin-resistant strains of *Enterococcus faecium* or *E. faecalis***: no reliably effective treatment.
- (VanA strains have UDP-N-acetylmuramyl pentapeptide with the terminal D-alanyl-D-alanine replaced with D-alanyl-D-lactate, which functions in cell wall synthesis but is not blocked by vancomycin.)

Prevention

Prophylactic use of penicillin and gentamicin in patients with damaged heart valves prior to intestinal or urinary tract manipulations

Gram-Positive Rods

Table I-2-17. Medically Important Gram-positive Rods

	Spore Formation	Aerobic Growth	Motility	Exotoxin	Intracellular Growth	IC Host?***
<i>Bacillus</i>	+	+	+	+	—	No
<i>Clostridium</i>	+	—	+	+	—	No
<i>Corynebacterium</i>	—	+	—	+	—	No
<i>Listeria</i>	—	+	+ tumbling, actin “jetting”	—	+	Yes

* Most *Bacillus* spp. are motile (except *Bacillus anthracis*, which is nonmotile).

** Most clostridia are motile (except for *Clostridium perfringens*, which is nonmotile).

*** Is organism a significant problem in immunocompromised (IC) hosts?

GENUS: *BACILLUS*

- Gram-positive rods
- Spore-forming
- Aerobic

Bacillus anthracis

Distinguishing Characteristics

- **Large**, boxcar-like, Gram-positive, **spore-forming rods**
- **Capsule is polypeptide (poly-D-glutamate)** and the only non-polysaccharide one. It is immunogenic.

Reservoir

Animals, skins, soil

Transmission

Contact with infected animals or inhalation of spores from animal hair and wool. Spores survive long after animal dies.

Pathogenesis

Capsule: Polypeptide, antiphagocytic

Anthrax toxin

Includes three protein components:

- **Protective antigen** (B component)—mediates entry of LF or EF into eukaryotic cells
- **Lethal factor**—kills cells
- **Edema factor** is an adenylate cyclase (calmodulin-activated like pertussis adenylate cyclase).

Disease

Anthrax (rare in humans)

Cutaneous anthrax: Papule → papule with vesicles (malignant pustules) → central necrosis (eschar) with erythematous border often with painful regional lymphadenopathy; fever in 50%

Pulmonary (Wool Sorter's Disease): Life-threatening **pneumonia**; cough, fever, malaise, and ultimately facial edema, dyspnea, diaphoresis, cyanosis, and shock with **mediastinal hemorrhagic lymphadenitis**

Treatment

Ciprofloxacin or doxycycline

Prevention

Vaccine: cell free vaccine for people in high-risk occupations

Bacillus cereus

- **Spores** found widely in nature, including food, and are not killed by boiling.
- **Food poisoning** associated with food held warm (not hot)
- Two possible toxins:
 - **Emetic toxin: fast (1–6 hours)**, similar to *S. aureus* with vomiting and diarrhea; associated with **fried rice**
 - **Diarrheal toxin** (meats, sauces): 18 hours, similar to *E. coli*; LT: increasing cAMP → watery diarrhea

GENUS: *LISTERIA*

- Gram-positive, nonspore-forming rods
- Facultative intracellular
- Tumbling motility

Listeria monocytogenes

Distinguishing Characteristics

- Small Gram-positive rods
- Beta hemolytic, nonspore-forming rod on BA
- Tumbling motility in broth; actin jet motility in cells
- Facultative intracellular parasite
- Cold growth

Reservoir

- Widespread: animals (gastrointestinal and genital tracts), **unpasteurized milk products**, plants, and soil
- Cold growth: soft cheeses, deli meats, cabbages (coleslaw)

Transmission

Foodborne, across the placenta, or by contact during delivery

Pathogenesis

- **Listeriolysin O, a β -hemolysin**: facilitates rapid egress from phagosome into cytoplasm, thus evading killing when lysosomal contents are dumped into phagosome; "jets" directly (by actin filament formation) from cytoplasm to another cell.
- **Immunologic immaturity predisposes to serious infection.**

Diseases

Listeriosis (human, peaks in summer)

- Healthy adults and children: generally asymptomatic or diarrhea with low % carriage
- Pregnant women: symptomatic carriage, septicemia characterized by fever and chills; can cross the placenta in septicemia.

Neonatal Disease

Early-onset: (granulomatosis infantiseptica) *in utero* transmission; sepsis with high mortality; disseminated granulomas with central necrosis.

Late-onset: 2–3 weeks after birth from fecal exposure; meningitis with septicemia.

Immunocompromised Patients (IC pts)

- **Septicemia and meningitis** (most common clinical presentation)
- *Listeria* meningitis is the most common cause of meningitis in **renal transplant patients and adults with cancer.**

Treatment

Ampicillin with gentamicin added for IC patients

Prevention

Precautions with food may reduce incidence.

GENUS: *CORYNEBACTERIUM*

- Gram-positive rods
- Nonspore-forming, nonmotile
- Aerobic

Corynebacterium diphtheriae

Distinguishing Characteristics

- Gray to black colonies of **club-shaped** Gram-positive rods arranged in V or L shapes on **tellurite medium**.
- **Granules (volutin)** produced on Loeffler's coagulated serum medium stain metachromatically
- Aerobic, nonspore-forming
- **Toxin-producing strains have β -prophage** carrying genes for the toxin (**lysogeny, β -corynephage**). The phage from one patient with diphtheria can infect the normal nontoxigenic diphtheroid of another person and cause diphtheria.

Transmission

Bacterium or phage via respiratory droplets from oropharynx of infected person

Pathogenesis

Organism **not invasive**; colonizes epithelium of oropharynx or skin in cutaneous diphtheria.

Diphtheria toxin (A-B component)—**inhibits protein synthesis by adding ADP-ribose to EF-2.**

- Effect on oropharynx:
 - **Dirty gray pseudomembrane** (made up of dead cells and fibrin exudate, bacterial pigment)
 - **Extension into larynx/trachea $\rightarrow$ obstruction**
- Effect of systemic circulation $\rightarrow$ **heart & nerve damage**

Disease

Diphtheria

Sore throat with **pseudomembrane**, **bull neck**, potential respiratory obstruction, **myocarditis**, cardiac dysfunction, **recurrent laryngeal nerve palsy**, and lower limb polyneuritis.

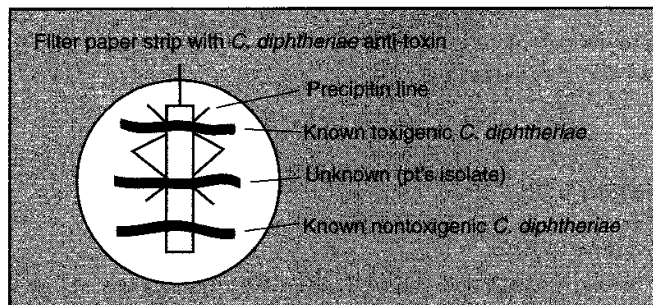


Figure I-2-7. Elek Test

Lab Notes

- Elek test to document toxin production.
- Toxin produced by Tox+ strains diffuses away from growth.
- Antitoxin diffuses away from the strip of filter paper.
- Precipitin lines form at zone of equivalence.

Treatment

Erythromycin and antitoxin

Prevention

Toxoid vaccine (formaldehyde-modified toxin is still immunogenic but with reduced toxicity), part of DTaP, DTP, or Td

GENUS: *ACTINOMYCES*

- Anaerobic BACTERIA
- Gram-positive rods to branching filaments
- Not acid fast

Actinomyces israelii

Distinguishing Characteristics

Anaerobic, Gram-positive branching rods

Reservoir

Human; normal flora of gingival crevices and female genital tract

Transmission

Endogenous

Pathogenesis

Invasive growth in tissues with compromised oxygen supply; anaerobic growth

Disease

Actinomycosis

- Generally not painful but **very invasive**, penetrating all tissues including bone
- **Tissue swelling** → **draining abscesses** (sinus tracts) with “**sulfur**” **granules** (hard yellow microcolonies) in exudate that can be used for microscopy or culture
- Only in tissues with low oxygenation (E_h)

Forms

- **Cervicofacial (lumpy jaw):** dental trauma or poor oral hygiene
- **Pelvic:** from thoracic or sometimes IUD's
- **Thoracic:** aspiration with contiguous spread
- **Abdominal:** surgery or bowel trauma
- **CNS:** **solitary brain abscess** most common

Treatment

Ampicillin or penicillin G and surgical drainage

In a Nutshell

Actinomyces

- Gram +
- Not acid fast
- Anaerobic
- Endogenous infection
- Penicillin

In A Nutshell

Nocardia

- Gram +
- Partially acid fast
- Aerobic
- Exogenous
- Sulfa drugs

GENUS: *NOCARDIA*

- BACTERIA
- Gram-positive filaments breaking up into rods
- Aerobic
- Partially acid-fast (some areas of smear will be blue and some red)

Nocardia asteroides

Distinguishing Characteristics

- Aerobic
- Gram-positive branching rods
- Partially acid-fast

Reservoir

Soil, dust

Transmission

Airborne or traumatic implantation

Pathogenesis

- No toxins or virulence factors known
- **Immunosuppression and cancer predispose** to pulmonary infection.

Diseases

Nocardiosis

- **Cavitary bronchopulmonary nocardiosis**
 - Sx: cough, fever, dyspnea
 - May spread hematogenously to brain (**brain abscesses**)
- **Cutaneous/subcutaneous nocardiosis**
 - Starts with traumatic implantation
 - Sx: **cellulitis** with swelling → **draining subcutaneous abscesses** with **granules** (mycetoma)

Treatment

Sulfonamides (high dose) or trimethoprim/sulfamethoxazole

Table I-2-18. Mycobacteria and Close Relatives

Genus	Aerobic Growth?	Acid Fast?	Morphology (Don't memorize!)
<i>Corynebacterium</i>	Yes	Not AF	Rods
<i>Actinomyces</i>	No, anaerobe	Not AF	Rods, filaments
<i>Nocardia</i>	Yes	Partially AF	Rods, filaments
<i>Mycobacterium</i>	Yes, obligate aerobe	Yes, AF	Rods

GENUS: MYCOBACTERIUM

- Acid fast rods with waxy cell wall
- Obligate aerobe

Cell Wall

- Unique: **high concentration of lipids** containing long chain fatty acids called mycolic acids
- Structural organization shown in Figure II-8.

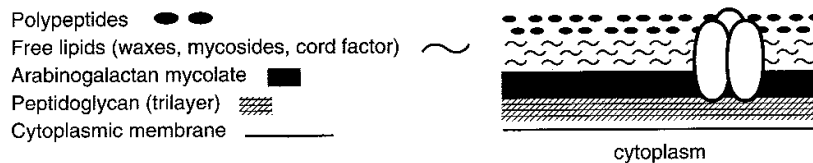


Figure I-2-8. Mycobacterial Cell Wall

- Wall makes mycobacteria highly resistant to
 - Desiccation
 - Many chemicals (including NaOH used to kill other bacteria in sputa before neutralizing and culturing)
- Mycobacteria are sensitive to UV

Mycobacterium tuberculosis

Distinguishing Characteristics

- Auramine-rhodamine staining bacilli (fluorescent apple green); no antibody involved (sensitive but not specific)
- Acid fast
- Aerobic, slow growing on Lowenstein-Jensen medium; new culture systems (broths with palmitic acid) faster
- Produces niacin

- Produces a **heat-sensitive catalase**:
 - Catalase negative at 68°C (standard catalase test)—(other mycobacterial catalases are heat insensitive)
 - Catalase active at body temperature.

Reservoir

Human lungs

Transmission

Respiratory droplets and droplet nuclei

Predisposing Factor

For active disease is poverty, HIV infection, or any CMI system immunosuppression.

Pathogenesis

Facultative Intracellular Organism

Sulfatides (sulfolipids in cell envelope)

- Inhibit the phagosome-lysosomal fusion allowing intracellular survival. (If fusion occurs, waxy nature of cell envelope reduces killing effect.)

Cord factor (trehalose di-mycolate)

- Causes **serpentine growth** *in vitro*
- **Inhibits leukocyte migration; disrupts mitochondrial respiration and oxidative phosphorylation**

Tuberculin (surface protein) along with mycolic acid → delayed hypersensitivity and CMI

- Granulomas and caseation mediated by **cell-mediated immunity** (CMI)
- No exotoxins nor endotoxin; damage done by immune system

Disease

Tuberculosis

- Causative agents: *Mycobacterium tuberculosis* and *M. bovis*
- Complex disease: pulmonary, urinary tract, any organ or miliary (disseminated) (see pathology notes).
- Primary infection: organisms replicate in naïve macrophages, killing macrophages until CMI is set up.
- Most people heal without disease; **some organisms walled off in the Ghon complex remain viable unless treated.**
- Post primary (reactivational TB) erosion of granulomas into airways (high oxygen) later in life under conditions of reduced T-cell immunity leads to mycobacterial replication and disease symptoms.



Figure I-2-9.
Cord Factor

Diagnosis

- Microscopy on specimen: auramine-rhodamine stain (fluorescent apple green); no antibody involved; very sensitive; if positive followed by acid fast stain.
- **PPD skin test (Mantoux): measure zone of induration at 48–72 hours; positive if zone of induration at 48–72 hours is:**
 - ≥ 5 mm in HIV+ or anyone with recent TB exposure; AIDS patients have reduced ability to mount skin test.
 - ≥ 10 mm in high-risk population: IV drug abusers, people living in poverty, or immigrants from high TB area.
 - ≥ 15 mm in low-risk population.
- **Positive skin test indicates only exposure but not necessarily active disease.**
- Slow-growing (3–6 weeks) colony on Lowenstein-Jensen medium (faster new systems)
- Organisms produce **niacin** and are **catalase-negative** (68°C).
- **No serodiagnosis**

Treatment

- **Multiple drugs critical** to treat infection
- Standard observed short-term therapy for uncomplicated pulmonary TB (rate where acquired <4%):
 - First 2 months: isoniazid + rifampin + pyrazinamide
 - Next 4 months: isoniazid and rifampin
- Ethambutol or streptomycin added for possible drug-resistant cases until susceptibility tests are back (if area acquired has >4% DRM TB)

Prevention

- **Isoniazid** taken for 6–9 months can prevent TB in persons with infection but no clinical symptoms.
- Bacille-Calmette-Guérin (BCG) **vaccine** containing live, attenuated organisms may prevent disseminated disease. Not commonly used in the U.S.
- UV lights or HEPA filters used to treat potentially contaminated air.

Mycobacteria Other Than Tuberculosis (MOTTs)

- (MOTTs) = Non-tuberculous Mycobacteria = atypical **Mycobacteria**
- **Non-contagious!**
- Found in **surface waters, soil**, cigarettes; most common in southeastern U.S.
- Runyon terminology still used, particularly for pigmented ones. Dr. Runyon grew two tubes of each in the dark and then exposed one of the pair to bright light for 2 hours after the cultures were grown up.

Table I-2-19. Runyon Grouping of MOTTs

Species*	Runyon Grouping	Pigment in Dark**	Pigment in Light	Growth
<i>M. kansasii</i>	Photochromogen	—	+	slow
<i>M. scrofulaceum</i>	Scotochromogens	+	+	slow
<i>M. avium-intracellulare</i>	Non-chromogens	—	—	slow

* Not classified: *M. tuberculosis*, *M. bovis*, and *M. ulcerans*. None produce pigments. Also not included is *M. leprae*, which cannot be grown in the lab.

** Pigments are carotenoids.

Note: Speciation now uses biochemical methods; some strains of *M. avium-intracellulare* complex (MAI or MAC) may produce pale yellow pigments.

Diseases

Pulmonary/Gastrointestinal/Disseminated

- Patients: AIDS (prophylaxis < 75 CD4+ cells/mm³), **cancer, chronic lung disease**
- *M. avium-intracellulare*, *M. kansasii*.

Mycobacterial lymphadenitis

- Usually solitary cervical lymph nodes (surgically removed) in kids.
- *M. scrofulaceum*.

Soft-Tissue Infections

M. marinum: cutaneous granulomas in tropical fish enthusiasts (fish tank granuloma) or scuba divers from abrasions on coral

Mycobacterium leprae

Distinguishing Characteristics

- Acid fast rods (seen in punch biopsy)
- Obligate intracellular parasite (cannot be cultured *in vitro*)
- Optimal growth at less than body temperature

Reservoir

- Human mucosa, skin, and nerves are the only significant reservoir.
- Some infected armadillos in Texas and Louisiana

Transmission

Nasal discharge from untreated lepromatous leprosy patients

Pathogenesis

- Obligate intracellular parasite
- Cooler parts of body, e.g., skin, mucous membranes, and peripheral nerves

Note

M. tuberculosis with normal CD4 count or with low CD4 count (disseminated).

MAI only late with low CD4 count.

Disease

Leprosy: A continuum of disease, which usually starts out with an indeterminate stage called "borderline."

Table I-2-20. Two Extreme Forms of Leprosy

	Tuberculoid		Lepromatous
Cell-mediated immune system	Strong CMI	B o r d e r l i n e	Weak CMI
Lepromin skin test	Lepromin test +		Lepromin test –
Number of organisms in tissue	Low		High (foam cells totally filled)
Damage from	Immune response (CMI killing infected cells) Granuloma formation → nerve enlargement/damage Loss of sensation → burns and trauma		Large number of intracellular organisms Nerve damage from overgrowth of bacteria in cells Loss of sensation → burns and trauma
Number of lesions and other symptoms	Fewer lesions: macular; nerve enlargement, paresthesia		Numerous lesions becoming nodular; loss of eyebrows; destruction of nasal septum Paresthesia Leonine facies

Laboratory Diagnosis

- **Punch biopsy or nasal scrapings; acid fast stain**
- **Lepromin skin test** is positive in the tuberculoid but not in the lepromatous form.
- (No cultures)

Treatment

Multiple-drug therapy with dapsone and rifampin, with clofazimine added for lepromatous

Prevention

Dapsone for close family contacts

GENUS: *CLOSTRIDIUM*

- Gram-positive rod
- Spore-forming
- Anaerobic

Clostridium tetani

Distinguishing Characteristics

- Large Gram-positive, spore-forming rods
- Anaerobes
- Produces tetanus toxin

Reservoir

Soil

Transmission

- Puncture wounds/trauma
- Requires low tissue oxygenation (E_h)

Disease

Tetanus; sx: risus sardonicus, opisthotonus

Pathogenesis

Spores germinate in the tissues producing **tetanus toxin** (an exotoxin also called **tetanospasmin**).

- Carried intra-axonally to CNS
- **Binds to ganglioside receptors**
- **Blocks release of inhibitory mediators (glycine and GABA) at spinal synapses**
- Excitatory neurons are unopposed → extreme muscle spasm
- One of the most toxic substances known

Diagnosis

Primarily a clinical diagnosis, organism is rarely isolated

Treatment

Of actual tetanus:

- **Hyperimmune human globulin (TIG) to neutralize toxin, + metronidazole or penicillin**
- **Spasmolytic drugs** (diazepam), debride, delay closure

Prevention

Toxoid Vaccines (DTP, DTaP, Td)

- Toxoid is formaldehyde-inactivated toxin.
- Important because disinfectants have poor sporucidal action
- Care of wounds: proper wound cleansing and care plus treatment

Table I-2-21. Wound Management

	Not Tetanus Prone linear, 1 cm deep cut, without devitalized tissue, without major contaminants, less than 6 hours old	Tetanus Prone blunt/missile, burn, frostbite, 1 cm deep; devitalized tissue present + contaminants (e.g., dirt, saliva), any wound 6 hours old
Not completed primary or vaccination history unknown	Vaccine	Vaccine and TIG*
Completed primary series	Vaccine if more than 10 years since last booster	Vaccine if more than 5 years since last booster

* TIG = tetanus immunoglobulin (human).

Clostridium botulinum

Distinguishing Characteristics

- Anaerobic, Gram-positive, spore-forming rods

Reservoir

- Soil/dust

Pathogenesis

Spores survive in soil and dust; **germinate** in moist, warm, nutritious but **non-acidic and anaerobic conditions**.

Botulinum toxin

- **A-B polypeptide neurotoxin** (actually a series of 7 antigenically different; Type A and B most common)
- **Coded for by a prophage** (lysogenized *Clostridium botulinum*).
- Highly toxic
- **Heat labile** (unlike staph), 10 minutes 60°C
- **Mechanism of action:**
 - **Absorbed by gut** and carried by blood to peripheral nerve synapses
 - **Blocks release of acetylcholine** at the myoneuronal junction resulting in a reversible **flaccid paralysis**

Table I-2-22. Forms of Botulism

Disease	Adult/food borne	Infant	Wound
Acquisition	Preformed toxin ingested (toxicosis) Poorly canned alkaline vegetables (green beans) Smoked fish	Spores ingested: household dust, HONEY Toxin produced in gut (toxi-infection)	Traumatic implantation of spores; <i>in vivo</i> production of toxin (toxi-infection) Debridement, no closure
Symptoms	1–2 day onset of Sx: weakness, dizziness, blurred vision, flaccid paralysis (reversible); $\pm$ diarrhea, nausea or vomiting	Constipation, limpness/flaccid paralysis (reversible): diplopia, dysphagia, weak feeding/crying; may lead to respiratory arrest	As for food without GI symptoms
Toxin demonstrated in	Suspected food or serum	Stool or serum	Serum
Treatment	Respiratory support Polyvalent antitoxin	Respiratory support in monitored intensive care; Hyperimmune human serum Antibiotics generally not used as may worsen or prolong	Rx: amoxicillin and antitoxin (respiratory support)
Prevention	Proper canning; Heat all canned foods	No honey first year	

Clostridium perfringens

Distinguishing Characteristics

- Large **Gram-positive, spore-forming** rods (spores rare in tissue), nonmotile
- **Anaerobic: “stormy fermentation”** in milk media
- **Double zone of beta hemolysis**

Reservoir

Soil and human colon

Pathogenesis

- **Spores** germinate under anaerobic conditions in tissue.
- Vegetative cells produce:
Alpha toxin (a.k.a., phospholipase C) is a **lecithinase**. It disrupts membranes, damaging RBC, platelets, WBC, endothelial cells $\rightarrow$ massive hemolysis, tissue destruction, hepatic toxicity.
- Identified by **Naegler’s reaction**: egg yolk agar plate—one side with anti- α -toxin. Lecithinase activity is detected on side with no antitoxin.
- 12 other toxins damage tissues.
- **Enterotoxin** produced in intestines in food poisoning; disrupts ion transport $\rightarrow$ watery diarrhea, cramps (similar to *E. coli*); resolution <24 hours.

Diseases

Gas Gangrene (Myonecrosis)

- Contamination of **wound with soil or feces**
- **Acute and increasing pain** at wound site
- **Tense tissue** (edema) and exudate
- Systemic symptoms include **fever** and **tachycardia** (disproportionate to fever), diaphoresis, pallor, etc.
- **Rapid, high mortality**
- Prevention: extensive **debridement** of the wound plus administration of penicillin decreases probability
- Treatment: debridement, delayed closure, clindamycin and penicillin, hyperbaric chamber

Food Poisoning

- **Reheated meat dishes**, organism grows to high numbers; 8–24 hour incubation
- **Enterotoxin** production in gut; self-limiting non-inflammatory, watery diarrhea
- Treatment: supportive for food poisoning

Clostridium difficile

Antibiotic-associated (clindamycin, cephalosporins, amoxicillin, ampicillin) **diarrhea**, colitis, or **pseudomembranous colitis** (yellow plaques on colon)

Two Toxins

Toxin A: Enterotoxin damaging mucosa leading to fluid increase; granulocyte attractant

Toxin B: Cytotoxin: cytopathic

Treatment

Metronidazole (vancomycin): use only if no other drug available; to avoid selecting for vancomycin-resistant normal flora.

Note**Oxidase**

Oxidase (cytochrome C oxidase) test: flood colony with phenylenediamine; in presence of oxidase, phenylenediamine turns black. Rapid test.

Major oxidase-negative Gram-negative group is Enterobacteriaceae.

GENUS: NEISSERIA

- Gram-negative
- Diplococci with flattened sides
- Oxidase positive

Table I-2-23. Medically Important *Neisseria*

Species	Capsule	Vaccine	Portal of Entry	Glucose Utilization	Maltose Fermentation	β -lactamase Production
<i>N. meningitidis</i>	+	Yes	Respiratory	+	+	Rare
<i>N. gonorrhoeae</i>	–	No	Genital	+	–	>16%

Neisseria meningitidis* (Meningococcus)*Distinguishing Characteristics**

- Gram-negative kidney bean-shaped diplococci
- Large capsule; latex particle agglutination (or CIE; counter immunoelectrophoresis) to identify *N. meningitidis* capsular antigens in CSF
- Grows on chocolate (not blood) agar in 5% CO₂ atmosphere
- Ferments maltose in contrast to gonococci

Reservoir

Human nasopharyngeal area

Transmission

- Respiratory droplets; oropharyngeal colonization, spread to the meninges via the bloodstream
- Disease occurs in only small percent of colonized.

Pathogenesis**Important Virulence Factors**

- Polysaccharide capsule: antiphagocytic, antigenic, 5 common serogroups: B is not strongly immunogenic (sialic acid), B strain is most common strain in USA. Used for: serogrouping, detection in CSF, and vaccine
- IgA protease allows oropharynx colonization.
- Endotoxin (LPS): fever, septic shock in meningococcemia, overproduction of outer membrane
- Pili and outer membrane proteins important in ability to colonize and invade
- Deficiency in late complement components (C5-8) predisposes to bacteremia.

Disease**Meningitis and Meningococcemia**

- Abrupt onset with fever, chills, malaise, prostration, and a rash that is generally petechial; rapid decline
- Fulminant cases: ecchymoses, DIC, shock, coma, and death (Waterhouse-Friderichsen syndrome)

Treatment

- Penicillin G or ceftriaxone
- Some β -lactamase production beginning to be seen

Prevention

- **Vaccine:** capsular polysaccharide of strains **Y, W-135, C, and A**
 - Type B (50% of the cases in USA) capsule not a good immunogen
- Prophylaxis of close contacts: **rifampin** (or ciprofloxacin)

Neisseria gonorrhoeae (*Gonococcus*)

Distinguishing Characteristics

- **Gram-negative** kidney bean-shaped **diplococci**
- (Intracellular **Gram (-) diplococci** in PMNs from urethral smear from symptomatic male is suggestive of *N.g.*)
- Commonly: diagnosis by **genetic probes** with amplification
- Culture (when done) on **Thayer-Martin medium**
 - Oxidase-positive colonies
 - Maltose not fermented
 - No capsule

Reservoir

Human genital tract

Transmission

Sexual contact, birth; sensitive to drying and cold.

Pathogenesis

Pili

- **Attachment** to mucosal surfaces
- **Inhibit phagocytic uptake**
- **Antigenic** (immunogenic) **variation:** >1 million variants

Outer Membrane Proteins

- OMP I: structural, antigen used in serotyping
- Opa proteins (opacity): **antigenic variation**, adherence
- **IgA protease:** aids in colonization and cellular uptake

Organism invades mucosal surfaces and causes inflammation.

Disease

Gonorrhea

- Male: **urethritis**, proctitis
- Female: **endocervicitis**, PID (contiguous spread), arthritis, proctitis
- Infants: **ophthalmia** (rapidly leads to **blindness if untreated**)

Treatment

- Ceftriaxone
- Test for *Chlamydia trachomatis* or treat with a tetracycline
- Penicillin-binding protein mutations led to gradual increases in penicillin resistance from the 50s to the 70s.
- Plasmid-mediated β -lactamase produces **high-level penicillin resistance**.

Prevention

- Adult forms: no vaccine; condoms
- Neonatal: silver nitrate or erythromycin ointment in eyes at birth

Moraxella catarrhalis

- Gram-negative diplococcus (close relative of neisseriae)
- Normal upper respiratory flora
- **Otitis media**
- Causes bronchitis and bronchopneumonia in elderly with COPD
- Drug resistance a problem; most strains produce a β -lactamase

GENUS: *PSEUDOMONAS*

- Gram-negative rod
- Oxidase-positive
- Aerobic

Pseudomonas aeruginosa

Distinguishing Characteristics

- Oxidase-positive, Gram-negative rods, non-fermenting
- Pigments: pyocyanin (blue-green) and fluorescein
- Grape-like odor
- Slime layer
- Non-lactose-fermenting colonies on EMB or MacConkey

Reservoir

Ubiquitous in water

Transmission

Water aerosols, raw vegetables, flowers

Pathogenesis

- Endotoxin causes inflammation in tissues and Gram-negative shock in septicemia.
- *Pseudomonas* exotoxin A ADP ribosylates EF-2, inhibiting protein synthesis (like diphtheria toxin)
- Liver is primary target.
- Capsule/slime layer: allows formation of pulmonary microcolonies; difficult to remove by phagocytosis

Compromising Condition/Opportunistic Infections

Normal People

- Transient GI tract colonization: loose stools (10% pop.)
- Hot tub folliculitis
- Eye ulcers: trauma, coma, or prolonged contact wear

Burn Patients: GI tract colonization → skin → colonization of eschar → cellulitis (blue-green pus) → septicemia

Neutropenic Patients: Pneumonia and septicemias—often superinfections (infections while on antibiotics)

Chronic Granulomatous Disease (CGD): Pneumonias, septicemias (*Pseudomonas* is catalase positive)

Note

Pseudomonas medical ecology

Pseudomonas aeruginosa is an ubiquitous water and soil organism that grows to very high numbers overnight in standing water (distilled or tap).

Sources for infections include:

- Raw vegetables, respirators, humidifiers, sink drains, faucet aerators, cut and potted flowers, and, if not properly maintained, whirlpools.
- Transient colonization of colons of about 10% of people. Bacteria get on skin from fecal organisms. Requires exquisitely careful housekeeping and restricted diets in burn units.

Note

Drug Resistance in
P. aeruginosa

Susceptibilities important.

Drug resistance very common:

Inherent resistance
(missing high affinity porin
some drugs enter through);
Plasmid-mediated β -
lactamases and acetylating
enzymes.

Septicemias: Fever, shock $\pm$ skin lesions (black necrotic center, erythematous margin (**ecthyma gangrenosum**))

Catheterized Patients: Urinary tract infections (UTI)

Cystic Fibrosis: Early pulmonary colonization, recurrent pneumonias. Always high **slime-producing strains**

Treatment

Antipseudomonal penicillin, third-generation cephalosporins

Prevention

- Pasteurization or disinfection of water-related equipment, hand washing; prompt removal of catheters.
- No flowers or raw vegetables in burn units.

GENUS: *LEGIONELLA*

- Weakly Gram-negative
- Pleomorphic rods requiring cysteine and iron
- Water organisms

Legionella pneumophila (and other legionellae)

Distinguishing Characteristics

- **Stain poorly** with standard Gram stain; **Gram-negative**
- **Fastidious** requiring increased **iron and cysteine** for laboratory culture (CYE, Charcoal Yeast Extract)
- **Facultative intracellular pathogens**
 - Diagnosis: **DFA** (Direct Fluorescent Antibody) on biopsy, (+) by Dieterle silver stain
 - Antigen urine test for serogroup 1 only
 - Fourfold increase in antibody

Reservoir

Rivers/streams/amoebae; air-conditioning water cooling tanks

Transmission

- Aerosols from contaminated **air-conditioning**
- **No human-to-human transmission**

Predisposing Factors

- **Smokers over 55 years with high alcohol intake**
- **Immunosuppressed patients**, e.g., renal transplant patients

Pathogenesis

- **Facultative intracellular pathogen**
- **Endotoxin**

Diseases

Seasonal: Associated with air-conditioning systems, now routinely decontaminated

Legionnaires' Disease ("Atypical Pneumonia")

- **Pneumonia**
- **Mental confusion**
- **Diarrhea** (no *Legionella* in gastrointestinal tract)

Pontiac Fever

- Pneumonitis
- No fatalities

Treatment

- Fluoroquinolone or azithromycin or erythromycin with rifampin for immunocompromised patients
- Drug must penetrate human cells.

Prevention

Routine decontamination of air-conditioner cooling tanks

GENUS: *BORDETELLA*

- Gram-negative small rods
- Strict aerobes

Bordetella pertussis

Distinguishing Characteristics

- Small **Gram-negative**, aerobic **rods**
- Fastidious/delicate: **Regan-Lowe** or Bordet-Gengou media; either direct cough plates or nasopharyngeal cultures.
- Difficult to culture from the middle of paroxysmal stage on
- Direct immunofluorescence (**DFA**) on nasopharyngeal smear
- PCR and serologic tests available
- Encapsulated organism

Reservoir/Transmission

Human (vaccinated); respiratory droplet

Pathogenesis

B. pertussis is a **Mucosal Surface Pathogen**

Attachment to nasopharyngeal ciliated epithelial cells

- Filamentous hemagglutinin
- Pertussis toxin (on outer membrane) aids in attachment

Toxins damage respiratory epithelium.

- **Adenylate cyclase toxin**: impairs leukocyte chemotaxis → inhibits phagocytosis and causes local edema
- **Tracheal cytotoxin**: interferes with ciliary action; kills ciliated cells
- Endotoxin
- **Pertussis toxin** (A and B component, OM protein toxin):
ADP ribosylation of G_i (inhibiting negative regulator of adenylyl cyclase) interferes with transfer of signals from cell surface to intracellular mediator system;
 - Lymphocytosis promotion
 - Islet-activation → hypoglycemia
 - Blocks immune effector cells
 - Increased histamine sensitivity

In A Nutshell

B. pertussis immunity

- Vaccine immunity lasts 5–10 years (and is primarily IgA)
- Babies born with little immunity.
- Vaccinated humans >10 yrs serve as reservoir.
- 12–20% of afebrile adults with cough >2 weeks have pertussis.
- Immunity to actual pertussis is life long.
- New vaccines (DTaP)
 - Acellular: components:
 - Immunogens vary by manufacturer
 - Pertussis toxoid
 - Filamentous hemagglutinin
 - Pertactin (OMP)
 - 1 other

Disease

Whooping Cough (Pertussis)

Three stages after a 7–10 day incubation; contagious

- (1–2 weeks) **catarrhal**: rhinorrhea, malaise, fever, sneezing; contagious
- (2–4 weeks) **paroxysmal**: repetitive cough with whoops, vomiting; anoxia and severity of cough cause neurological damage and eye hemorrhages; organism present at beginning disappears
- (>3 weeks) **convalescence**: less cough, secondary complications manifest: pneumonia, seizures, encephalopathy

Treatment

Supportive care; hospitalization if <6 months old, erythromycin

Prevention

- Vaccine: DTaP; immunity wanes 5–7 years
- Babies born with little or no immunity (IgA) from mom

GENUS: *FRANCISELLA*

- Gram-negative small rods
- Facultative intracellular pathogen

Francisella tularensis

Distinguishing Characteristics

- Small Gram-negative rod
- Serodiagnosis; culture is hazardous

Reservoir

Many species of wild **animals**, especially rabbits, deer, and rodents

Transmission

- **Tick bite** (*Dermacentor*) → **ulceroglandular** disease, characterized by **fever**, ulcer at bite site, and regional lymph node enlargement and necrosis
- **Traumatic implantation** while skinning rabbits → ulceroglandular disease
- Aerosols (skinning rabbits) → pneumonia
- Ingestion (of undercooked, infected meat or contaminated water) produces typhoidal tularemia.

Pathogenesis

- **Facultative intracellular pathogen** (localizes in reticuloendothelial cells)
- Granulomatous response

Disease

Tularemia

- Endemic in every state of U.S.
- Arkansas and Missouri highest

Treatment

Streptomycin

Prevention

- Protect against tick bites, gloves while butchering rabbits
- Live, attenuated vaccine for persons in high-risk occupations

Note

Zoonotic organisms

- *Brucella*
- *Bacillus anthracis*
- *Listeria monocytogenes*
- *Salmonella enteritidis*
- *Campylobacter*
- Q fever (*Coxiella burnetii*)
- *Chlamydia psittaci*
- *Francisella tularensis*

GENUS: BRUCELLA

- Gram-negative rods
- Zoonotic
- Facultative intracellular pathogen

Brucella species

Distinguishing Characteristics

- Small **Gram-negative** rods, aerobic
- Facultative intracellular
- Serological confirmation of disease most common
- Culture is hazardous.

Reservoir

Domestic livestock

Transmission

- Unpasteurized dairy products
- Direct contact with the animal, work in slaughterhouse

Pathogenesis

- Endotoxin
- Facultative intracellular parasite (localizes in cells of reticuloendothelial system, RES) → septicemia
- Granulomatous response with central necrosis

Disease

Brucellosis (undulant fever)

- *B. abortus*: cattle
- *B. suis*: pigs
- *B. melitensis*: goats

Acute Septicemias

- Fever 100–104°F (often in evening)
- Influenza-like symptoms, including arthralgias, myalgia, back pain
- Sweating
- Hepatomegaly

Undulant Form: Milder, often a result of incomplete treatment

Chronic Form

(Disease for more than one year.)

- Usually in older people (veterinarians)
- Cyclic bouts of depression and sweating
- Fever rare
- Ocular complications (uveitis) in 5–10%

Prevention

- Vaccinate cattle; vaccinate high-risk humans.
- Pasteurize milk, especially goat milk.

GENUS: *CAMPYLOBACTER*

- Gram-negative curved rod with polar flagella
- Microaerophilic

Campylobacter jejuni

Distinguishing Characteristics

- Motile, curved Gram-negative rods ("gulls' wings")
- Microaerophilic, grows well at 42°C on selective media (Campy medium or Skirrow's agar); oxidase positive.



Figure I-2-10. *Campylobacter*

Reservoir

Intestinal tracts of humans, cattle, sheep, dogs, cats, **poultry**

Transmission

Fecal-oral, primarily from **poultry**

Pathogenesis

- **Low infectious dose (as few as 500)**
- **Invades** mucosa of the colon, destroying mucosal surfaces; blood and **pus** in stools (inflammatory diarrhea)
- Rarely penetrates to cause septicemia

Disease

- Common cause of infectious diarrhea worldwide
- In U.S., *Campylobacter* enteritis > (*Salmonella* + *Shigella*)
- **Ten to more stools/day, may be frankly bloody**
- Abdominal pain, fever, malaise, nausea, and vomiting
- Generally **self-limiting in 3–5 days** but may last longer
- Complications:
 - **Guillain-Barré syndrome** (~30% of the GBS in the U.S.).
 - Reactive arthritis

Treatment

Erythromycin, fluoroquinolones, penicillin-resistant

Helicobacter pylori

Distinguishing Characteristics

- Gram-negative spiral gastric bacilli with flagella
- Microaerophilic, 37°C growth (Campy medium or Skirrow's agar); oxidase positive.

Reservoir

Humans

Transmission

Fecal-oral, oral-oral

Pathogenesis

- **Motile**
- **Urease-positive:** ammonium cloud neutralizes stomach acid, allowing survival in stomach acid during transit to border.
- **Mucinase** aids in penetration of mucous layer (rapid shift down to neutral as it penetrates).
- **Invasive** into stomach lining where pH is neutral.
- Inflammation is prominent.
- Two biotypes (I and II); type I produces vacuolating cytotoxin.

Disease

- Causes chronic **gastritis and duodenal ulcers**.
- Associated with several forms of **stomach cancer**. (Atrophic gastritis rather than duodenal ulcers correlates with risk.)
- Now classed by WHO as **Type I carcinogen**.

Lab Diagnosis

- Serologic test
- Biopsy with culture; histology with Giemsa or silver stain
- Breath test: ^{13}C -urea swallowed; ammonia + ^{13}C -CO₂ produced

Treatment

(Myriad of regimens.)

Omeprazole + amoxicillin + clarithromycin

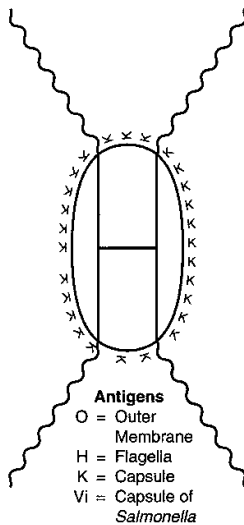


Figure I-2-11. Antigens of Enterobacteriaceae

FAMILY: ENTEROBACTERIACEAE

- Gram-negative rods
- Facultative anaerobes
- Ferment glucose
- Cytochrome C oxidase negative
- Reduce nitrates to nitrites
- Catalase positive

Pathogenesis (overall for family)

- Endotoxin
- Some also produce exotoxins.
- Antigens
 - O = cell envelope or O antigen
 - H = flagellar (motile cells only) antigen
 - K = capsular polysaccharide antigen
 - Vi = *Salmonella* capsular antigen (virulence)

Lab Diagnosis

Grow on two media:

- Blood
- Eosin-methylene blue or MacConkey agar (differentiate lactose fermentation)

Genera in Family

- Lactose fermenters: CEEK ("colored" colonies)
 - Citrobacter*
 - Escherichia*
 - Enterobacter*
 - Klebsiella*
- Nonlactose fermenters: ShYPS ("colorless" colonies [same color as original media])
 - Shigella* } nonmotile
 - Yersinia* } non H₂S producers
 - Proteus* } motile
 - Salmonella* } H₂S producers

Slow lactose fermenter: *Serratia*, which produces a salmon red pigment and is an opportunist. Each genus is characterized by a whole series of chemical tests to detect the presence of certain enzymes or pathways.

GENUS: *ESCHERICHIA*

- Gram-negative rod
- *Enterobacteriaceae*
- Ferments lactose

Escherichia coli

Distinguishing Characteristics

- Gram-negative rod
- Facultative anaerobic, oxidase negative
- *E. coli* is **lactose-fermenter**: colonies with iridescent green sheen on EMB

Reservoirs

- **Human colon**; may colonize vagina or urethra
- Contaminate crops where human fecal fertilizer is used
- **Enterohemorrhagic strains**: bovine feces

Diseases

Urinary Tract Infections (UTIs)

- *E. coli* is most common cause.
- Transmission: from own fecal flora → urethra
Predisposing factors: female anatomy; strictures, stones, or abnormality in urine flow; indwelling urinary catheters
- Pathogenesis
 - Motility aid
 - Adherence to uroepithelium important: **P-pili**: pyelonephritis-associated pili (PAP or P-pili); X-adhesins
 - β -hemolytic (many)
- Treatment: ampicillin or sulfonamides for UTI

Neonatal Septicemia and Meningitis

- Second most common cause after Group B streptococci
- Maternal fecal flora → vagina contaminates baby
 - **Capsule**: most strains have K1 serotype. Capsule allows the organism to evade phagocytic uptake in the blood and spleen so organism can get to blood/brain barrier.
 - **Endotoxin**: triggers **shock**; **inflammation** at blood/brain barrier **facilitates invasion**.

Septicemia

- Indwelling IV lines predispose to invasion from skin.
- Cytotoxic drugs damage intestinal mucosa, allowing escape.
- **Endotoxin** triggers Gram-negative shock.

Note

***E. coli* Identification from Stool**

- Isolation of *E. coli* from feces is not by itself significant.
- Sorbitol MacConkey screen.
- Most *E. coli* ferment sorbitol.
- Most EHEC do not (colorless).

Animal models and tissue culture assays may be used, but other methods of differentiating pathogenic *E. coli* from normal flora are more commonly:

- Immunoassay looking for specific protein antigens (on or excreted by the bacterium)
- Serotyping since certain serotypes are more often pathogenic
- DNA probes for specific genes in a culture
- PCR on clinical specimen

Note

Mnemonic:

Toxins ↑ cAMP

c = cholera
A = anthrax
Σ = *E. coli* LT
P = pertussis

Gastroenteritides/Diarrheas

- **Most strains:** fecal-oral spread; poor sanitation/water; use of human feces on vegetables or fruits.
- **Enterohemorrhagic strains:** bovine fecal contamination: undercooked hamburger, raw milk, apple juice (from fallen apples), alfalfa sprouts.

ETEC = Enterotoxigenic *E. coli*

Major cause of "Traveler's Diarrhea" and diarrhea in <3 year olds in developing countries.

- Two enterotoxins are produced:
LT: The **heat labile toxin** stimulates adenylate cyclase by **ADP ribosylation of G_s** resulting in increased cAMP causes outflow of chloride ions and water in small intestine → watery diarrhea.
ST: The **heat stable toxin** causes diarrhea by stimulating guanylate cyclase.
- Enterotoxin production identified by immunoassay, bioassay, or DNA probe assays.
- **Capsule** impedes phagocytosis
- **Colonizing factor adhesins** (CFAs): bind to small intestine.

EPEC = Enteropathogenic *E. coli*

- Second most common cause of infantile diarrhea after rotavirus
- **Prolonged, watery diarrhea** in babies <1 year in developing countries
- Plasmid containing virulence genes called the *E. coli* adherence factor (EAF); EPEC adhere to M cells, causing rearrangement of actin and effacement of the brush border microvilli.

EIEC = Enteroinvasive *E. coli*

- **Invades** large bowel, similar to shigellosis, including the formation of actin "jet trails"
- Often manifests as **watery diarrhea** (with **excess of leukocytes**, however) with **fever** and **abdominal pain**. Only about 10% progress on to dysentery.

EHEC = Enterohemorrhagic *E. coli*

- Also known as **verotoxin producing *E. coli*** (VTEC)
- **O157:H7** is the most common serotype.
- EHEC disease ranges from **mild diarrhea** without blood to **hemorrhagic colitis ± hemolytic uremic syndrome (HUS)**. **Fever is generally absent**, distinguishing it from shigellosis. HUS most common in kids <5 years old.
- Plasmid-associated **verotoxins = Shiga-like toxin 1 and 2**. Both toxins **inhibit protein synthesis in the large intestine by nicking the 60S ribosome**.
- Not invasive, no inflammatory response; afebrile
- Focus of infection is the large intestine.
- Excess fecal leukocytes are **not** seen. Damage is due to toxin activity so does not promote inflammatory response.
 Lab screen: most **EHEC** are sorbitol non-fermenters so have colorless colonies on sorbitol MacConkey; DNA probe for verotoxin genes.
- Antibiotics may increase risk of HUS and kidney damage, especially early and in children. Check before using.

EaggEC = Enteraggregative *E. coli*

- Important cause of **persistent diarrhea** with vomiting, low-grade fever in the developing world.
- **Fimbriae** create a “stacked brick-like” biofilm by attaching to each other and enterocytes.
- Also produce enterotoxin (EAST1).

Treatment

- Depends on strain, severity of diarrhea, presence of fever, blood, and pus
- Rehydration for traveler’s diarrhea; trimethoprim-sulfamethoxazole may shorten duration of symptoms
- No antibiotic currently recommended for 0157:H7 (generally bloody diarrhea without fever)
- Fluoroquinolones for bloody diarrhea with pus and fever >101°F and tenesmus
- Antibiotic resistance primarily mediated by plasmid-encoded enzymes, e.g., β -lactamase, and aminoglycoside modifying enzymes

GENUS: *SHIGELLA*

- Gram-negative rod
- *Enterobacteriaceae*
- Nonlactose fermenter
- Nonmotile

Shigella species

S. dysenteriae (most severe disease), *S. sonnei* (most common in U.S.), etc.

Distinguishing Characteristics

- Gram-negative rods, non-motile
- Facultative anaerobes, **non-lactose fermenting** (colorless colonies on EMB or MacConkey)
- Identified by biochemical reactions or by serology with anti-O antibody in agglutination test.

Reservoir

Human colon only (no animal carriers)

Transmission

- Fecal-oral spread, person to person
- As with all fecal-oral spread, the "Four Fs": fingers, food, feces, flies

Pathogenesis

- Endotoxin triggers inflammation.
- No H antigens
- Shigellae **invade M cells** (membrane ruffling and macropinocytosis); get into the cytoplasm, replicate and then **polymerize actin jet trails to go laterally** without going back out into the extracellular milieu. This produces **very shallow ulcers** and rarely causes invasion of blood vessels.
- Shiga toxin:
 - Produced by *S. dysenteriae*, type 1
 - Three activities: **neurotoxic, cytotoxic, enterotoxic**
 - **AB component toxin** is internalized in human cells; **inhibits protein synthesis by clipping 60 ribosomal subunit**.

Disease

Enterocolitis/Shigellosis (most severe form is dysentery)

- Few organisms required to start infection (1–10) (extremely acid resistant)
- 1–4 day incubation
- Organisms invade producing bloody diarrhea.
- **Fever (generally >101°F), lower abdominal cramps, tenesmus; diarrhea first watery, then bloody, invasive but rarely septicemia; shallow ulcers.**
- Severity depends on the age of patient and the strain; *S. dysenteriae* Type 1 with toxin most severe.

Treatment

- Mild cases: fluid and electrolyte replacement only
- Severe cases: antibiotics
- Resistance is mediated by plasmid-encoded enzymes

Prevention

- Proper sanitation (sewage, clean drinking water, hand washing)

Note

Comparative Microbiology

- Invasive bacteria: PMN in stool: *Shigella*, *Salmonella*, *Campylobacter*, EIEC.
- Toxigenic bacteria: ETEC, *V. cholera*, *Cl. perfringens*, EHEC.

In A Nutshell

Comparative Microbiology:
Major encapsulated
organisms

Some Killers Have Pretty Nice
Capsules:

Strep pneumoniae

Klebsiella pneumoniae

Haemophilus influenzae Type b
(a-d)

Pseudomonas aeruginosa

Neisseria meningitidis

Cryptococcus neoformans (the
yeast)

(Not a complete list, just the
big ones!)

GENUS: *KLEBSIELLA*

- Gram-negative rod
- *Enterobacteriaceae*
- Major capsule

Klebsiella pneumoniae

Distinguishing Characteristics

- Gram-negative rods with large polysaccharide capsule
- Mucoid, lactose-fermenting colonies on MacConkey agar
- Oxidase negative

Reservoir

Human colon and upper respiratory tract

Transmission

All commonly from own flora

Pathogenesis

- Capsule: impedes phagocytosis
- Endotoxin: causes fever, inflammation, and shock (septicemia)

Transmission/Disease

Pneumonia

- Community-acquired, most often in older males; most commonly in patients with either chronic lung disease or alcoholism, or diabetes (but not the most common cause of pneumonia in alcoholics! *S. pneumoniae* is.)
- Endogenous; assumed to reach lungs by inhalation of respiratory droplets from upper respiratory tract.
- Frequent abscesses make it hard to treat; fatality rate high
- Sputum is generally thick and bloody (currant jelly) but not foul smelling as in anaerobic aspiration pneumonia.

Urinary Tract Infections: Catheter-related (nosocomial) from fecal contamination of catheters

Septicemia: In immunocompromised patients may originate from bowel defects or invasion of IV lines

Treatment

Antibiotic sensitivity testing must be done

Prevention

Good catheter care; limit catheter use

GENUS: *SALMONELLA*

- Gram-negative rod (*Enterobacteriaceae*)
- Non-lactose fermenter
- Motile

2,000 serotypes of salmonellae; serotype names are still in use.

Diseases

Enteric or Typhoid Fever (*S. typhi*)

Gastroenteritis

Septicemia

Salmonella typhi

Distinguishing Characteristics

- **Gram-negative rods**, highly motile with the Vi capsule
- Facultative anaerobe, **non-lactose fermenting**
- **Produces H₂S**
- Species identification with biochemical reactions
- **Sensitive to acid**

Reservoir

- **Humans only; NO ANIMAL RESERVOIRS**
- Only the typhoid Mary's of the world!

Transmission

- **Fecal-oral route from human carriers (gall bladder)**
- **Decreased stomach acid** or impairment of mononuclear cells such as in **sickle cell disease predisposes** to *Salmonella* infections

Pathogenesis/Disease

Typhoid Fever (Enteric Fever), *S. typhi*

(milder form: paratyphoid fever; *S. paratyphi*)

- Organism ingested (large number if stomach acid is normal).
- **Infection begins in ileocecal region**; constipation common.
- Host cell membranes "ruffle" from *Salmonella* contact.
- *Salmonella* reach basolateral side of M cells then **mesenteric lymph nodes and blood** (transient 1st septicemia)
- At 1 week: patients have 80% **positive blood cultures**; 25% have **rose spots** (trunk/abdomen)
- **Liver and spleen** are infected with additional release of bacteria to bloodstream → signs of **septicemia** (mainly fever)

- *S. typhi* survives intracellularly and replicates in macrophages; **resistant to macrophage killing due to:**
 - Decreased fusion of lysosomes with phagosomes
 - Defensins (proteins) allow it to withstand oxygen-dependent and -independent killing.
- Released from the macrophages. The Vi capsular antigen (*S. typhi* only) withstands complement-mediated killing.
- **Biliary system** (liver, gall bladder) is infected, organisms enter intestinal tract in bile.
- **By week 3: 85% of stool cultures** are positive.
- Symptoms: **fever**, headache, abdominal pain, constipation more common than diarrhea
- Complications of untreated: **necrosis of Peyer's patches** with perforation (local endotoxin triggered damage), thrombophlebitis, cholecystitis, pneumonia, abscess formation, etc.

Treatment

Ciprofloxacin or ceftriaxone in seriously ill; some drug resistance

Prevention

Sanitation; 3 vaccines: attenuated oral vaccine of *S. typhi* strain 21 (Ty21a), parenteral heat killed *S. typhi*, and parenteral Vi polysaccharide capsular vaccine

Nontyphoidal Salmonellae: *S. enteritidis*, *S. typhimurium*, and Other Species

Distinguishing Characteristics

- Facultative Gram-negative rods, non-lactose-fermenting on EMB, MacConkey medium
- Produces H_2S , **motile** (unlike *Shigella*)
- Speciated with biochemical reactions and serotyped with O, H, and Vi antigens
- **Antibodies** to O, Vi, and H antigens **in patient's serum** can be detected by agglutination (**Widal test**).

Reservoir

Enteric tracts of human and domestic animals, e.g., **chickens** and turtles

Transmission

Largely through chicken products (raw chicken and eggs) in the kitchen

Pathogenesis

- Sensitive to stomach acid (infectious dose 10^5 organisms)
- Lowered stomach acidity (antacids or gastrectomy) increases risk.
- Endotoxin in cell wall; no exotoxin
- **Invades** the mucosa in the ileocecal region, invasive to lamina propria → inflammation → increased PG → increased cAMP → loose diarrhea; shallow ulceration.
- Spread to septicemia not common with *S. enteritidis* (the most common) but may occur with others.

Diseases

Enterocolitis/Gastroenteritis

Second most common after *Campylobacter*: 6–48 hour incubation; nausea, vomiting, only occasionally bloody, loose stools, fever, abdominal pain, myalgia, headache

Septicemia

- *S. cholerae-suis*, *S. paratyphi*, and *S. dublin*
- When it occurs, it is usually in very young or elderly.
- Endocarditis or arthritis complicate about 10%.

Osteomyelitis: Sickle cell disease predisposes to osteomyelitis. *Salmonella* is the most common causative agent of osteomyelitis in sickle cell disease (not trait) patients (>80%).

Treatment

- Antibiotics **generally not effective so not recommended** for uncomplicated enterocolitis.
- Antibiotics for septicemia depend on sensitivity tests.

Prevention

Proper sanitation (sewage, clean drinking water, hand washing, particularly food handlers)

GENUS: *YERSINIA*

- Gram-negative rod
- *Enterobacteriaceae* (oxidase negative)

Yersinia pestis (*Enterobacteriaceae*)

Distinguishing Characteristics

- Small Gram-negative rods with **bipolar staining**
- **Facultative intracellular parasite**
- **Coagulase +**
- Clinical specimens and **cultures are hazardous**
- **Serodiagnosis** or direct **immunofluorescence**

Reservoir

U.S.: desert southwest, rodents, e.g., prairie dogs, chipmunks, squirrels, field mice, and voles

Transmission

- Wild rodents **flea bite** → sylvatic plague
- Human-to-human transmission by **respiratory droplets**

Pathogenesis

- **Coagulase**-contaminated mouth parts of flea
- **Endotoxin** and an exotoxin
- **Two antigens (V and W)**
- **Envelope antigen (F-1) inhibits phagocytosis**

Disease

Bubonic Plague

- Flea bites infected animal and then later uninfected human: coagulase role-contaminated mouth parts
- Symptoms:
 - **Rapidly increasing fever**
 - **Regional buboes**
 - **Conjunctivitis**
 - Leads to septicemia and death if untreated

Pneumonic Plague

- Arises from septic pulmonary emboli in bubonic plague or **inhalation of organisms from infected individual**.
- **Highly contagious!**

Treatment

- Streptomycin with tetracycline
- **Strict quarantine for 72 hours** after starting antibiotics

Prevention

- Animal control; avoid sick and dead animals.
- **Killed vaccine** is available for high-risk occupations.

Yersinia enterocolitica

- **Zoonotic, unpasteurized milk, pork**
- Enterotoxin
- **Multiplies in the cold**
- **Enterocolitis** in northern climates (Michigan, Scandinavia)
- Presentations may vary with age
 - Very young: febrile diarrhea (blood and pus)
 - Older kids/young adults: **pseudoappendicitis**
 - Adults: enterocolitis with post-infective sequelae like reactive arthritis
- **Blood transfusion-associated infections**

GENUS: *PROTEUS*

- Gram-negative rod
- *Enterobacteriaceae*
- Peritrichous flagella
- Non-lactose-fermenting
- Urease positive

Proteus mirabilis, *Proteus vulgaris*

Distinguishing Characteristics

- Gram-negative rods
- **Highly motile**; “swarming” motility on surface of blood agar
- **Urease** produced
- Facultative anaerobe (*Enterobacteriaceae*), oxidase negative

Reservoir

Human colon and environment (water and soil)

Disease

Urinary tract infection and septicemia

Pathogenesis

- **Urease raises urine pH to cause kidney stones** (staghorn renal calculi)
- **Motility** may aid entry into bladder.
- Endotoxin causes fever and shock when septicemia occurs.

Treatment

Do susceptibilities.

Prevention

Promptly remove urinary tract catheters.

Note

Weil-Felix test: antigens of OX strains of *Proteus vulgaris* cross-react with rickettsial organisms.

GENUS: *VIBRIO*

- Gram-negative curved rod with polar flagella
- Oxidase positive (*Vibrionaceae*)

Vibrio cholerae

- *Vibrio cholerae* O1 divided into biotype El Tor (predominant now) and Cholerae (classical)
- *Vibrio cholerae* O139 also produces cholera toxin.

Distinguishing Characteristics

- "Shooting star" motility inactivated by specific serum.
- Oxidase-positive, which distinguishes them from *Enterobacteriaceae*
- Growth on alkaline but not acidic media
(TCBS = Thiosulfate Citrate Bile salt Sucrose medium)

Reservoir

Human colon; **no vertebrate animal carriers.** (Copepod or shellfish may be contaminated by water contamination.) Human carriage may persist after untreated infection for months after infection; permanent carrier state rare.

Transmission

- Fecal-oral spread; sensitive to stomach acid
- Requires high dose ($>10^7$ organisms), if stomach acid is normal.

Pathogenesis

- Motility, mucinase, and toxin co-regulated pili (Tcp) aid in attachment to the intestinal mucosa.
- Cholera enterotoxin (choleragen) similar to *E. coli* LT. ADP ribosylates (G_s alpha) activating adenylate cyclase $\rightarrow$ increased cAMP $\rightarrow$ efflux of Cl^- & H_2O (persistent activation of adenyl cyclase).

Disease

Cholera: Rice water stools, tremendous fluid loss; hypovolemic shock if not treated.

Treatment

- Fluid and electrolyte replacement
- Doxycycline or ciprofloxacin shortens disease and reduces carriage.

Prevention

Proper sanitation; new vaccine; tetracycline to reduce transmission

Vibrio parahaemolyticus

- Food poisoning associated with **undercooked or raw seafood**
- **Marine**
- 5–94 hours incubation (mean 24 hours)
- Self-limiting watery diarrhea with cramping, abdominal pain

Vibrio vulnificus

- Brackish water; **oysters**; warm months
- **Cellulitis** when it gets into cuts; hard to treat
- **Gastroenteritis** when ingested
- **Septicemia** in patients with preexisting liver disease (50% fatal)

GENUS: PASTEURELLA

- Small Gram-negative rods
- Facultative anaerobic rods

Pasteurella multocida

Distinguishing Characteristics

Small **Gram-negative** rods

Reservoir

Mouths of many animals, especially cats and dogs

Transmission

Animal bites; particularly from cat bites

Disease

Wound infection leading to **cellulitis with lymphadenitis**

Pathogenesis

Endotoxin, capsule; spreads rapidly within skin, no exotoxins known

Lab Diagnosis

Rarely cultured because routine prophylaxis is common.

Treatment/Prevention

Amoxicillin/clavulanate for cat bites. Amoxicillin/Clavulanate is standard prophylaxis and treatment for most bites (human included), along with thorough cleaning.

GENUS: *HAEMOPHILUS*

- Gram-negative, pleomorphic rod
- Requires growth factors

Haemophilus influenzae

Distinguishing Characteristics

- Encapsulated, Gram-negative rod
- Fastidious: requires factors X (hemin) and V (NAD) for growth on nutrient or blood agar
- Grows near *S. aureus* on BA = "Satellite" phenomenon
- Chocolate agar provides both X and V factors

Reservoir

Human nasopharynx

Transmission

Respiratory droplets, shared toys

Pathogenesis

- Polysaccharide capsule (type b capsule is polyribitol phosphate) most important virulence factor
- 95% of invasive disease is caused by capsular type b. Capsule important in diagnosis. Antigen screen on CSF (e.g., latex particle agglutination); serotype all isolates by Quellung.
- IgA protease is a mucosal colonizing factor.

Diseases

Meningitis: *H. influenzae*; type b encapsulated strains.

Epidemic in unvaccinated children ages 3 months to 2 years

- After maternal antibody has waned
- Before the immune response of the child is adequate
- *H. influenzae* was most common cause of meningitis in 1 to 5-year-old children (mainly younger than 2) up to 1990.
- Still a problem if child is <2 years and not vaccinated.

Vaccination effective to prevent type b disease.

- **Polyribitol capsule conjugated to protein:**
(diphtheria toxoid or *N. meningitidis* outer membrane proteins)
making it a T cell dependent vaccine.
- Vaccine: 2, 4, 6 months; booster 15 months; 95% effective

Otitis Media

Usually nontypeable strains

Bronchitis

Exacerbations of acute bronchitis in smokers with COPD

Pneumonia

1–24 months; smokers

Epiglottitis

Rare in vaccinated kids; seen in unvaccinated toddlers. *H. influenzae* was the major causative agent.

Treatment

Cefotaxime or ceftriaxone for empirical therapy of meningitis. Check nasal carriage before releasing; use rifampin if still colonized.

Prevention

- **Conjugate capsular polysaccharide-protein vaccine**
- Rifampin reduces oropharynx colonization and prevents meningitis in unvaccinated, close contacts <2 years.

Haemophilus ducreyi

- Sexually transmitted disease.
- **Chancroid** (genital ulcers): **soft, painful chancre** (“You do cry with ducreyi.”)
- **Slow to heal without treatment**
- **Open lesions increase transmission of HIV**
- Diagnosis: DNA probe

GENUS: *BACTEROIDES/PREVOTELLA*

- Gram-negative rod
- Anaerobic
- Modified LPS with reduced activity

Bacteroides fragilis

Distinguishing Characteristics

- Anaerobic, Gram-negative rods
- Anaerobes are identified by biochemical tests and gas chromatography.

Reservoir

Human colon; the genus *Bacteroides* is the predominant anaerobe.

Transmission

Endogenous from bowel defects (e.g., from cytotoxic drug use, cancer), surgery, or trauma

Pathogenesis

- Modified LPS (missing heptose and 2-keto-3 deoxyoctonate) has reduced endotoxin activity.
- Capsule is antiphagocytic.

Disease

Septicemia, peritonitis (often mixed infections), and abdominal abscess

Treatment

- Metronidazole, clindamycin, or cefoxitin. Abscesses should be surgically drained.
- Antibiotic resistance is common (penicillin G, some cephalosporins, and aminoglycosides).

Prevention

Prophylactic antibiotics for gastrointestinal or biliary tract surgery

Bacteroides melaninogenicus* = *Prevotella melaninogenica

- Melanin-producing (black) *Bacteroides*
- Normal gingival flora
- Anaerobe
- Oral abscesses
- Heparinase leads to clotting in brain.

GENUS: *TREPONEMA*

- Spirochetes: spiral with axial filament (endoflagellum)
- Poorly visible on Gram stain but basically Gram-negative

Treponema pallidum

Distinguishing Characteristics

- **Thin spirochete**, not reliably seen on Gram stain
- Basically a Gram-negative cell envelope
- Outer membrane has endotoxin-like lipids.
- Axial filaments = endoflagella = periplasmic flagella
- Cannot culture in clinical lab; serodiagnosis
- Is an **obligate pathogen (but not intracellular)**

Reservoir/Transmission

Human genital tract; transmitted **sexually** or **across the placenta**.

Pathogenesis

Disease characterized by **endarteritis resulting in lesions**. Strong tendency to chronicity.

Disease

Syphilis: Progression in untreated syphilis: **incubation:** 10–90 days.

Primary

- Nontender chancre(s) at site of inoculation
- Margins generally “clean,” distinctly indurated edge
- **Contagious (but you still cannot culture!)**
- Chancre is good source of material for microscopy. (This is important as only 50% of those with chancres will be positive by nontreponemal serologic tests.)
- Heals spontaneously in 3–6 weeks

Secondary

- 1 to 3 months later (*T. pallidum* has spread early via bloodstream.)
- **Maculopapular** (often copper colored) **rash on skin including palms and soles** ± patchy alopecia
- Flat wart-like perianal **condylomata lata** and **mucous membrane lesions, both highly infectious**
- May “heal” spontaneously, regress (several times), and finally heal.
- Serology is almost always strongly reactive.

Latent syphilis

- Positive serology only

Tertiary

- May be years later in about one-third of the untreated patients
- Tertiary lesions consist of gummas, aortitis, or central nervous system inflammation.
- VDRL may be negative.

Congenital syphilis

- Commonly in babies of IV drug abusing women
- Sx: stillbirths, multiple fetal abnormalities (keratitis, 8th nerve deafness, notched teeth), or sometimes asymptomatic (or snuffles) at birth until two.

Laboratory Diagnosis

Visualize organisms by immunofluorescence or dark-field microscopy.

Serology important: Two types of antibody:

1. Nontreponemal antibody (= reagin)

- **Binds to cardiolipin**
 - An antigen found in mammalian mitochondrial membranes and in treponemes (but probably are from the host since treponemes don't make).
 - Cheap source of antigen is cow heart, which is used in screening tests (VDRL, RPR, ART).
- **Screening tests: Nontreponemal antibody tests:**
 - Venereal Disease Research Lab (VDRL)
 - Rapid plasma reagin (RPR)
 - Automated Reagin Test (ART)
 - Any one may be used for screening (inexpensive).
 - Very sensitive in primary (except early) and secondary; titer may decline in tertiary and with treatment.
 - But not specific; confirm with FTA-ABS

2. Treponemal antibody

Earliest antibody; binds to spirochetes

Specific tests for *Treponema* (more expensive)

- Fluorescent Treponemal Antibody-Absorption (FTA-ABS; most widely used test)
- *Treponema pallidum* Microhemagglutination
- These tests are more specific and positive earlier; usually remains positive for life. But positive in patients with other treponeme disease (bejel) and may be positive in Lyme disease.

TORCH screen for neonates no longer includes syphilis test; must order separately.

Treatment

Benzathine penicillin (long-acting form) for primary and secondary syphilis (no resistance to penicillin)

Prevention

Benzathine penicillin is given to contacts, no vaccine is available.

Jarisch-Herxheimer Reaction

- Starts generally during the first 24 hours of antibiotic treatment
- Increase in temperature, decrease in blood pressure; rigors, leukopenia
- May occur during treatment of any of the spirochete diseases

GENUS: *BORRELIA*

- Larger spirochetes
- Gram-negative
- Microaerophilic

Borrelia burgdorferi

Distinguishing Characteristics

Spirochete, not seen well on Gram-stained smear; can be cultured

Reservoir

Two reservoirs: **white-footed mice (nymphs)** and **white-tailed deer (adult ticks)**

Transmission

By *Ixodes* (deer) ticks and nymphs; worldwide but in three main areas in the U.S.:

- *Ixodes scapularis* (*I. dammini*) in Northeast (e.g., Connecticut), Midwest (e.g., Wisconsin)
- *Ixodes pacificus* on West Coast (e.g., California)

Pathogenesis

- *B. burgdorferi* invades skin and spreads via the bloodstream to involve primarily the heart, joints, and central nervous system.
- Arthritis is caused by immune complexes.

Disease

Lyme Disease

Initial symptoms:

- Erythema (chronicum) migrans: spreading annular skin lesion with an erythematous leading edge and central clearing, "bull's eye" seen in 85% of cases.
- Malaise, headache, severe fatigue, fever, chills
- Musculoskeletal pain, lymphadenopathy

One to several weeks dissemination:

- Neurologic: severe headache, meningitis, cranial nerve palsies (**Bell's palsy**)
- Cardiac: arrhythmias, myocarditis, pericarditis

Late and lasting for months or years:

- Arthralgias, arthritis (sx in 80% within few weeks to 2 years)

Laboratory Diagnosis

- Serodiagnosis by detecting IgM or IgG antibody (many false negatives/some false positives)
- Amplification/probes and cultures (Kelly medium) available

Treatment

- Doxycycline, amoxicillin, or azithromycin/clarithromycin (primary)
- Ceftriaxone for secondary
- Doxycycline or ceftriaxone for arthritis

Prevention

- DEET; avoid tick bite
- New vaccine: ospA flagellar antigen

GENUS: *LEPTOSPIRA*

- Spirochetes: thin, with hooks
- Too thin to visualize, but Gram-negative cell envelope

Leptospira interrogans

Distinguishing Characteristics

- Spirochetes with tight terminal hooks
 - seen on **dark-field microscopy** but not light microscopy
 - can be cultured *in vitro*; aerobic
- Generally diagnosed by serology

Reservoir

Wild and domestic animals

Transmission

- Via animal urine in water
- In U.S., via **dog, livestock, and rat urine** through contaminated **recreational waters** (jet skiers) or occupational exposure (sewer workers)

Pathogenesis

No toxins or virulence factors known.

Disease

Leptospirosis

- Influenza-like disease $\pm$ GI tract symptoms
- Progressing on to hepatitis and renal failure if not treated

Laboratory Diagnosis

Serodiagnosis and dark-field microscopy

Treatment

Penicillin G or doxycycline

Prevention

- Doxycycline effective for short-term exposure
- Vaccination of domestic livestock and pets; rat control

Table I-2-24. Comparison of the Genera *Rickettsia*, *Chlamydia*, and *Mycoplasma* with Typical Bacteria

	Typical Bacteria (<i>S. aureus</i>)	<i>Chlamydia</i>	<i>Rickettsia</i>	<i>Mycoplasma</i>
Obligate intracellular parasite?	Most no	Yes	Yes	No
Make ATP?	Normal ATP	No ATP	Limited ATP	Normal ATP
Peptidoglycan layer in cell envelope?	Normal peptidoglycan	Modified* peptidoglycan	Normal peptidoglycan	No peptidoglycan

*Chlamydial peptidoglycan lacks muramic acid and is considered by some as modified, by others as absent.

Table I-2-25. Infections Caused by Rickettsiae and Close Relatives

Group Disease	Bacterium	Arthropod Vector	Reservoir Host
<u>Spotted Fevers:</u>			
Rocky Mountain Spotted Fever	<i>Rickettsia rickettsii</i>	Ticks	Dogs, rodents, ticks
Rickettsialpox	<i>Rickettsia akari</i>	Mites	Mice
<u>Typhus Group:</u>			
Epidemic	<i>Rickettsia prowazekii</i>	Human louse	Humans
Endemic	<i>Rickettsia typhi</i>	Fleas	Rodents
Scrub	<i>Rickettsia tsutsugamushi</i>	Mites	Rodents
<u>Others:</u>			
Q fever	<i>Coxiella burnetii</i>	None	Cattle, sheep, goats
Bacillary angiomatosis in AIDS	<i>Bartonella quintana</i> and <i>B. henselae</i> (<i>Rochalimaea</i>)	Human louse	Humans
Cat Scratch Fever	<i>Bartonella henselae</i> (<i>Rochalimaea</i>)	None	Cats
Human granulocytic or monocytic Ehrlichiosis*	<i>Ehrlichia</i>	<i>Ixodes</i> ticks + ?	?

* New tick-borne rickettsial disease caused by *Ehrlichia*.

Note: Co-infections from a single tick infected with more than one agent may occur; N. East U.S.A.: *Babesia* and *B. burgdorferi*.

N. Central U.S.A.: *B. burgdorferi* and *Ehrlichia*.

GENUS: *RICKETTSIA*

Obligate intracellular bacteria

Rickettsia rickettsii

Distinguishing Characteristics

- **Obligate intracellular** bacteria that divide by binary fission and cannot make sufficient ATP.
- Not seen well on Gram-stained smear (too small), but have **Gram-negative cell envelope**.
- Cross-reaction of *Rickettsia* antigens with OX strains of *P. vulgaris* (Weil-Felix reaction)

Reservoir

Small wild rodents and larger wild and domestic animals (dogs)

Transmission

Hard ticks: *Dermacentor* (also reservoir hosts because of transovarian transmission)

Pathogenesis

Rickettsia invades **endothelial lining of capillaries, causing vasculitis**.

Disease

Rocky Mountain Spotted Fever

- Prevalent on **East Coast** (2–12 day incubation)
- Headache, fever, malaise, myalgias, toxicity, vomiting, and confusion
- **Rash** (maculopapular → **petechial**) **starts** (by day 6 of illness) **on ankles and wrists** and then spreads to the trunk, **palms, soles, and face (centripetal rash)**.
- **Ankle and wrist swelling** also occur.
- Dx may be confused by gastrointestinal symptoms, periorbital swelling, stiff neck, conjunctivitis and arthralgias.

Diagnosis

- Clinical symptoms (above) and tick bite
- **Start treatment without laboratory confirmation.**
- Serodiagnosis by complement fixation or Weil-Felix test

Treatment

Doxycycline

Prevention

Tick protection and prompt removal; doxycycline effective in exposed persons

GENUS: *COXIELLA*

- Obligate intracellular bacteria
- Two antigenic phases, one resistant to drying

Coxiella burnetii

Distinguishing Characteristics

- Obligate intracellular bacterium
- Not seen well on gram-stained smear

Reservoir

Domestic livestock: **pregnant animals have high titers.**

Transmission

- Inhalation of aerosols of urine, feces, amniotic fluid, or placental tissue
- Survives drying: can be infective miles away
- No significant arthropod vector in human infection

Disease

Q fever

- Febrile illness with **NO RASH**
- Pneumonia with hepatitis

Pathogenesis

Intracellular

Lab Diagnosis

Serodiagnosis; **Weil-Felix test is negative.**

Treatment

Doxycycline or erythromycin

Prevention

Vaccine for high-risk occupations

GENUS: *EHRlichia*

- Obligate intracellular bacteria
- Rickettsial family

Ehrlichia chaffeensis

- Reservoir: ticks and deer
- Transmitted by the Lone Star tick (*Amblyomma*)
- Infects monocytes and macrophages
- **Human monocytic ehrlichiosis**

***Ehrlichia* (near relative of *E. equi*)**

- Reservoir: ticks, deer, mice
- Transmitted by the *Ixodes* tick
- Infects primarily neutrophils
- **Human granulocytic ehrlichiosis**

Disease

Ehrlichiosis

- **Similar to Rocky Mountain spotted fever** but generally without rash
- Leukopenia or low platelets
- Morulae (mulberry-like structures inside infected cells)

GENUS: *CHLAMYDIA*

- Obligate intracellular bacteria
- Elementary body/reticulate body
- Not seen on Gram stain
- Cannot make ATP
- Cell wall lacks muramic acid

Chlamydia trachomatis

Distinguishing Characteristics

- Obligate intracellular bacterium; cannot make ATP.
- Found in cells as metabolically active, replicating reticulate bodies.
- Infective form: inactive, extracellular elementary body
- Not seen on Gram stain; peptidoglycan layer lacks muramic acid.

Reservoir

Human genital tract and eyes

Transmission

Sexual contact and at birth. Trachoma is transmitted by hand-to-eye contact. Flies (trachoma).

Pathogenesis

Infection of nonciliated columnar or cuboidal epithelial cells of mucosal surfaces leads to granulomatous response and damage.

Diseases

STDs in U.S.

- Serotypes D-K (This is the most common bacterial STD in U.S. Herpes and HPV are more common.)
- Nongonococcal urethritis, cervicitis, PID, and major portion of infertility (no resistance to reinfection)
- Inclusion conjunctivitis
- Inclusion conjunctivitis and/or pneumonia in neonates/infants (staccato cough)

Lymphogranuloma venereum

- Serotypes L1, 2, 3
 - STD is prevalent in Africa, Asia, and South America
 - Swollen lymph nodes leading to genital elephantiasis in late stages
 - Tertiary: ulcers, fistulas, genital elephantiasis.

Trachoma

- Leading cause of preventable infectious blindness, Serotypes A, B, Ba, and C
- Follicular conjunctivitis leading to conjunctival scarring and inturned eyelashes leading to corneal scarring and blindness.

Lab Diagnosis

- DNA probes in U.S.
- **Cytoplasmic inclusions seen on Giemsa-, iodine-, or fluorescent-antibody-stained smear or scrapings**
- **Cannot be cultured on inert media**
- Is cultured in **tissue cultures or embryonated eggs**
- Serodiagnosis: complement fixation or microimmunofluorescence test

Treatment

Doxycycline or azithromycin

Prevention

- Erythromycin is effective in infected mothers to prevent neonatal disease.
- Treat neonatal conjunctivitis with systemic erythromycin to prevent pneumonia.

Chlamydia pneumoniae

- Respiratory infections. Atypical pneumonia (single lobe) is very common.
- Infect smooth muscle, endothelial cells of coronary artery and macrophages.
- Potential association with atherosclerosis.

Chlamydia psittaci

Distinguishing Characteristics

- *Chlamydia* associated with birds.
- No glycogen in the inclusion bodies.

Reservoir

Birds: parrots (psittacine), turkeys, others

Transmission

- Dust of dried bird secretions and feces
- Turkeys are a major U.S. reservoir.

Pathogenesis

Intracellular growth

Disease

Psittacosis (Atypical Pneumonia)

- **Pneumonia often occurs with hepatitis.** Fever, chills, rash, myalgia but generally mild/moderate while X-ray may look severe.
- Cough may be absent; when present, non-productive initially; then scant mucopurulent.

Laboratory Diagnosis

- Cytoplasmic inclusions seen on Giemsa or fluorescent-antibody-stained sputum or biopsy
- Organism can be isolated from sputum in tissue culture, but rarely done.
- Serodiagnosis by complement fixation test

Treatment

Doxycycline

Prevention

No vaccine or drug is available.

GENUS: *MYCOPLASMA*

- Smallest free-living (extracellular) bacteria
- Missing peptidoglycan (no cell wall)
- Sterols in membrane

Mycoplasma pneumoniae (Eaton's agent)

Distinguishing Characteristics

- Extracellular, tiny, flexible
- No cell wall. Not seen on Gram-stained smear
- Membrane with cholesterol but **does not synthesize cholesterol**
- Requires cholesterol for *in vitro* culture

Reservoir

Human respiratory tract

Transmission

Respiratory droplets; close contact: families, military recruits, medical school classes, college dorms

Pathogenesis

- Surface parasite: not invasive
- **Attaches to respiratory epithelium via P1 protein**
- **Inhibits ciliary action**
- **Produces hydrogen peroxide, superoxide radicals, and cytolytic enzymes**, which damage the respiratory epithelium, leading to necrosis and a bad hacking cough (walking pneumonia).

Diseases

Pneumonia

- Pharyngitis
- May develop into an atypical pneumonia with persistent hack (little sputum produced)
- Most common pneumonia (along with viruses) in young adults

Lab Diagnosis

- Primarily clinical diagnosis; PCR/nucleic acid probes
- Microscopy not useful
- **Fried egg colonies** on Mycoplasma or Eaton's media (have sterols); 10 days
- **Positive cold agglutinins** (autoantibody to red blood cells) test is not very specific and positive in only 65%.
- Complement fixation test for antibodies to *Mycoplasma pneumoniae* is more specific.

Treatment

Erythromycin, azithromycin, clarithromycin. **No cephalosporins nor penicillins.**

Prevention

None

Ureaplasma urealyticum

Distinguishing Characteristics

- Belongs to the *Mycoplasma* family
- **Produces a urease**

Diseases

- **Urethritis** (half of the non-Neisserial, non-Chlamydial), prostatitis
- Can cause renal calculi

Chapter Summary

The morphology of Gram-positive and Gram-negative bacteria is compared. Only the Gram-negative cells have an outer membrane and as a consequence a periplasmic space. The cell wall of Gram-positive cells is relatively thick and has teichoic acid filaments bound to it. Acid-fast cells contain mycolic acid in their cell walls.

The Gram stain is a four-step procedure. Cells are treated with crystal violet and then with Gram's iodine. Cells are then washed with acetone or alcohol, which removes the purple/blue stain from Gram-negative cells. Safranin is used as a counterstain, leaving Gram-positive cells purple/blue but staining Gram-negative cells red/pink.

The Ziehl-Neelsen/Kinyoun acid-fast stain is used to identify cells that retain a carbol-fuchsin red dye after washing with acetone or alcohol. *Mycobacterium* and *Legionella micdadei* are acid fast, *Nocardia* is partially acid fast, and all other bacteria are not acid fast. Cysts of *Cryptosporidium* and *Isospora* are also acid fast.

In processing sputa for *Mycobacterium*, samples are first screened using an auramine-rhodamine fluorescent stain. Samples showing fluorescence are then stained with the acid-fast stain to confirm the presence of acid-fast bacteria.

The internal structure of bacterial cells includes the circular, intron- and histone-free, DNA-containing nucleoid region; 70S ribosomes, consisting of 50S and 30S subunits; various granules; and, in many cases, plasmids. Bacterial cells have no mitochondria, chloroplasts, or any other membrane-bound organelle.

Two Gram-positive genera, *Bacillus* and *Clostridium*, are endospore producers. These spores have no reproductive function but permit survival under adverse conditions.

Bacterial cells reproduce by binary fission. When cultured, they initially enter a short lag phase with no cell division and then enter a logarithmic (exponential) growth phase, followed by a stationary and finally a death phase.

The types of culture media required for growth of the various microbes are summarized in a series of tables (Tables 1-2-6, 1-2-7, and 1-2-8).

Children should be vaccinated against diphtheria, tetanus, and pertussis with either the DTP or the newer DTaP (acellular) vaccine and against *Haemophilus influenzae* type b with the Hib vaccine.

It is recommended that senior citizens get vaccinated against *Streptococcus pneumoniae*. Other vaccines with specialized uses include those against *Neisseria meningitidis*, *Salmonella typhi*, *Yersinia pestis*, *Bacillus anthracis*, and Bacille Calmette Guérin (BCG).

Table 1-2-9 lists the Gram-positive pathogens, Table 1-2-10 lists the non-Gram staining pathogens, and Table 1-2-11 lists the Gram-negative pathogens.

The remainder of this chapter is an inventory of the various medically important pathogenic bacteria arranged by genus. The diseases associated with each genus and the important properties for the various species in each genus are succinctly described.

Review Questions

1. Staphylococci are routinely differentiated from streptococci by
 - A. Coagulase test
 - B. Test with hydrogen peroxide
 - C. Polymerase chain reaction
 - D. Protein A immune assay
 - E. Growth in 6.5% sodium chloride

2. An atherosclerotic 80-year-old man develops a pelvic abscess following a ruptured appendix. What is/are the most likely causative agent(s)?
 - A. *Bacteroides* species and microaerophilic streptococci
 - B. *Candida albicans*
 - C. *Enterobacter aerogenes*
 - D. *Haemophilus influenzae* Group B
 - E. *Streptococcus viridans*

3. A 21-year-old student was seen by his family physician with complaints of pharyngitis. Examination of the pharynx revealed patchy erythema and exudates on the tonsillar pillars. Throat smear showed Gram-positive cocci in chains and Gram-negative diplococci. He admitted having been sexually active. What is the significance of the Gram stain smear in this case?
 - A. It provides a rapid means of diagnosing the infection
 - B. It should have been examined by an experienced microbiologist
 - C. It is not useful as it is not possible to make a diagnosis this way
 - D. It strongly suggests gonococcal pharyngitis
 - E. It is evidence of infection with hemolytic streptococci and neisseriae

4. With an appropriately performed acid-fast staining procedure, *Staphylococcus epidermidis* will appear
 - A. Blue
 - B. Red
 - C. Purple
 - D. Colorless
 - E. Brown

5. *Treponema pallidum* can be identified in a syphilitic lesion (either primary or secondary stage) by
 - A. Culture on Fletcher's serum semi-solid medium
 - B. Immunofluorescent stain of smear made from the active lesion
 - C. Gram stain
 - D. Special culture using hemoglobin and yeast extract
 - E. Rapid plasma reagin (RPR) assay

6. Which of the following properties is shared by *Legionella pneumophila* and *Mycobacterium avium-intracellulare*, an atypical *Mycobacterium*?
 - A. Common cause of venereal disease
 - B. Acid fast
 - C. Requiring iron and cysteine for growth
 - D. Not readily transmitted from person to person
 - E. Inability to grow in laboratory culture media
7. What laboratory test is most useful for diagnosis of Lyme disease?
 - A. Blood culture on sheep blood agar plate
 - B. Spinal fluid culture on Thayer-Martin agar
 - C. Detection of IgM/IgG antibodies to the spirochete
 - D. Detection of specific antibody to *Ixodes* tick
 - E. Documentation of fever and arthritis
8. With which of the following diseases is strict isolation indicated for the hospitalized patient?
 - A. Botulism
 - B. *Y. pestis* pneumonia
 - C. Pneumococcal pneumonia
 - D. *Mycobacterium kansasii* pulmonary infection
 - E. Cervical-facial actinomycosis
9. A previously healthy 5-month-old infant now with apparent upper body weakness including droopy eyes, head lag, drooling, and inability to sit unassisted. The most likely infectious form is
 - A. Elementary body
 - B. Reticulate body
 - C. Endospore
 - D. Exotoxin
 - E. Vegetative cell
10. Sixteen residents in a retirement home have fever, malaise, and anorexia. These residents have taken their meals prepared by the same kitchen. Blood cultures from 11 of these residents grow *Salmonella typhi*. The primary reservoir of this organism is
 - A. Hen's egg
 - B. Dogs and cats
 - C. Turkeys
 - D. People
 - E. Water

11. If a culture is inoculated to a density of 5×10^2 cells/ml at time 0 and has both a generation time and lag time of 10 minutes, how many cells/ml will there be at 40 minutes?
 - A. 1.5×10^3
 - B. 2×10^3
 - C. 4×10^3
 - D. 6×10^3
 - E. 4×10^6

12. A 6-year-old girl crashed on a toboggan ride and complained of pain in the perineal area. Exam showed only bruising of the area. Two days later, she developed fever, prostration, discoloration of the buttock and blebs of the skin in the area. After admission to the hospital, she developed progressive involvement of the leg, thigh, and buttock with extension to the lower abdomen. She went into shock and died before surgery could be performed. At autopsy, a 1-inch piece of wood was found in the perineum, which had perforated the anus. The most likely causative agent
 - A. Requires an elevated oxidation reduction potential
 - B. Is a Gram-negative coccobacillus
 - C. Is a marked lecithinase producer
 - D. Is non-hemolytic on blood agar
 - E. Is non-fermentative

13. A 71-year-old male was admitted from his extended care facility (nursing home) because of recent aggravation of an exfoliative skin condition that has plagued him for several years. He had been receiving a variety of topical antibiotic regimens over the last year or two. He now has a temperature of 38.9°C (102°F). The skin of upper chest, extremities, and neck shows erythema with diffuse epidermal peeling and many pustular lesions. Cultures obtained from these lesions were reported back from the laboratory as yielding a Gram-positive organism that is highly salt (NaCl) tolerant. What lab result is used to confirm the species of the causative agent?
 - A. Bacitracin sensitivity
 - B. Bile solubility
 - C. Catalase production
 - D. Coagulase production
 - E. Optochin sensitivity

14. Eight of 10 family practice residents who had a potluck 4 days ago now have diarrhea with abdominal cramps, general malaise, and fever ranging from 37.5° to 38.7°C . Stools from three are blood tinged. Laboratory studies revealed the causative agent was a microaerophilic Gram-negative, curved rod with polar flagella often in pairs to give a "seagull" appearance. It grew on special media at 42°C . The original contamination probably was found in
 - A. Poultry
 - B. Improperly canned food
 - C. Fried rice
 - D. Fish
 - E. Vegetables

15. In the screen for bacterial meningitis (most commonly a latex particle agglutination test), what chemical component are we searching for in the cerebrospinal fluid?
 - A. Cellular proteins
 - B. Endotoxin
 - C. Immunoglobulins
 - D. Polysaccharide
 - E. Ribosomal RNA

16. What percentage of the time does tetracycline resistance accompany methicillin resistance in methicillin-resistant *Staphylococcus aureus*?
 - A. Less than 5% of the time
 - B. 5–24% of the time
 - C. 25–49% of the time
 - D. 50–74% of the time
 - E. Greater than 75% of the time

17. The structure that is found in Gram-negative but not in Gram-positive bacteria is
 - A. Capsule
 - B. Cell wall
 - C. Cytoplasmic membrane
 - D. Endospore
 - E. Outer membrane

18. Since prokaryotes do not possess mitochondria, oxidative phosphorylation and electron transport in these cells take place in association with the
 - A. Polysaccharide gel layer
 - B. Lipopolysaccharide layer
 - C. Peptidoglycan layer
 - D. Periplasmic space
 - E. Lipoprotein bilayer

19. A 5-year-old child of an Eastern European immigrant family is brought to your pediatric clinic. The child is afebrile, but weak and exhausted from a week of paroxysmal coughing with inspiratory whoops, frequently associated with vomiting. The parents profess religious objections to childhood vaccinations, but permit withdrawal of a blood sample, which reveals a lymphocytosis of 44,000/mm³. Production of lymphocytosis, insulin secretion, and histamine sensitization are all results of which attribute of this organism?
 - A. Filamentous hemagglutinin
 - B. Adenylate cyclase toxin
 - C. Beta-hemolysin
 - D. Anaerobic growth
 - E. Pertussis toxin
 - F. Tracheal cytotoxin
 - G. Motility throughout the circulation

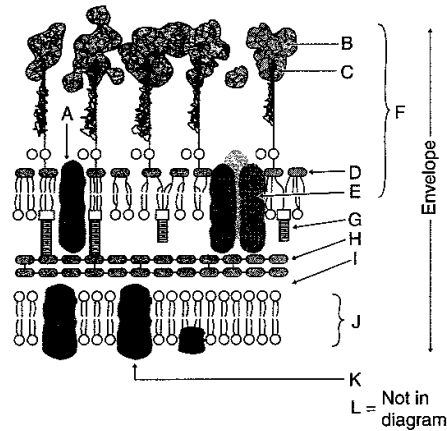
20. The earliest definitive diagnosis of shigellosis in the U.S. is routinely made by clinical findings and
 - A. Culture of the stool
 - B. Identification of Shiga toxin in stools
 - C. Positive blood cultures
 - D. Isolation of a sorbitol fermenting Enterobacteriaceae either from stool or blood
 - E. Demonstration of fecal PMNs
 - F. PCR for H antigens
21. Which of the following bacterial structures or products is notoriously anti-phagocytic?
 - A. Teichoic acid of *Streptococcus pyogenes*
 - B. Bound coagulase of *Staphylococcus aureus*
 - C. Lipopolysaccharide (LPS)
 - D. Pili of gonococci
 - E. Peptidoglycan of rough strains of pneumococci
22. What structure is most responsible for triggering Gram-negative shock?
 - A. Capsule
 - B. Heat shock proteins
 - C. Outer membrane
 - D. Periplasmic space
 - E. Peptidoglycan-teichoic acid fragments
 - F. Sex pili
23. Pneumococcal pneumonia or meningitis rarely occurs in the absence of what virulence factor?
 - A. Capsule
 - B. Heat shock proteins
 - C. Outer membrane
 - D. Periplasmic space
 - E. Peptidoglycan-teichoic acid fragments
 - F. Sex pili
24. What antigen is most useful in identifying nephritogenic strains of Group A streptococci which may induce glomerulonephritis?
 - A. Capsular antigen
 - B. Cell wall carbohydrates
 - C. M proteins
 - D. Outer membrane proteins
 - E. P protein
 - F. Teichoic acids

25. What toxin stimulates adenylate cyclase by catalyzing the transfer of ADP-ribose to the inhibitory subunit of the G protein?
- A. Pertussis toxin
 - B. Cholera toxin
 - C. Diphtheria toxin
 - D. *Escherichia coli* labile toxin
 - E. *Escherichia coli* verotoxin
 - F. *Shigella dysenteriae* Shiga toxin
26. What toxin blocks the function of the inhibitory neurotransmitter at synapses in the spinal cord leading to spasm?
- A. *Bordetella pertussis* neurotoxin
 - B. Cholera toxin
 - C. *Clostridium botulinum* toxin
 - D. Diphtheria toxin
 - E. *Escherichia coli* verotoxin
 - F. Shiga toxin
 - G. Tetanus toxin
27. What toxin inhibits acetylcholine release at the neuromuscular junction?
- A. *Bordetella pertussis* neurotoxin
 - B. Cholera toxin
 - C. *Clostridium botulinum* toxin
 - D. Diphtheria toxin
 - E. *Escherichia coli* verotoxin
 - F. Shiga toxin
 - G. Tetanus toxin
28. What toxin inhibits protein synthesis in mammalian cells by catalyzing the ADP-ribosylation of the elongation factor 2 (EF2)?
- A. Pertussis toxin
 - B. Cholera toxin
 - C. Botulinum toxin
 - D. Diphtheria toxin
 - E. *Escherichia coli* verotoxin
 - F. Shiga toxin
 - G. Tetanus toxin

29. What toxin continually stimulates adenylate cyclase to overproduce cAMP by catalyzing the binding of ADP-ribose to the Gs protein leading to severe fluid loss?
 - A. Pertussis toxin
 - B. Cholera toxin
 - C. *Escherichia coli* stable toxin
 - D. *Escherichia coli* verotoxin
 - E. Shiga toxin
 - F. Tetanus toxin
30. Which organism grows best in low O₂ concentrations but requires oxygen?
 - A. *Bacteroides melaninogenicus* (*Prevotella melaninogenica*)
 - B. *Campylobacter jejuni*
 - C. *Escherichia coli*
 - D. *Mycobacterium leprae*
 - E. *Mycobacterium tuberculosis*
31. What is the major chemical component giving bacterial cells protection from osmotic damage?
 - A. Lipopolysaccharide-phospholipid
 - B. Peptidoglycan
 - C. Phospholipid
 - D. Polysaccharide
 - E. Protein
 - F. Teichoic acid
32. A 27-year-old female, returning home from her honeymoon, has developed urinary frequency, dysuria, and urgency. Her urine is grossly bloody. Which lab data are most likely to define the causative agent?
 - A. A Gram-negative diplococcus, which is oxidase positive but does not ferment maltose
 - B. A Gram-positive coccus, which is catalase positive and coagulase negative
 - C. An optochin-resistant, catalase-negative, Gram-positive coccus
 - D. A Gram-positive bacillus grown on a low oxidation-reduction medium
 - E. A Gram-negative bacterium capable of reducing nitrates to nitrites
33. Two days after eating a meal that included home-canned green beans, three people developed various degrees of visual problems, including double vision and difficulties focusing. Describe the Gram reaction of the organism most likely to be isolated from the left-over beans and lab findings, which will be used in its identification.
 - A. A Gram-positive coccus, which is catalase-positive and grows in a high salt environment
 - B. A Gram-positive aerobic bacillus, which sporulates
 - C. A Gram-positive coccus, which is catalase-negative and optochin-resistant
 - D. A Gram-positive bacillus grown on a low oxidation-reduction medium
 - E. A Gram-negative bacillus capable of reducing nitrates to nitrites

34. A 16-year-old has pneumonia with a dry, hacking cough. The X-ray pattern shows a light, diffuse infiltrative pattern. The most likely organism producing these symptoms is
 - A. A non-Gram-staining bacterium requiring sterols
 - B. A bacillus showing granules when stained with methylene blue
 - C. A bacitracin-sensitive, catalase-negative Gram-positive coccus
 - D. A coagulase positive, Gram-positive coccus in clusters; catalase positive
 - E. A Gram-positive bacillus grown on a low oxidation-reduction medium
35. Because of cold growth of this fairly common animal fecal bacterium potentially contaminating some deli meats and soft cheeses, renal transplant patients should only have hot deli sandwiches. What is this causative agent of meningitis in transplant patients?
 - A. *Brucella* spp
 - B. *Francisella tularensis*
 - C. *Leptospira interrogans*
 - D. *Listeria monocytogenes*
 - E. *Streptococcus pneumoniae*
36. A 55-year-old woman had her rheumatic heart valve replaced with a prosthetic valve. Six blood cultures became positive after 3 days of incubation. An optochin-resistant, catalase-negative Gram-positive coccus that was alpha-hemolytic was isolated. What was the most likely causative agent?
 - A. *Candida albicans*
 - B. *Pseudomonas aeruginosa*
 - C. *Serratia marcescens*
 - D. *Staphylococcus aureus*
 - E. *Streptococcus pneumoniae*
 - F. *Streptococcus pyogenes*
 - G. *Streptococcus viridans*
37. Which of the following organisms is killed by oxygen and ferments in absence of oxygen?
 - A. *Bacteroides melaninogenicus* (*Prevotella melaninogenica*)
 - B. *Campylobacter jejuni*
 - C. *Escherichia coli*
 - D. *Mycobacterium leprae*
 - E. *Mycobacterium tuberculosis*
38. A virulence factor that causes *Mycobacterium tuberculosis* to clump together and grow in a "serpentine-like" fashion is
 - A. Endotoxin
 - B. M protein
 - C. PPD (purified protein derivative)
 - D. Slimy capsule
 - E. Trehalose-6,6-dimycolate
 - F. Wax D

39. From the diagram below, pick the structure that is associated with a passive transport across the membrane.



40. Calcium dipicolinate is found in
- Aspergillus*
 - Bacillus*
 - Escherichia*
 - Mycobacterium*
 - Rickettsia*
 - Vibrio*
41. What are the trimeric structures involved in transport of materials across the outer membrane of the Gram-negative bacteria?
- GTP-binding proteins
 - Lipopolysaccharides
 - Outer membrane proteins
 - Periplasmic space
 - Porin proteins
 - Prion proteins
42. What is the function of penicillin-binding proteins when there is no penicillin present in the bacterium's environment and the cell is actively replicating?
- They are involved in microtubule formation and cell division
 - They have enzymatic activity: transpeptidases and carboxypeptidases
 - They are involved in protein elongation
 - They are involved in the supercoiling of DNA
 - They are transcriptional regulators

43. What is an organism called that respire in the presence of oxygen and ferments in the absence of oxygen?
- A. Aerobe
 - B. Anaerobe
 - C. Facultative aerobe
 - D. Facultative anaerobe
 - E. Microaerophile
 - F. Obligate aerobe
44. A 15-day-old male presents with purulent conjunctivitis. Iodine staining bodies are seen in conjunctival scrapings. The most likely infectious form is
- A. Elementary body
 - B. Reticulate body
 - C. Endospore
 - D. Exotoxin
 - E. Vegetative cell
 - F. Virus resistant to alcohol
 - G. Virus sensitive to alcohol
45. What organism is most likely responsible for bacterial pneumonia in persons with alcohol intoxication?
- A. *Haemophilus influenzae*
 - B. *Proteus vulgaris*
 - C. *Pseudomonas aeruginosa*
 - D. *Staphylococcus aureus*
 - E. *Streptococcus pneumoniae*
 - F. *Streptococcus viridans*
46. What organism is most likely responsible for bacterial meningitis in infants during the first month of life?
- A. *Enterococcus faecalis* (*Streptococcus faecalis*)
 - B. *Haemophilus influenzae*
 - C. *Staphylococcus aureus*
 - D. *Streptococcus agalactiae*
 - E. *Streptococcus pneumoniae*
 - F. *Streptococcus pyogenes*

47. What organism is most likely responsible for bacterial endocarditis in men following urological instrumentation?
 - A. *Enterococcus faecalis*
 - B. *Pseudomonas aeruginosa*
 - C. *Streptococcus pyogenes*
 - D. *Streptococcus viridans*
 - E. *Ureaplasma urealyticum*
48. An AIDS patient with septicemia and a target-shaped necrotic lesion on the buttock with a black center and an erythematous margin. Which causative agent is most likely?
 - A. *Bacillus anthracis*
 - B. *Enterococcus faecalis*
 - C. *Pseudomonas aeruginosa*
 - D. *Staphylococcus aureus*
 - E. *Streptococcus pyogenes*
49. What causative agent is most likely responsible for edema, hematuria, proteinuria in a patient who had impetigo 3 weeks ago?
 - A. *Clostridium perfringens*
 - B. *Pseudomonas aeruginosa*
 - C. *Staphylococcus aureus*
 - D. *Staphylococcus epidermidis*
 - E. *Streptococcus agalactiae*
 - F. *Streptococcus pyogenes*
50. Patient was admitted to the hospital because of bleeding duodenal ulcer. Culture at 37°C grew urease-positive curved bacteria. The most likely causative agent is
 - A. *Campylobacter jejuni*
 - B. *Entamoeba histolytica*
 - C. *Enterococcus faecalis* (*Streptococcus faecalis*)
 - D. *Helicobacter pylori*
 - E. *Pseudomonas aeruginosa*

Answers

1. **Answer: B.** The catalase test is carried out with hydrogen peroxide. The other four tests do not differentiate.
2. **Answer: A.** Atherosclerosis leads to poor circulation to the lower extremities, which in turn lowers the oxidation-reduction potential of the tissues. All this predisposes to infections caused by anaerobic M-Os, in this case, *Bacteroides* and *Streptococci*. The patient is suffering from anaerobic cellulitis or possibly myonecrosis.
3. **Answer: C.** Gram-positive cocci (alpha hem. Strep) and Gram-negative cocci (neisseriae) are normally present in the throat. There is no way to differentiate pathogens from non-pathogens by the Gram stain.
4. **Answer: A.** Students need to remember that the primary stain in the acid-fast stain is carbol-fuchsin, which stains acid-fast organisms red. The counterstain is methylene blue, which stains everything else blue. They also need to know the three important genera of acid-fast organisms: *Mycobacterium*, *Nocardia*, and *Cryptosporidium*.
5. **Answer: B.** *Treponema* cannot be cultured. Fletcher's medium is for *Leptospira*. The spirochete that will take Gram stain is *Borrelia*. RPR is for detection of antibody.
6. **Answer: D.** Neither one is common for venereal disease. *Legionella* is not acid fast. Iron and cysteine requirement refers to *Legionella*. Both can grow in artificial media.
7. **Answer: C.** *Borrelia burgdorferi*, the causative agent of Lyme disease, grows in a complex medium, not on sheep blood agar. Thayer-Martin is for pathogenic neisseriae. There is no purpose to detect antibody to the tick. ELISA or some rapid test for IgG/IgM is currently used for diagnosis of Lyme disease.
8. **Answer: B.** *Y. pestis* causes bubonic plague, which is not contagious through respiratory droplets; however, the bubonic form can progress via septic emboli to pneumonia, which is contagious. Adult botulism is a toxemia, thereby not contagious. Pneumococcal pneumonia is caused by *S. pneumoniae*, which is not considered highly infectious and colonizes many people without causing disease. Choice D (*Mycobacterium kansasii*) and choice E (*Actinomyces*) are both environmental M-O: not contagious.

Remember that as a rule-of-thumb, M-O that have the environment (water or soil) as reservoirs, and those that are zoonotic or arthropod-borne, and those that are normal human flora are generally not contagious from person to person. *Y. pestis* is an exception.
9. **Answer: C.** Infant botulism is a toxi-infection started by the ingestion of *Clostridium botulinum* endospores from the environment. The spores germinate in the immature flora of the GI tract and the toxin is produced *in vivo* in contrast to adult botulism where the preformed toxin is ingested.
10. **Answer: D.** The reservoir for *S. typhi* is people (humans). Other species of *Salmonella* have animals as their reservoirs.
11. **Answer: C.** Explanation: Remember that each cell divides into two at each generation following the single lag phase. So at the end of the first 10 minutes there is still 5×10^2 , and then at the end of the first 20 minutes (total) there are 10×10^2 . At the end of 30 minutes total time there will be 20×10^2 , and at the end of the total time, 40×10^2 , which is written 4×10^3 in proper scientific notation.

12. **Answer: C.** The description suggests strongly that she has myonecrosis. Therefore, the causative agent (at least one) is *C. perfringens*. *C. perfringens* is an anaerobe, therefore choice A is wrong. Clostridia are all Gram-positive, therefore choice B is wrong. *C. perfringens* have concentric areas of beta hemolysis, therefore choice D is wrong. *C. perfringens* is a marked lecithinase producer; therefore choice C is correct.
13. **Answer: D.** The patient has the "scalded skin" syndrome caused by *S. aureus*. The GENUS *Staphylococcus* would be distinguished from Strep by Staphylococcal production of catalase. But the SPECIES (*S. aureus*) would be distinguished from *S. epidermidis* on the basis of *S. aureus* production of coagulase. Bacitracin sensitivity and bile solubility are species characteristics of Strep pneumoniae.
14. **Answer: A.** The clue is Gram-negative curved rods with polar flagella often in pairs to give a "seagull" appearance and the microaerophilic on special media and growing at 42°C. That description is most compatible with *Campylobacter jejuni*. Poultry are one of the most important reservoirs so choice A is the correct response.
15. **Answer: D.** The diagnostic test looks for capsular material, which is a polysaccharide (choice D). The only non-polysaccharide one is anthrax, which is a polypeptide. Immunoglobulins would not be found this early in the CSF and may never be formed in some severely immunocompromised patients.
16. **Answer: E.** Methicillin-resistant strains of *Staph aureus* are generally resistant to all available antibiotics except for vancomycin, teicoplanin, and fusidic acid. Therefore, the answer is generally 100%, making choice E the best answer.
17. **Answer: E.** Capsules, cell wall, and cytoplasmic membranes are found in both Gram-positive and Gram-negative bacteria. Endospores (choice D) occur with certain Gram-positive bacteria, e.g., *Bacillus* and *Clostridium*. Only Gram-negatives have an outer membrane.
18. **Answer: E.** The site of most metabolic processes in the prokaryotic cell is the cytoplasmic membrane, which is best described as a lipoprotein bilayer (choice E). Of the distractors: Polysaccharide (choice A) describes the capsule layer, which is antigenic and antiphagocytic and outside the cell; lipoprotein (choice B) describes the outer membrane of Gram-negative bacteria, which also is basically outside of the cell; peptidoglycan (choice C) refers to the cell wall layer of all bacteria and periplasmic space; choice D refers to the space between the inner membrane and cell wall of Gram-negative bacteria. None of the distractors (A)–(D) are significant sites of action of any of the metabolic enzymes.
19. **Answer: E.** The disease here is whooping cough, caused by *Bordetella pertussis*. The pertussis toxin (also known as the lymphocytosis-promoting toxin) is not believed to be directly cytotoxic, but stimulates adenylate cyclase by ribosylating regulatory proteins. It causes a variety of effects depending on the cell type involved: insulin secretion, lymphocytosis, and alteration of immune effector cells. Of the distractors: the filamentous hemagglutinin (choice A) mediates attachment; the adenylate cyclase toxin (choice B) stimulates local edema; the organism produces only a small zone of hemolysis around its colonies, so choice C is not true; it is an aerobe and does not grow anaerobically (choice D); and is non-motile (choice G). All systemic manifestations of the disease arise from the circulation of the toxins, not the organism itself.

20. **Answer: A.** Shigellosis is an invasive disease confined to the intestine. Definitive diagnosis is made by prompt stool culture (usually positive by week 2). The finding of PMNs is not definitive for *Shigella*. In the U.S., the most common organism is *Shigella sonnei* or *flexneri*, neither of which produce Shiga toxin, making choice B false. Although *Shigella* is invasive, it does not invade the vasculature so blood cultures would not be positive. Sorbitol fermentation is used in isolation of most strains of VTEC. Remember that *Shigellae* are nonmotile; thus, there are no H antigens.
21. **Answer: D.** The correct answer to this question is the pili of *Neisseria gonorrhoeae*. Another antiphagocytic component is any bacterial capsule (missing in rough strains of *Pneumococcus* and the peptidoglycan does not inhibit phagocytosis, making choice E not a proper choice). Other correct answers would have been M-protein of *Strep pyogenes*, and the A proteins of *Staph aureus*.
22. **Answer: C.** Lipid A is the actual component of LPS, which is responsible for triggering Gram-negative shock. Since it is not listed, then you need to think about where it is found as it is a structural toxin. It is located in the outer membrane, and thus this is the correct answer. Peptidoglycan-teichoic acid fragments are found in Gram-positive bacteria only and are responsible for triggering septic shock when a Gram-positive organism is in the bloodstream, or inflammatory response when it is in the CSF or in tissues.
23. **Answer: A.** The most important virulence factor for *S. pneumoniae* is the capsule. Without it the strains are avirulent, except in seriously immunocompromised patients.
24. **Answer: C.** Remember first that Group A Strep is also called *Streptococcus pyogenes* and that it has a nonimmunogenic capsule of hyaluronic acid. The important antigen here is a surface protein called the M protein; it is used in "typing" GAS. Certain M types are more commonly involved in glomerulonephritis with M12 the most common in AGN. The large number of M-protein types and nonimmunogenic capsules (hyaluronic acid) are why many people get repeated Strep infections.
25. **Answer: A.** Pertussis toxin activates adenylate cyclase by turning off G_i protein via ADP-ribosylation. Cholera toxin and the labile toxin of *E. coli* are the other two that ADP ribosylate G proteins, but their ADP ribosylation is of G_s . Both verotoxin and Shiga toxin clip the 60S ribosomal subunit.
26. **Answer: G.** Tetanus toxin blocks the release of inhibitory factor (glycine) at the spinal cord. If one forgets the mode of action, think of the clinical presentation of tetanus, which is spasm rather than paralysis, as in botulism.
27. **Answer: C.** This is the mode of action of botulism toxin, and the result is paralysis of muscles.
28. **Answer: D.** Diphtheria toxin inactivates EF2 and thus inhibits protein synthesis. Students need to remember *Pseudomonas* exotoxin will do the same—a question frequenting USMLE exams.
29. **Answer: B.** Cholera toxin locks G_s protein in the "on" position via ADP-ribosylation. Adenylate cyclase is thus continually being stimulated. Most severe fluid loss leading to hypovolemic shock occurs with cholera.
30. **Answer: B.** *Campylobacter* is the only microaerophile on the list.

31. **Answer: B.** Although the outer membrane (which is chemically a lipopolysaccharide-phospholipid) plays a minor role in protecting Gram-negative bacteria from osmotic damage, it is not as important as the cell wall. CW is the peptidoglycan net structure that offers major protection from osmotic damage and confers cell shape.
32. **Answer: E.** *E. coli*-induced cystitis is highest in sexually active females. It generally reduces nitrates and is also a lactose fermenter. Choice A = *Neisseria gonorrhoeae*; choice B = *Staphylococcus saprophyticus*; choice C = *Enterococcus faecalis* is one possibility; choice D = *Clostridium*.
33. **Answer: D.** This case history describes botulism (key words: home-canned green beans and visual problems). Foods classically associated are those with a neutral or alkaline pH. *C. botulinum*, the agent of botulism, is an anaerobe and thus has a low oxidation-reduction requirement. The other Gram-positive bacillus (aerobic) would be *Bacillus cereus*. Choice A = *Staph aureus*; choice B = *Bacillus cereus*; choice C = *S. viridans*; choice D = *Clostridium botulinum*; choice E = *E. coli*.
34. **Answer: A.** The disease is most likely mycoplasma pneumonia caused by *Mycoplasma pneumoniae*, which is non-Gram staining and requires cholesterol for growth. Choice B = *C. diphtheriae*; choice C = *S. viridans*; choice D = *Staph aureus*; choice E = *Clostridium*.
35. **Answer: D.** Meningitis in renal transplant patients and many cancer patients is frequently *Listeria* acquired from foods. This is also a problem if pregnant women are infected; it may cause meningitis in neonates.
36. **Answer: G.** A patient with a history of rheumatic fever and blood cultures growing an alpha hemolytic coccus, which is catalase-positive and optochin-resistant.
37. **Answer: A.** This is one of the dark-pigmented *Bacteroides*, now called *Prevotella*. They are obligate anaerobes.
38. **Answer: E.** Cord factor (trehalose dimycolate) causes serpentine-like clumps of TB organism.
39. **Answer: E.** This refers to the porins that passively (with the aid of electrostatic charges) allow entry of materials into the periplasmic space, from whence they are actively transported across the cytoplasmic membrane.
40. **Answer: B.** Calcium dipicolinate is found only in the core of bacterial spores. It plays a role in the dehydration and stabilization of the DNA in the spores. The two bacterial genera that form endospores are *Bacillus* and *Clostridium*.
41. **Answer: E.** The trimeric structures spanning the outer membrane are porins, a series of proteins forming a pore, permitting passive transfer of materials into the periplasmic space. They are outer membrane proteins. Please remember that this is a "best answer test," and if an answer is more specific to the question, it would be better.
42. **Answer: B.** Penicillin-binding proteins (PBPs) are involved in the final cross linkage of new pieces of peptidoglycan. PBPs are cellular transpeptidases and carboxypeptidases. They are located in the cytoplasmic membrane.
43. **Answer: D.** This describes a facultative anaerobe. Facultative aerobe is a term that does not exist.

44. **Answer: A.** The patient has inclusion conjunctivitis caused by *Chlamydia trachomatis*. The only form of this bacterium that has the ability to bind to the membranes and infect is the elementary form.
45. **Answer: E.** *S. pneumoniae* is the most common cause of community-acquired pneumonia. One of the predisposing factors is alcohol intoxication; another one is influenza virus infection.
46. **Answer: D.** *Strep agalactiae* is one of the most prominent neonatal pathogens. The other common causative agent is *E. coli*, which is not on the list.
47. **Answer: A.** Several organisms on this list can cause endocarditis under various conditions. *E. faecalis* is the one associated with urological manipulation. *S. viridans* is associated with oral manipulation.
48. **Answer: C.** *Pseudomonas aeruginosa*. Target-shaped necrotic lesion with a black center and an erythematous margin depicts ecthyma gangrenosum caused by *Pseudomonas* associated with immunodeficiency.
49. **Answer: F.** Edema, hematuria, and proteinuria are pathognomonic for acute glomerulonephritis following *S. pyogenes* skin infections. Impetigo can be caused by *S. aureus* and *S. pyogenes* but only *S. pyogenes* leads to acute glomerulonephritis.
50. **Answer: D.** This patient had *Helicobacter*, which is urease positive. *Campylobacter jejuni* also grows at 42°C, but it is urease negative.

Medically Important Fungi

3

What the USMLE Requires You to Know

- Basic morphology of fungi (hyphae, yeast, dimorphic, and various types of conidia)
- Basic chemistry, particularly that involved in antifungals or that distinguishes fungi from other groups
- Scientific names of the fungal pathogens and opportunists found in the United States
 - What are the four dimorphic fungi?
 - Recognize their tissue and environmental forms.
 - What are the three dermatophytes and what tissues do they invade?
 - What is the most common cause of meningitis in AIDS patients?
 - What does it look like? How do you diagnose?
 - What is a common cause of interstitial pneumonitis in AIDS patients?
 - Why is it considered a fungus?
 - Which medically important fungus has a capsule?
 - Which is found most commonly inside cells of the reticuloendothelial system?
- The diseases they cause and:
 - How acquired (geography, route)
 - Common presenting symptoms, most common sites of dissemination (if they disseminate commonly)
 - What two fungi are a problem in IV lines?
 - Most common cause of fungal septicemia and the clues used in the cases (germ tube test positive, pseudohyphae and true hyphae as well as yeast forms)

MYCOLOGY

Mycology is the study of fungi (molds, yeasts, and mushrooms).

All fungi are

- **Eukaryotic** (e.g., true nucleus, 80S ribosomes, mitochondria, as are humans).
- **Complex carbohydrate cell walls: chitin, glucan, and mannan.**
- **Ergosterol = Major membrane sterol**
 - Imidazole antifungals inhibit synthesis of ergosterol.**
 - Polyene antifungals bind more tightly to ergosterol than cholesterol.**
- Heterotrophic (require organic carbon)
 - Saprophytic or saprobic (fungus living on dead organic material)
 - Parasitic (fungus living on another living organism)

FUNGAL MORPHOLOGY

Hyphae = filamentous cellular units of molds and mushrooms



Figure I-3-1.
Nonseptate Hyphae

Nonseptate Hyphae

- No cross walls
- Broad hyphae with irregular width
- Broad angle of branching

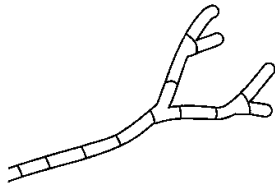


Figure I-3-2.
Septate Hyphae

Septate Hyphae

- With cross walls
- Width is fairly regular (tube-like).

Hyphal Coloration

- Dematiaceous: dark colored (gray, olive)
- Hyaline: clear

Mat of hyphae = mycelium

Yeasts = single celled (round to oval) fungi

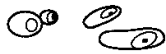


Figure I-3-3. Yeasts

Dimorphic Fungi

- Fungi able to convert from hyphal to yeast or yeast-like forms.
- Thermally dimorphic: in the "cold" are the mold form.

Key Dimorphic Fungi

Histoplasma
Blastomyces
Coccidioides
Sporothrix

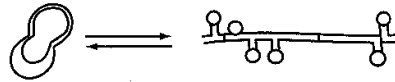


Figure I-3-4. Dimorphic Fungi

Pseudohyphae (*Candida albicans*)

Hyphae with constrictions at each septum



Figure I-3-5. *Candida*
Pseudohyphae

Spore Types

Conidia

- Asexual spores
- Formed off of hyphae
- Common
- Airborne

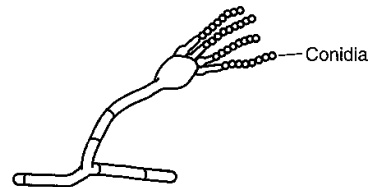


Figure I-3-6. Conidia

Blastoconidia: “Buds” on yeasts (asexual budding daughter yeast cells)



Figure I-3-7. Blastoconidia

Arthroconidia: Asexual spores formed by a “joint”

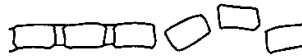


Figure I-3-8. Arthroconidia

Spherules and Endospores (*Coccidioides*): Spores inside the spherules in tissues

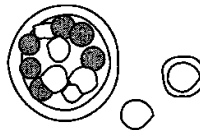


Figure I-3-9. Endospores and Spherules

Diagnosis

Table I-3-1. Microscopic Methods/Special Fungal Stains

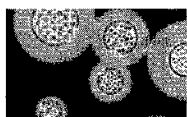
Preparation	Fungal Color	Notes
KOH wet mount (KOH degrades human tissues releasing hyphae and yeasts)	Colorless (hyaline) refractive green or light olive to brown (dematiaceous) fungal elements	Heat gently; let set 10 minutes; dissolves human cells
PAS	Hot pink	
Silver stain	Old rose gray to black	
Calcofluor white (Can be done on wet mounts.)	Bright blue-white on black	Scrapings or sections, fluorescent microscope needed
India ink wet mount of CSF sediment	Colorless cells with halos (capsule) on a black particulate background (<i>Cryptococcus neoformans</i>)	Only “rules in.” Insensitive; misses 50%. 

Figure I-3-10.
Cryptococcus neoformans

Culture

(May take several weeks.) Special fungal media: inhibitory mold agar is modification of Sabouraud's with antibiotics.

- Sabouraud's agar
- Blood agar
- Both of the above with antibiotics

Identification from Cultures

- Fungal **morphology**
- PCR with nucleic acid probes

Serology

(E.g., antibody screen, complement fixation, etc.) Looking for patient antibody.

Fungal Antigen Detection: (CSF, serum)

Cryptococcal capsular polysaccharide detection by latex particle agglutination (LPA) or counter immunoelectrophoresis

Skin Tests

- Most useful for **epidemiology** or **demonstration of anergy** to an agent you know patient is infected with (grave prognosis)
- Otherwise, like tuberculosis, a skin test **only indicates exposure** to the agent.

NONSYSTEMIC FUNGAL INFECTIONS

Superficial Infections (Keratinized Tissues)

Malassezia furfur (Fungus Name)

Normal skin flora (lipophilic yeast)

Diseases

- **Pityriasis or tinea versicolor**
 - Superficial infection of keratinized cells
 - **Hypopigmented spots on the chest/back** (blotchy suntan)
 - KOH mount of skin scales: spaghetti and meatballs
 - **Yeast clusters & short curved septate hyphae**
 - Treatment is topical selenium sulfide; recurs.
- **Fungemia in premature infants** on intravenous lipid supplements

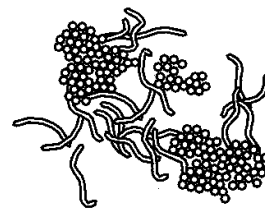


Figure I-3-11.
Malassezia furfur

Cutaneous Fungal Infections (without systemic disease)

Fungal or dermatophytic infections.

Yeast Skin Infections

- Commonly **cutaneous or mucocutaneous candidiasis**
- May disseminate in compromised patients
- Discussed with opportunistic fungi

Dermatophytes (Group of Fungi)

- Filamentous fungi (**monomorphic**)
- Infect **only skin and hair and/or nails** (do not disseminate)
- Three genera:
 - Trichophyton* - Infects skin, hair and nails
 - Microsporum* - Infects hair and skin
 - Epidermophyton* - Infects nail and skin



Figure I-3-12. Dermatophyte

Diseases

Dermatophytic Infections = **Tineas** (Ringworms)

- If **highly inflammatory**, generally from **animals** (zoophilic) (i.e., *Microsporum canis*: cats or dogs)
- If **little inflammation**, generally from humans (**anthropophilic** tinea capitis: *M. audouinii*)
- Tinea capitis = ringworm of the scalp
- The most serious of the tinea capitis is **favus** (tinea favosa), which causes **permanent hair loss** and is very contagious.
- Tinea barbae = ringworm of the bearded region
- Tinea corporis = dermatophytic infection of the glabrous skin
- Tinea cruris = jock itch
- Tinea pedis = athlete's foot

Diagnosis

- *Microsporum fluoresces* (Wood's lamp)
- **KOH mount** of nail or skin scrapings should show **arthroconidia and hyphae**.

Treatment

- Topical imidazoles or tolnaftate
- Oral imidazoles or griseofulvin where hairs are infected, or skin contact hurts
- Keep areas dry.
- **ID reaction (Dermatophytid) = Allergic response to circulating fungal antigens**

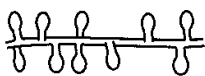


Figure I-3-13.
Sporothrix Hyphae

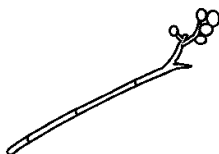


Figure I-3-14.
Sporothrix

Subcutaneous Mycoses

Sporothrix schenckii

Dimorphic Fungus

- **Environmental form:** on **plant material**, world wide as **hyphae with rosettes and sleeves of conidia**
- **Traumatic implantation** (rose or plum tree thorns, wire/sphagnum moss)
- **Tissue form:** **cigar-shaped yeast** in tissue

Diseases

- **Sporotrichosis** (rose gardener's disease): **subcutaneous or lymphocutaneous lesions**. Treatment: itraconazole or **potassium iodide in milk**
- **Pulmonary** (acute or chronic) **sporotrichosis**. Urban alcoholics, particularly homeless (alcoholic rose-garden-sleeper's disease)

DEEP FUNGAL INFECTIONS

Classical Pathogens

Three important classical pathogens in the U.S.A.:

Histoplasma

Coccidioides

Blastomyces

All three cause

- **Acute pulmonary** (asymptomatic or self-resolving in about 95% of the cases)
- **Chronic pulmonary**, or
- **Disseminated infections**

Diagnosis

(Most people never see a doctor.)

- **Sputum cytology** (calcofluor white helpful)
- **Sputum cultures** on blood agar and **special fungal media** (inhibitory mold agar, Sabouraud's)
- **Peripheral blood cultures are useful for *Histoplasma*** since it circulates in RES cells.

Histoplasma capsulatum

Dimorphic Fungus

- Environmental form: hyphae with microconidia and tuberculate macroconidia
- Endemic region: Eastern Great Lakes, Ohio, Mississippi, and Missouri River beds

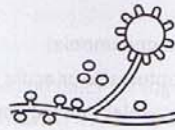


Figure I-3-15. *Histoplasma* Environmental Form

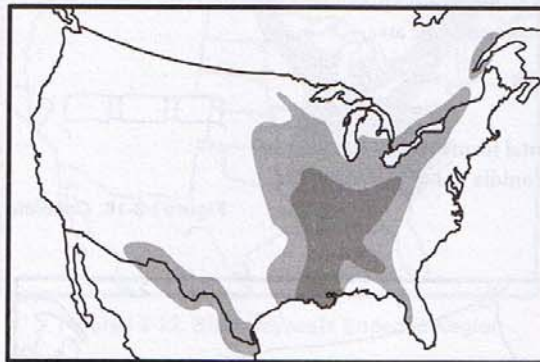


Figure I-3-16. *Histoplasma* Endemic Region

- Found in soil (dust) enriched with bird or bat feces
- Spelunking (cave exploring), cleaning chicken coops, or bulldozing starling roosts
- Tissue form: small intracellular yeasts with narrow neck on bud; **no capsule**
- Facultative intracellular parasite found in reticuloendothelial (RES) cells (tiny; can get 30 or so in a human cell)

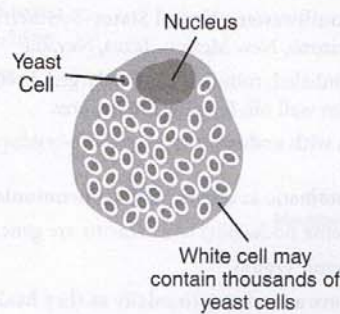


Figure I-3-17. Human RES Cell

Disease

Fungus flu (a pneumonia)

- Asymptomatic or acute (but self-resolving) pneumonia with flu-like symptomatology
- **Hepatosplenomegaly may be present** even in acute pulmonary infections (facultative intracellular RES)
- Very common in summer in endemic areas: kids or newcomers (80% of adults are skin-test positive in some areas)
- Lesions have a tendency to **calcify as they heal**.
- Relapse potential increases with T cell immunosuppression.
- **Disseminated infections:** Mucocutaneous lesions are common; also common in AIDS patients in endemic area.

Coccidioides immitis

Dimorphic Fungus

- **Environmental form:** hyphae breaking up into arthroconidia found in desert sand.

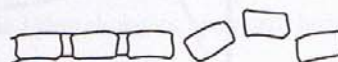


Figure I-3-18. *Coccidioides immitis*

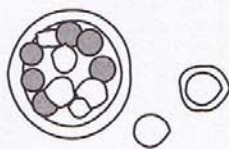


Figure I-3-20. *Coccidioides immitis* Spherules

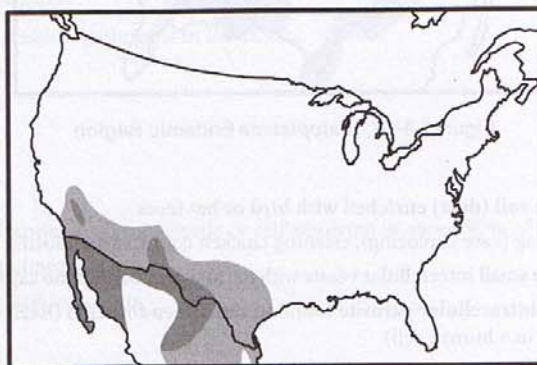


Figure I-3-19. *Coccidioides* Endemic Region

- Endemic region: **Southwestern United States**—Southern California (especially San Joaquin Valley), Arizona, New Mexico, Texas, Nevada
- Arthroconidia are inhaled, round up, and enlarged, becoming spherules inside which the cytoplasm wall off, forming endospores.
- **Tissue form: spherules with endospores**

Disease: Valley Fever (asymptomatic to self-resolving pneumonia)

- **Desert bumps** (erythema nodosum) and arthritis are generally good prognostic signs.
- Very common in endemic region
- **Pulmonary lesions have a tendency to calcify as they heal.**
- **Systemic infections are a problem in AIDS and immunocompromised patients in endemic region** (meningitis, mucocutaneous lesion).
 - Cocci has a tendency to **disseminate in third trimester of pregnancy.**

Blastomyces dermatitidis

Dimorphic Fungus

Environmental form: hyphae with nondescript conidia (i.e., no fancy arrangements)

- Association not definitively known, appears to be associated with rotting wood such as beaver dams
- Endemic region: Upper Great Lakes, Ohio, Mississippi River beds plus the southeastern seaboard of the U.S. and northern Minnesota into Canada

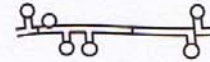


Figure I-3-21.
Blastomyces dermatitidis
Hyphae with Conidia



Figure I-3-22. Blastomycosis Endemic Region

Tissue form: broad-based budding yeasts and a double refractile cell wall (not capsule)

Disease: Blastomycosis

- Acute and chronic pulmonary disease
- Considered less likely to self-resolve than *Histoplasma* or *Coccidioides*, so many physicians will treat even acute infections with ketoconazole.
- Disseminated disease



Figure I-3-23.
Blastomyces dermatitidis
Broad-Based Budding Yeasts

Opportunistic Fungi

Aspergillus fumigatus

Monomorphic filamentous fungus

- Dichotomously branching
- Generally acute angles
- Septate
- One of our major recyclers: compost pits, moldy marijuana



Figure I-3-24.
Aspergillus Showing
Monomorphic Filamentous
Fungus

Diseases/Predisposing Conditions

- **Allergic bronchopulmonary aspergillosis/asthma**, allergies
(growing in mucous plugs in the lung but not penetrating the lung tissue)
- **Fungus ball**: free in preformed lung cavities (surgical removal to reduce coughing, which may induce pulmonary hemorrhage)
- **Invasive aspergillosis/severe neutropenia**, CGD, CF, burns
Invades tissues causing infarcts and hemorrhage.
Nasal colonization → **pneumonia or meningitis**
Cellulitis/in burn patients; may also disseminate.

Treatment

- Depends on severity of disease and underlying conditions: Itraconazole or amphotericin B

Candida albicans (and other species of *Candida*)

- **Yeast** endogenous to our mucous membrane normal flora
- *C. albicans* yeasts form germ tubes at 37°C in serum.
- Form **pseudohyphae** and **true hyphae** when it invades tissues (nonpathogenic *Candida* do not).

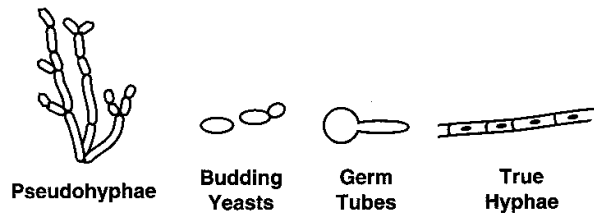


Figure I-3-25. *Candida albicans*

Diseases/Predisposing Conditions

- **Perlèche**: crevices of mouth/malnutrition
- **Oral thrush**/prematurity, antibiotic use, immunocompromised (IC) host, AIDS
- **Esophagitis**/antibiotic use, IC host, AIDS
- **Gastritis**/antibiotic use, IC host, AIDS
- **Septicemia** (with endophthalmitis and macronodular skin lesions)/immunocompromised, cancer and intravenous (IV) patients
- **Endocarditis** (with transient septicemias)/**IV drug abusers**
- **Cutaneous infections**/obesity and infants; patients with rubber gloves
- **Yeast vaginitis**/particularly a problem in diabetic women
- Chronic mucocutaneous candidiasis/endocrine defects; anergy to *Candida*

Diagnosis

- KOH: pseudohyphae, true hyphae, budding yeasts
- Septicemia: culture lab identification: biochemical tests/formation of germ tubes

Treatment

- Topical imidazoles or oral imidazoles; nystatin
- Disseminated: Amphotericin B or fluconazole

Cryptococcus neoformans

Encapsulated Yeast (Monomorphic)

Environmental Source: Soil enriched with pigeon droppings

Diseases/Predisposing Conditions

- **Meningitis/Hodgkin's, AIDS (the dominant meningitis)**
- **Acute pulmonary** (usually asymptomatic)/pigeon breeders

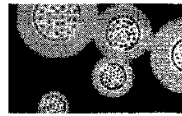


Figure I-3-26. *Cryptococcus neoformans*

Diagnosis of Meningitis: CSF

- Detect capsular antigen in CSF (by latex particle agglutination or counter immunoelectrophoresis)
- India ink mount (misses 50%) of CSF sediment to find budding yeasts with capsular "halos"
- Cultures (urease positive yeast)

Treatment: AMB+5FC until afebrile and culture negative, then fluconazole

Mucor, Rhizopus, Absidia (Zygomycophyta)

Nonseptate filamentous fungi

Environmental Source: Soil; sporangiospores are inhaled

Disease

- **Rhinocerebral infection** caused by *Mucor* (or other zygomycophyta)
- (Old names: Mucormycosis = Phycomycosis = Zygomycosis)
- Characterized by paranasal swelling, necrotic tissues, hemorrhagic exudates from nose and eyes, and mental lethargy
- Occurs in **ketoacidotic diabetic patients and leukemic patients.**
- These fungi penetrate without respect to anatomical barriers, progressing rapidly from sinuses into the brain tissue.

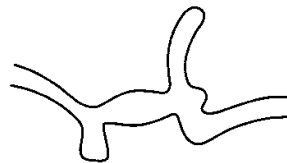


Figure I-3-27. Nonseptate Hyphae with Broad Angles

Diagnosis: KOH of tissue; broad ribbon-like nonseptate hyphae with about 90° angles on branches.

Treatment

- Debride necrotic tissue and start Amphotericin B fast
- High fatality rate because of rapid growth and invasion

Pneumocystis carinii

Fungus (based on molecular techniques like ribotyping)

- Obligate extracellular parasite
- Silver stained cysts in tissues



Figure I-3-28. *Pneumocystis*

Disease: Interstitial pneumonia

- Pneumonia in AIDS patients even with prophylaxis (mean CD4+/mm³ of 26), malnourished babies, premature neonates, and some other IC adults and kids
- Symptoms: fever, cough, shortness of breath; sputum nonproductive except in smokers
- *Pneumocystis* attaches to and kills Type I pneumocytes, causing excess replication of Type II pneumocytes and damage to alveolar epithelium. Serum leaks into alveoli, producing an exudate with a foamy or **honeycomb appearance on H & E stain**.
(Silver stain reveals the holes in the exudate are actually the cysts and trophozoites, which do not stain with H & E.)
- X-ray: **patchy infiltrative** (ground glass appearance), the lower lobe periphery may be spared.

Diagnosis: Silver-staining cysts in bronchial alveolar lavage fluids or biopsy

Treatment: Trimethoprim/sulfamethoxazole

Chapter Summary

Mycology is the study of fungi. The fungi include molds, yeasts, and mushrooms. All fungi are eukaryotic, have complex cell walls, have ergosterol in their cell walls, and are heterotrophic, either as free-living saprophytes or as parasites.

Yeasts are single-celled fungi. Some fungi grow as extended filamentous units called hyphae and can be classified according to the presence (septate-types) or absence (nonseptate-types) of cross walls and color. A mat of hyphae is called a mycelium.

Dimorphic fungi can convert between hyphal and yeast forms. Important examples of such fungi are *Histoplasma*, *Blastomyces*, *Coccidioides*, and *Sporothrix*.

Pseudohyphal forms present in *Candida albicans* have constrictions in the hyphae that resemble budding yeasts.

Most fungi reproduce by forming asexual spores called conidia. Variant forms of conidia include blastoconidia and arthroconidia. Some form sexual endospores inside of spherules in host tissues.

Cutaneous fungi are generally diagnosed by using a KOH scraping of skin. Other fungal preparations also may be stained using periodic acid-Schiff (PAS) stain, silver stain, calcofluor, or India ink.

Fungi may also be cultured on Sabouraud's or blood agar with antibiotics and then identified by cell morphology or by polymerase chain reaction (PCR) using specific nucleic acid probes.

Nonsystemic fungal infections include pityriasis (tinea) versicolor or fungemia in premature infants, caused by *Malassezia furfur*; cutaneous or mucocutaneous candidiasis; and nail and hair infections caused by species of *Trichophyton*, *Microsporum*, or *Epidermophyton*.

Sporotrichosis is a subcutaneous or lymphatic (rose gardener's disease) or pulmonary (alcoholic rose-garden-sleeper's disease) mycosis.

In the United States, deep fungal infections are caused by *Histoplasma capsulatum* (endemic eastern Great Lakes, Mississippi, and Missouri river beds); *Coccidioides immitis* (endemic southwestern United States, northern Mexico), or *Blastomyces dermatitidis* (endemic Great Lakes, Saint Lawrence, Ohio, and Mississippi river beds, southeastern United States). The disease states, modes of recognition, and other salient properties of each of these pathogens are described.

Opportunistic fungi include *Aspergillus fumigatus*, *Candida albicans*, *Cryptococcus neoformans*, *Mucor* species, *Rhizopus* species, *Absidia* species, and *Pneumocystis carinii*. The compromising condition leading to infection, the diseases caused, the modes of identification, and other significant features describing each of these organisms are described.

Review Questions

1. An obese 32-year-old diabetic woman presents with complaint of red and painful skin in her abdominal skin folds. Examination reveals a creamy white material at the base of the fold. It is erythematous underneath and extends beyond the creamy material. Microscopic examination of the exudate reveals oval budding structures ($3 \times 6 \mu\text{m}$) mixed with more budding elongated forms. The most likely causative agent is
 - A. *Aspergillus fumigatus*
 - B. *Candida albicans*
 - C. *Epidermophyton floccosum*
 - D. *Microsporum canis*
 - E. *Sporothrix schenckii*
2. What fungus causes tinea capitis or ringworm infection of the scalp?
 - A. *Aspergillus fumigatus*
 - B. *Microsporum canis*
 - C. *Epidermophyton floccosum*
 - D. *Candida albicans*
 - E. *Sporothrix schenckii*
3. An 18-year-old high school student in rural north Mississippi develops fever, cough, and chest pain. The cough, associated with weight loss, persisted. Because of poor performance at football practice he was advised to see a physician. Lymph node biopsies stained with H and E studies revealed granulomatous inflammation and macrophages engorged with oval structures measuring $2\text{--}4 \mu\text{m}$. Cultures incubated at room temperature grew powdery white colonies, which on microscopic study had tuberculate spores. The high school student most likely acquired the infection from
 - A. Desert sand
 - B. Cat feces
 - C. Soil enriched with bird excrement
 - D. Another human via respiratory secretions
 - E. Contaminated drinking water
4. The most common portal of entry in *Blastomyces dermatitidis* infection is
 - A. Mouth
 - B. Circulatory system
 - C. Skin
 - D. Respiratory tract
 - E. Central nervous system

5. For which pathogen is infection triggered by traumatic contact with plants on which the organism is growing?
 - A. *Candida albicans*
 - B. *Coccidioides immitis*
 - C. *Cryptococcus neoformans*
 - D. *Histoplasma capsulatum*
 - E. *Sporothrix schenckii*
6. Hematoxylin and eosin stain of a biopsy specimen from an AIDS patient shows spherules with endospores. The most likely organism is
 - A. *Blastomyces dermatitidis*
 - B. *Candida albicans*
 - C. *Coccidioides immitis*
 - D. *Cryptococcus neoformans*
 - E. *Histoplasma capsulatum*
 - F. *Sporothrix schenckii*

Answers

1. **Answer: B.** Cutaneous candidiasis is a problem in skin folds of obese individuals. It is an even greater problem in diabetic patients because of the high sugar levels. Only the members of the genus *Candida* would produce a creamy surface growth. The erythematous base is due to the production of a cytotoxin. *Aspergillus*, *Epidermophyton*, and *Microsporum* are all monomorphic filamentous fungi and would not fit the description. *Sporothrix* is found as cigar-shaped budding yeasts but would not clinically present like this. It is traumatically implanted to start subcutaneous infections.
2. **Answer: B.** Tinea capitis or fungal infection of the scalp hair is caused by two genera of the dermatophytes: *Microsporum* and *Trichophyton*. *Epidermophyton* causes infections of the skin and nails but not the hair. *Microsporum* (the correct answer here) infects hair and nails. The other three fungi do not cause hair infection.
3. **Answer: C.** The clues here are the geography, weight loss, granulomatous inflammation, and macrophages engorged with oval structures (RES disease). The colonial appearance and tuberculate spores strongly suggests the causative agent to be *Histoplasma capsulatum*. *Histoplasma* is acquired from dusty environments containing bird (most often chicken or starling) or bat feces. The areas of highest endemicity are in the great central river beds with bat caves, chicken coops, and starling roosts having extremely high levels.
4. **Answer: D.** Like most of the systemic fungal agents, this fungus is transmitted by respiratory route. Direct inoculation via skin is possible but not the most common route, even though the species name of this fungus suggests skin.
5. **Answer: E.** Transmission of *Sporothrix* is by break or cut of the skin, whereby the fungus in the contaminated plants or wood is introduced.
6. **Answer: C.** *C. immitis* is the only fungus with endosporulating spherules, demonstrated by either KOH prep or histopathologic stain.

Medical Parasitology

4

What the USMLE Requires You to Know

The USMLE generally does not have many parasitology questions but you will be expected to know the following.

- Name of organism (scientific and common) and major parasite type (e.g., nematode or flagellate).
- Name of disease (common names are frequently used).
- Route of spread, including vector names and reservoir hosts.

For the following organisms you should also know symptoms and understand the pathogenicity:

Entamoeba

Giardia

Plasmodium

Toxoplasma

Cryptosporidium

Enterobius

Ascaris

Hookworms: *Necator* and *Ancylostoma*

Trichinella

Schistosoma

- Review additional bolded material in the following tables.

CLASSIFICATION OF PARASITES

Medical Parasitology is the study of the invertebrate animals and the diseases they cause. Parasites are classified as protozoans or metazoans. The most important organisms in the U.S.A. are identified in the following two tables in boldface type.

Table I-4-1. Protozoans

Common Name	Anebae	Flagellates	Ciliates	Apicomplexa
Important Genera	Entamoeba <i>Naegleria</i> <i>Acanthamoeba</i>	LUMINAL (GUT, UG) Trichomonas Giardia HEMOFLAGELLATES <i>Leishmania</i> <i>Trypanosoma</i>	<i>Balantidium</i>	BLOOD/TISSUE Plasmodium Toxoplasma <i>Babesia</i> INTESTINAL Cryptosporidium <i>Isospora</i>

Pneumocystis, which was formerly classified as a protozoan, has been determined to be a fungus through ribotyping and other molecular biologic techniques.

Table I-4-2. Metazoans: Worms*

Phylum	Flat worms (Platyhelminthes)		Roundworms
Classes: Common name:	Trematodes (flukes)	Cestodes (tapeworms)	Nematodes ** (roundworms)
Genera:	<i>Fasciola</i> <i>Fasciolopsis</i> <i>Paragonimus</i> <i>Opisthorchis</i> (<i>Clonorchis</i>) <i>Schistosoma</i>	<i>Diphyllobothrium</i> <i>Hymenolepis</i> <i>Taenia</i> <i>Echinococcus</i>	Necator Enterobius (W) <i>uchereria/Brugia</i> Ascaris and Ancylostoma <i>Toxocara</i> , <i>Trichuris</i> & <i>Trichinella</i> <i>Onchocerca</i> <i>Dracunculus</i> Eye worm (<i>Loa loa</i>) <i>Strongyloides</i>

* Metazoans also include the Arthropoda, which serve mainly as intermediate hosts (the crustaceans) or as vectors of disease (the Arachnida and Insecta).

** Nematodes mnemonic.

HOSTS

The infected host is classified as

- **Intermediate**—host in which **larval or asexual stages develop**.
- **Definitive**—host in which the **adult or sexual stages occur**.

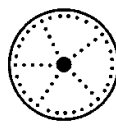
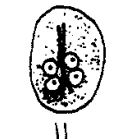
VECTORS

Vectors are **living transmitters** (e.g., a fly) of disease and may be

- **Mechanical**, which transport the parasite but there is no development of the parasite in the vector.
- **Biologic**, in which some stages of the life cycle occur.

IMPORTANT PROTOZOAN PARASITES

Table 1-4-3. Protozoan Parasites

Species	Disease / Organs Most Affected	Form/Transmission	Diagnosis	Treatment
<i>Entamoeba histolytica</i>	Amebiasis: dysentery Inverted flask-shaped lesions in large intestine with extension to peritoneum and liver, lungs, brain, and heart. Blood and pus in stools. Liver abscesses.	Cysts Fecal-oral transmission—water, fresh fruits, and vegetables	Trophozoites:  or cysts in stool:  Serology Nuclei have sharp central karyosome and fine chromatin "spokes". 	Metronidazole followed by iodoquinol
<i>Giardia lamblia</i>	Giardiasis: Ventral sucking disk attaches to lining of duodenal wall, causing a fatty, foul-smelling diarrhea (diarrhea → malabsorption duodenum, jejunum)	Cysts Fecal (human, beaver, muskrat, etc.), oral transmission—water, food, day care, oral-anal sex	Trophozoites or cysts in stool or fecal antigen test (replaces "string" test)  "Falling leaf" motility Acid fast oocysts in stool: Biopsy shows dots (cysts) in intestinal glands 	Metronidazole
<i>Cryptosporidium</i> sp.	Cryptosporidiosis: transient diarrhea in healthy, severe in immunocompromised hosts	Cysts Undercooked meat, water; not killed by chlorination	Ciliated trophozoites, cysts in feces  	Tetracycline
<i>Balantidium coli</i>	Dysentery: infection of colon with penetration	Cysts Contaminated food or water	Motile trophozoites in methylene blue wet mount 	Metronidazole
<i>Trichomonas vaginalis</i> (urogenital)	Trichomoniasis: often asymptomatic or frothy vaginal discharge	Trophozoites Sexual		

Free Living Amebae

- Occur in polluted water or soil (*Naegleria*, *Acanthamoeba*)
- Occur in contact lens saline solutions (*Acanthamoeba*): cysts from dust contaminate

Table 1-4-4. Free Living Amebae That Occasionally Infect Humans

Species	Disease / Locale	Form / Transmission	Diagnosis	Treatment
<i>Naegleria</i>	Primary amebic meningoencephalitis (PAM): severe pre-frontal headache, nausea, high fever, often an altered sense of smell; often fatal.	Free-living amebae picked up while swimming or diving in very warm fresh water.	Motile trophozoites in CSF Culture on plates seeded with Gram-negative bacteria. Amebae will leave trails.	Amphotericin B (rarely successful)
<i>Acanthamoeba</i>	Keratitis; Granulomatous amebic encephalitis (GAE) in immunocompromised patients: insidious onset but progressive to death.	Free living amebae in contaminated contact lens solution (airborne cysts) Not certain for GAE: inhalation or contact with contaminated soil or water.	Star-shaped cysts on biopsy; rarely seen in CSF. Culture as above.	Keratitis: topical miconazole and propamidine isothionate GAE: sulfadiazine (rarely successful)

Plasmodium Species

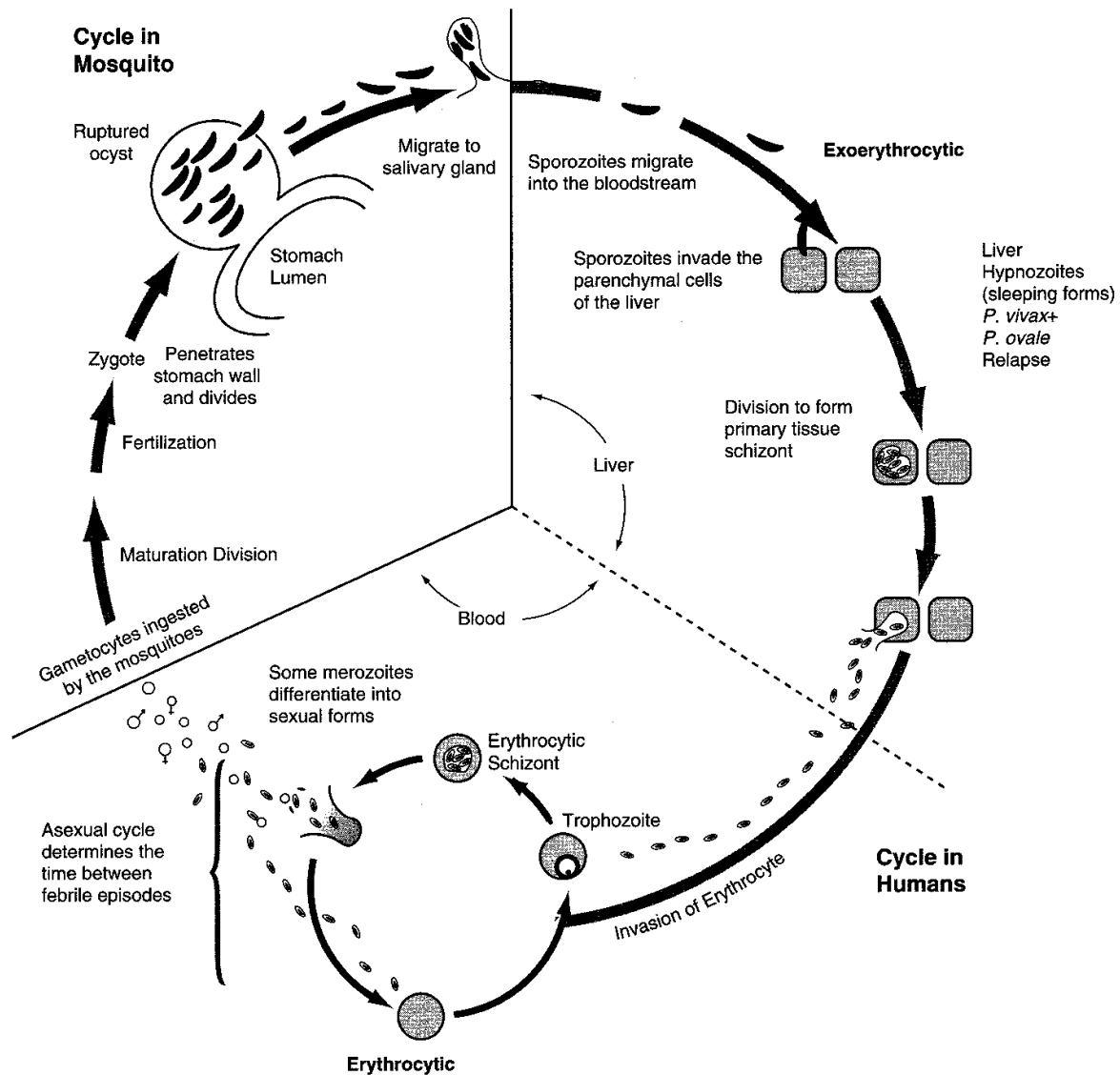


Figure I-4-1. *Plasmodium* Life Cycle

Each *Plasmodium* has two distinct hosts.

- A vertebrate such as the human where asexual phase (schizogony) takes place in the liver and red blood cells.
- An arthropod host (*Anopheles* mosquito) where gametogony (sexual phase) and sporogony take place.

Cause disease by a wide variety of mechanisms, including metabolism of hemoglobin and lysis of infected cells leading to anemia and to agglutination of infected RBC.

Cause paroxysms (chills, fever spike, and malarial rigors) when the infected RBC are lysed, liberating a new crop of merozoites.

Table 1-4-5. *Plasmodium* Species

Species	Disease	Important Features	Blood Smears	Liver Stages	Treatment**
<i>Plasmodium vivax</i>	Benign tertian	48-hour fever spikes	Enlarged host cells; amoeboid trophozoites	Persistent hypnozoites Relapse*	Chloroquine PO ₄ then primaquine
<i>Plasmodium ovale</i>	Benign tertian	48-hour fever spikes	Oval, jagged, infected RBCs	Persistent hypnozoites Relapse	Chloroquine PO ₄ then primaquine
<i>Plasmodium malariae</i>	Quartan or malarial	72-hour fever spikes; recrudescence*	Bar and band forms; rosette schizonts	No persistent stage*	Chloroquine PO ₄ (no radical cure necessary)
<i>Plasmodium falciparum</i>	Malignant tertian	Irregular fever spikes; causes cerebral malaria	Multiple ring forms crescent-shaped gametes	No persistent stage*	Chloroquine resistance a problem***

*Recrudescence is a reoccurrence of symptoms from low levels of organisms remaining in red cells (recrudescence). Relapse is an exacerbation from liver stages (hypnozoites).

**Treatment:

1. Suppressive (to avoid infection)
2. Therapeutic (eliminate erythrocytic)
3. Radical cure (eliminate exoerythrocytic)
4. Gametocidal (destruction of gametocytes)

Successful treatment is accomplished with chloroquine followed by primaquine. Chloroquine therapy is suppressive, therapeutic, and gametocidal, whereas primaquine eliminates the exoerythrocytic form.

***Use quinine sulfate plus pyrimethamine-sulfadoxine.

Hemoflagellates (Trypanosomes and Leishmanias)

Hemoflagellates infect blood and tissues.

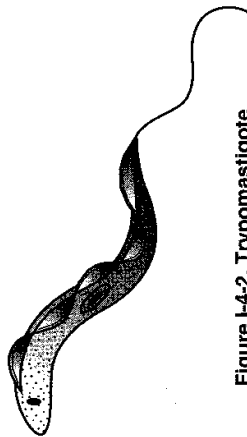


Figure I-4-2. Trypomastigote

Trypanosomes are found

- In human blood as trypomastigotes with flagellum and undulating membrane
- In tissues as amastigotes (oval cells having neither the flagellum nor undulating membrane)

Leishmania found always as amastigotes in macrophages.

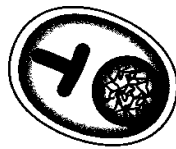
Table I-4-6. Hemoflagellates

Species	Disease	Vector/Forn/Transmission	Reservoirs	Diagnosis	Treatment
<i>Trypanosoma cruzi</i> *	Chagas' disease (American trypanosomiasis) Latin America Swelling around eye: (Romana's sign) common early sign Cardiac muscle, liver, brain often involved	Reduviid bug (kissing or cone bug; genus <i>Triatoma</i>) passes trypomastigote (flagellated form) in feces as it bites. Scratching implants in bite site.	Cats, dogs, armadillos, opossums Poverty housing	Blood films	Nifurtimox
<i>Trypanosoma brucei</i> <i>Trypanosoma b. gambiense</i> <i>Trypanosoma b. rhodesiense</i> <i>Leishmania donovani</i> **	African sleeping sickness (African trypanosomiasis) Antigenic variation	Trypomastigote in saliva of tsetse fly contaminates bite	Humans, some wild animals	Blood films, CSF High immunoglobulin levels in CSF	Acute: suramin Chronic: melarsoprol
<i>Leishmania</i> (About 15 different species)	Visceral Leishmaniasis Kala-azar	Sandfly bite	Urban: humans Rural: rodents and wild animals	Amastigotes in macrophages in bone marrow, liver, spleen	Stibogluconate sodium (from CDC)
<i>Leishmania</i> <i>Leishmania braziliensis</i> complex	Cutaneous Leishmaniasis (Oriental sore, etc.) Mucocutaneous Leishmaniasis	Sandfly bite	Urban: humans Rural: rodents and wild animals	Amastigotes in macrophages in cutaneous lesions	Stibogluconate sodium
		Sandfly bite	Urban: humans Rural: rodents and wild animals	Same	Stibogluconate sodium

**T. cruzi*: An estimated 1/2 million Americans are infected, creating some risk of transfection transmission in U.S. In babies, acute infections often serious involving CNS. In older children and adults, mild acute infections but may become chronic with the risk of development of cardiomyopathy and heart failure.

***Leishmania* all: Intracellular, sandfly vector, stibogluconate.

Figure I-4-3. Amastigote



Miscellaneous Apicomplexa Infecting Blood or Tissues

Table I-4-7. Miscellaneous Apicomplexa Infecting Blood or Tissues

Species	Disease/Locale of Origin	Transmission	Diagnosis	Treatment
<i>Babesia</i> (primarily a disease of cattle) Humans: <i>Babesia microti</i> , WA1, & MO1 strains	Babesiosis (hemolytic, malaria-like) Same range as Lyme NE, N Central, California and NW USA	<i>Ixodes</i> tick Co-infections with <i>Borrelia</i>	Giemsa stain of thin smear or hamster inoculation	Clindamycin + quinine
<i>Toxoplasma gondii</i>	See below	Cat is essential definitive host. Many other animals are intermediate host. Mode: 1) Raw meat in US #1 = pork 2) Contact with cat feces	Serology High IgM or rising IgM acute unfection	Pyrimethamine + sulfadiazine

Toxoplasmosis

Most common parasitic disease.

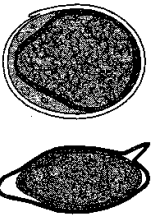
- *Toxoplasma* acquired **after birth** is most commonly asymptomatic or mild, nonspecific with lymphadenopathy and fever. May mimic infectious mononucleosis.
- **Once infected**, as immunity develops, **bradyzoites encyst**, but **generally remain viable as evidenced by a positive serotiter**.
- Unless prophylactic drugs are given, **AIDS** patients who are seropositive for *Toxoplasma* will have reactivational infections.
- Produces **severe disease** in **AIDS** or immunocompromised patients.
- **Maternal antibodies protect the fetus**, even if the mother is reinfected during pregnancy.
- If **primary maternal infection** occurs **during pregnancy**, the **fetus may be infected**.
- If *Toxoplasma* crosses **placenta early**, **severe congenital infections**: intracerebral calcifications, chorioretinitis, hydro- or microcephaly, convulsions.
- If later, may be inapparent; **untreated inapparent congenital infections lead to progressive blindness**.

IMPORTANT METAZOAN PARASITES

Trematodes

- Are commonly called flukes.
- Are leaf-shaped worms, which are generally flat and fleshy.
- Are hermaphroditic except for *Schistosoma*, which has separate male and female.
- Have complicated life cycles occurring in two or more hosts.
- Have operculated eggs (except for *Schistosoma*), which contaminate water, perpetuating the life cycle, and which are also used to diagnose infections.
- **The first intermediate hosts are snails.**

Table I-4-8. Trematode (Fluke) Diseases

Organism	Common Name	Reservoir Host	Acquisition	Progression in Humans	Important Ova	Treatment
<i>S. mansoni</i> <i>Schistosoma japonicum</i>	Intestinal schistosomiasis	Cats, dogs, cattle, etc.	Contact with water; skin penetration	Skin penetration (itching) → mature in veins of mesentery → eggs cause granulomas in liver (liver enlargement in chronic cases)		Praziquantel
<i>Schistosoma haematobium</i>	Vesicular schistosomiasis	Primates	Contact with water; skin penetration	Skin penetration (itching) → Mature in bladder veins; chronic infection has high association with bladder carcinoma in Egypt and Africa		Praziquantel
Non-human schistosomes	Swimmer's itch	Birds (Great Lakes U.S.)	Contact with water; skin penetration	Penetrate skin producing dermatitis without further development in humans; itching is most intense at 2 to 3 days		Trimeprazine Calamine Sedatives
<i>Clonorchis sinensis</i>	Chinese liver fluke	Dogs, cats, humans	Raw fish ingestion		Operculated eggs	Praziquantel
<i>Fasciola hepatica</i>	Sheep liver fluke	Sheep, cattle, humans	Ingestion of aquatic plants; water cress		Operculated eggs	Praziquantel
<i>Fasciolopsis buski</i>	Giant intestinal fluke	Pigs, dogs, rabbits, humans	Ingestion of aquatic plants; water chestnuts		Operculated eggs	Praziquantel
<i>Paragonimus westermani</i>	Lung fluke	Humans, cat family, canines, pigs	Raw crabs, crayfish		Operculated eggs	Praziquantel

Cestodes

- Are the tapeworms.
- Consist of three basic portions: the head or scolex; a "neck" section, which produces the proglottids; and the segments or proglottids, which mature as they move away from the scolex. (The combination of the neck and proglottids is called the strobila.)
- Are hermaphroditic with each proglottid developing both male and female reproductive organs, and mature eggs developing in the most distal proglottids.
- Adhere to the mucosa via the scolex, which is knobby looking and has either suckers or a sucking groove.
- Have no gastrointestinal (GI) tract, they absorb nutrients from the host's GI tract.
- Are diagnosed by finding eggs or proglottids in the feces.
- Have for the most part complex life cycles involving extraintestinal larval forms in intermediate hosts. When humans are the intermediate host, these infections are generally more serious than the intestinal infections with adult tapeworms.

Gastrointestinal Cestodes (Tapeworms)

Table I-4-9. Gastrointestinal Cestodes (Tapeworms)

Cestode (Common name) Intermediate Host (IH) Definitive Host (DH)*	Form/Transmission	Humans Are:	Disease/Organ Involvement/Symptoms (Sx)	Diagnosis	Treatment
<i>Taenia saginata</i> (beef tapeworm) IH: cattle DH: humans	Rare beef containing the cysticerci is ingested	DH	Intestinal tapeworm/Small intestine Sx: Asymptomatic or vague abdominal pains	Proglottids or eggs in feces	Praziquantel
<i>Taenia solium</i> (pork tapeworm) IH: swine; Rarely: humans DH: humans Developing and Slavic countries	Water, vegetation, food contaminated with eggs Autoinfection	IH	Cysticercosis/eggs → larva develop in brain, eye, heart, lung, etc.	Biopsy	Praziquantel; surgery in some sites
	Rare/raw pork containing the cysticerci is ingested by humans	DH	Intestinal tapeworm Sx: as for <i>Taenia saginata</i>	Proglottids or eggs in feces	Praziquantel
<i>Diphyllobothrium</i> <i>latum</i> (fish tapeworm) IH (2): crustaceans → fish; Rare: humans DH: humans/mammals Cool lake regions	Drinking pond water w/ → copepods (crustaceans) carrying the larval forms or frog/snake poultices Rare, raw, pickled fish → containing a sparganum	IH	Sparganosis/larvae penetrate intestinal wall and encyst	Biopsy	Praziquantel
<i>Echinococcus</i> <i>granulosus</i> IH: herbivores; Rare: humans DH: carnivores in sheep- raising areas	Ingestion of eggs	DH	Intestinal tapeworm (up to 10 meters)/Small intestine megaloblastic anemia	Proglottids or eggs in feces	Praziquantel
<i>Echinococcus</i> <i>multilocularis</i> IH: rodents DH: canines and cats Northern areas	Ingestion of eggs	IH	Hydatid cyst disease liver & lung where cysts containing brood capsules develop.	Imaging; Serology	Surgery; albendazole
		IH	Alveolar hydatid cyst disease	Tough, as above but no protoscolices	Surgical resection

* Definitive host = adult tapeworm develops in; intermediate host = cysticerci or larvae develop in; cysticerci = encysted larvae found in intermediate host.

Nematodes

- Are the roundworms
- Cause a wide variety of diseases (pinworms, whipworms, hookworms, trichinosis, threadworms, filariasis, etc.)
- Have round unsegmented bodies
- Are transmitted by:
 - ingestion of eggs (*Enterobius*, *Ascaris*, or *Trichuris*);
 - direct invasion of skin by larval forms (*Necator*, *Ancylostoma*, or *Strongyloides*);
 - ingestion of meat containing larvae (*Trichinella*); or
 - infection involving insects transmitting the larvae with bites (*Wuchereria*, *Loa loa*, *Mansonella*, *Onchocerca*, and *Dracunculus*).

Table 1-4-10. Round Worms (Nematodes) Transmitted by Eggs

Species	Disease/Organs Most Affected	Form/Transmission	Diagnosis	Treatment
<i>Enterobius vermicularis</i> Most frequent helminth parasite in U.S.	Pinworms, large intestine, perianal itching	Eggs/person to person autoinfection	Sticky swab of perianal area Ova have flattened side with larvae inside	Albendazole Treat entire family
<i>Trichuris trichiura</i>	Whipworm cecum, appendicitis, and rectal prolapse	Eggs ingested	Barrel-shaped eggs with bipolar plugs in stools	Albendazole
<i>Ascaris lumbricoides</i> Most common helminth worldwide Largest roundworm	Ascariasis Ingest egg → larva migrate thru lungs (cough) and mature in small intestine; may obstruct intestine or bile duct	Eggs ingested	Bile stained, knobby eggs Adults 6-12" roundworms	Supportive therapy during pneumonitis; surgery for ectopic migrations; albendazole
<i>Toxocara canis or cati</i> (dog/cat Ascariids)	Visceral Larva Migrans Larvae wander aimlessly until they die, cause inflammation	Eggs ingested/from handling puppies or from eating dirt in yard (pica)	Clinical findings and serology	Albendazole; self-limiting in most cases

Table 1-4-11. Roundworms (Nematodes) Transmitted By Larvae

Species	Disease/Organs	Form/Transmission	Diagnosis	Treatment
<i>Necator americanus</i> New World hookworm	Hookworm infection Lung migration → pneumonia bloodsucking → anemia	Filariform larva penetrates intact skin of bare feet	Fecal larvae (up to 13 mm) and ova: oval, transparent with 2-8 cell- stage visible inside Occult blood fecal may be present	Meibendazole and iron therapy
<i>Ancylostoma braziliense</i> <i>Ancylostoma caninum</i> (dog and cat hookworms)	Cutaneous Larva Migrans/intense skin itching	Filariform larva penetrates intact skin but cannot mature in humans	Usually a presumptive diagnosis; exposure	Thiabendazole
<i>Strongyloides stercoralis</i>	Threadworm strongyloidiasis: Early: pneumonia, abdominal pain, diarrhea Later: malabsorption, ulcers, bloody stools	Filariform larva penetrates intact skin; Autoinfection leads to indefinite infections unless treated.	Larvae in stool, serology	Thiabendazole
<i>Trichinella spiralis</i>	Trichinosis: larvae encyst in muscle → pain	Viable encysted larvae in meat are consumed: wild game meat.	Muscle biopsy; clinical findings: fever, myalgia, splinter hemorrhages, eosinophilia	Steroids for severe symptoms + mebendazole

Wuchereria bancrofti and *Brugia* are filarial worms causing elephantiasis. Both are transmitted by mosquitoes.

Loa loa is the eye worm transmitted by biting flies.

Onchocera volvulus causes river blindness characterized by itchy "leopard" rash and worms in eye. Blackfly transmits.

Dracunculus medinensis is the guinea worm. It is transmitted by drinking infected copepods (cyclops) in water.

Matching

- (A) *Ancylostoma braziliensis*
 (B) *Ascaris lumbricoides*
 (C) *Balanitidium coli*
 (D) *Diphyllobothrium latum*
 (E) *Clonorchis sinensis*
 (F) *Cryptosporidium parvum*
 (G) *Dermacentor andersoni*
 (H) *Echinococcus multilocularis*
 (I) *Entamoeba coli*
 (J) *Entamoeba histolytica*
 (K) *Enterobius vermicularis*
 (L) *Giardia lamblia*
 (M) *Ixodes scapularis* (*I. dammini*)
 (N) *Leishmania braziliensis*
 (O) *Plasmodium vivax*
 (P) *Pneumocystis carinii*
 (Q) *Pediculus humanus*
 (R) *Sarcoptes scabiei*
 (S) *Schistosoma haematobium*
 (T) *Strongyloides stercoralis*
 (U) *Toxoplasma gondii*
 (V) *Trichinella spiralis*
 (W) *Trichuris trichiura*
 (X) *Trichomonas vaginalis*
 (Y) *Trypanosoma cruzi*
 (Z) *Wuchereria bancrofti*

- ____ 1. Invasive amebae causing dysentery, which is noted for causing extraintestinal abscesses
 ____ 2. Chronic infections associated with bladder carcinoma
 ____ 3. Vector for *Babesia microti*
 ____ 4. Untreatable or at least poorly treatable causative agent of chronic diarrhea in AIDS patients, which is diagnosed by finding acid fast cysts in the stool
 ____ 5. By ribotyping, now considered to be a fungus
 ____ 6. Critically careful surgery is major component of therapy
 ____ 7. Ciliate causative agent of diarrhea
 ____ 8. Filarial worm maturing in the lymphatics and causing elephantiasis
 ____ 9. Adult females live inside adult male groove
 ____ 10. Carrier of epidemic typhus and Trench Fever
 ____ 11. Infection results in enteritis and eosinophilia with flu-like symptoms, periorbital swelling, petechial hemorrhages, and ultimately muscle pain; later in life X-ray may show fine calcifications in the muscle
 ____ 12. Fatty diarrhea associated with malabsorption syndrome

Answers

1. J The organism causing diarrhea but most noted for its ability to leave the gastrointestinal tract is *Entamoeba histolytica*. Remember that *Entamoeba coli* is not a pathogen but rather a commensal organism.
2. S Chronic infections with *Schistosoma haematobium* are notably associated with bladder carcinoma.
3. M The vector for babesiosis, Lyme disease, and ehrlichiosis is the *Ixodes* tick, *Ixodes scapularis* (*I. dammini*) in the midwest and east, with *Ixodes pacificus* in the western United States.
4. F *Cryptosporidium* treatment is still experimental and not highly effective.
5. P *Pneumocystis carinii* has always stained like a fungus but until ribotyping was considered a protozoan parasite.
6. H Hydatid cyst disease requires delicate surgery to remove them without breaking and releasing the larvae.
7. C *Balantidium* is the only ciliate on the list (and the only one you need to know).
8. Z *Wuchereria* is one causative agent of elephantiasis. *Brugia* is another.
9. S Schistosomes are not hermaphroditic like other flukes (trematodes). They instead have separate sexes, but they live permanently together with the female in a groove of the male.
10. Q The human body louse is a *Pediculus* and carries both epidemic typhus and Trench Fever.
11. V Classical description of trichinosis, causative agent *Trichinella spiralis*.
12. L *Giardia* is probably most noted among the parasites for causing diarrhea with fat malabsorption.

Chapter Summary

Table I-4-1 identifies the important genera in each of the four groups of protozoa, and Table I-4-2 identifies the important genera in each of the three phyla of metazoa.

Table I-4-3 describes the diseases, affected organs, cellular forms, methods of transmission, modes of diagnosis, and treatments for the important pathogenic protozoan species. Table I-4-4 does the same for free-living species that occasionally infect humans.

The life cycle of malaria-causing species is illustrated in Figure I-4-1, and the properties of individual *Plasmodium* species are described in Table I-4-5.

The morphology and properties, as well as the diseases caused by, the vectors responsible for transmission of, the modes of transmission of, the reservoirs for, the diagnoses of, and the treatments for infection by the various species of the *Trypanosoma* and *Leishmania* hemoflagellates are described in Table I-4-6 and Figures I-4-2 and I-4-3.

The most common parasitic disease, toxoplasmosis, is caused by *Toxoplasma gondii* and is transmitted by raw meat or contact with cat feces. It and *Babesia* are described in Table I-4-7.

The trematode organisms, the common names of the diseases they cause, their reservoir hosts, their modes of acquisition, the progression of the diseases they cause in humans, their ova, and treatments for their infections are described in Table I-4-8.

The tapeworms (Cestodes) are described in Table I-4-9; the roundworms (Nematodes) transmitted by eggs are described in Table I-4-10, and the roundworms transmitted by larvae are described in Table I-4-11.

Review Questions

1. B.E., a 44-year-old, returns home to New York following a 2-week camera safari to East Africa. She started chloroquine anti-malarial prophylaxis 2 weeks prior to her departure for Kenya and continued throughout her foreign travel. She stopped taking the pills on her arrival home because they made her nauseated. Two weeks after her return she develops paroxysmal fever and diaphoresis and is quickly hospitalized with febrile convulsions, jaundice, and anemia. Blood smears reveal red blood cells multiply infected with delicate ring-like trophozoites and rare sausage-shaped gametocytes. The stage of the parasite life cycle that is responsible for the appearance of the parasites 2 weeks after departure from the malarious area is the
 - A. Hypnozoite
 - B. Sporozoite
 - C. Exoerythrocytic schizont
 - D. Erythrocytic schizont
 - E. Merozoite

2. At a school nurse's request, a clinic in rural South Carolina sees a 9-year-old girl who appears listless and inattentive, although hearing and visual testing has been within normal limits. The physician finds the child thin, with the "potbelly" of malnutrition, and orders a fecal exam and CBC. The CBC reveals a microcytic, hypochromic anemia, and the fecal exam detects brown, oval nematode eggs approximately 65 microns in size, too numerous to count. What was the most likely means by which this child was infected?
 - A. Ingestion of eggs
 - B. Ingestion of larvae
 - C. Ingestion of cysts in muscle
 - D. Skin penetration by larvae
 - E. Mosquito transmission of sporozoites
3. An HIV-positive patient with a CD4+ count of 47 presents with diarrhea. Acid-fast oocysts are found in the stool. From this finding, what is the proper care and prognosis with that care?
 - A. Infection is short lasting and self-resolving and requires no treatment
 - B. If treated with antibiotics, the infection should resolve in 3–6 days
 - C. Infection will resolve only with a combination of anti-tuberculous drugs and then it may take weeks
 - D. Infection could have been prevented by avoiding cat feces and undercooked or raw meat
 - E. Even with the best treatment, the infection may be unrelenting
4. A 24-year-old primiparous woman in her eighth month of gestation develops a positive IgM titer to *Toxoplasma gondii* for the first time. She should be advised by her physician
 - A. That this child and all future fetuses are likely to be infected
 - B. That a newborn with a positive anti-Toxoplasma IgG response should be treated with anti-parasitics
 - C. That future infections can be avoided by proper vaccination and worming of cats
 - D. That retinochoroiditis can be prevented by drug treatment of an infant with a positive IgM response
 - E. That major organ damage can be reversed by prompt treatment of the newborn
5. A 35-year-old Captain in the Army Reserves has been plagued by a painful, erosive lesion near his ear lobe since his return from Operation Desert Storm several years ago. He denies exposure to the toxic by-products of burning oil fields. Punch biopsy of the leading edge of the erosion reveals macrophages distended with oval amastigotes. How was this infection acquired?
 - A. Contact with contaminated drinking water
 - B. Bite of infected *Anopheles* mosquito
 - C. Bite of infected reduviid bug
 - D. Fecal contamination of food
 - E. Direct human contact in barracks
 - F. Bite of sandfly
 - G. Bite of tsetse fly

6. A group of six college students undertake to climb Mt. Rainier outside Seattle on their spring break. They pack food and camping provisions except for water, which they obtain from the many fresh water mountain streams that arise at the summit. The adventure takes a little over a week to accomplish, and all return safely and in good spirits to their classes the following week. Within the first week after their return, 5 of the 6 students report to the infirmary with profuse diarrhea and tenesmus. Each affected student experiences weakness and weight loss and stool samples submitted to the lab are yellow, greasy, and foul smelling. What attribute of this parasite imparts its pathogenicity?
 - A. Lytic enzymes
 - B. Flagella
 - C. Ventral sucking disc
 - D. Encystment
 - E. Toxic metabolites

Answers

1. **Answer: C.** B.F. is suffering from *Plasmodium falciparum* malaria acquired shortly before her departure from Kenya. Liver stages of *Plasmodium* are not susceptible to chloroquine killing. Because she did not continue the prophylaxis after her return to the States, this allowed those parasites to initiate all of the erythrocytic stages of the life cycle. Any erythrocytic stages generated out of the liver phase of the life cycle while she remained on prophylaxis would have been killed. Thus, the late onset of her symptoms was due to survival of exoerythrocytic stages that had not yet left the liver at the time she ceased prophylaxis. Hypnozoites are responsible for relapse of symptoms in *P. vivax* and *P. ovale* malarias, but do not exist in *P. falciparum*, and it is clear that she has *falciparum* malaria due to the delicate ring forms multiply infecting erythrocytes and the sausage-shaped gametocytes. Sporozoites are the infectious forms injected by mosquitoes and would not have been available in this country to initiate the symptoms on the time course described. Erythrocytic schizonts and merozoites would have been killed by prophylaxis before she left Africa and could not be responsible for the late onset of symptoms.
2. **Answer: D.** This child has the typical symptoms of hookworm disease, caused in this country usually by *Necator americanus*. The infection is acquired by penetration of the filariform larvae through the skin of the feet or buttocks, after contamination of soil with the eggs of the agent deposited in human feces. Of the other distractors, choice A would be most likely if the infection were due to ascarids, pinworms, or whipworms. Choice C would describe infection with either *Taenia* or *Trichinella*, and choice E would be the means of infection with *Plasmodium*.
3. **Answer: E.** The described infection is most likely to be *Cryptosporidium*, which is a very difficult infection in AIDS patients even though it is self-resolving in normal noncompromised individuals. In AIDS patients it is most commonly unrelenting, even with treatment. *Cryptosporidium* is usually acquired from water; it is *Toxoplasma* that's from cats.
4. **Answer: D.** The positive IgM titer arising in the eighth month means that this woman has become acutely infected with *Toxoplasma*. Infections acquired at this time have a high likelihood of infecting the fetus and are most likely to be manifested by the development of retinochoroiditis. A mother can transmit this parasite to her fetus only during an acute infection; therefore, all future fetuses will be protected from the disease. Since IgG antibodies cross the placenta, presence of the anti-*Toxoplasma* antibodies of this class in the neonate may simply reflect the infection of the mother—only a positive IgM response in

the neonate is proof of the child's infection, which should therefore be treated. There is no way to reverse major organ damage when it occurs in utero, but it would not be expected to occur with an acute infection beginning in the third trimester.

5. **Answer: F.** *Leishmania* spp. are transmitted by the bite of sandflies. They cannot be transmitted from person to person by trivial means, so unless organ transplantation is occurring in the barracks, direct human contact (**choice E**) is not a possibility. To survive outside the human host, they must be in the vector (sandfly), so transmission by food or water (**choices A or D**) are not possible. Of the distractors that involve true vectors: *Anopheles* mosquitoes (**choice B**) transfer malaria; reduviid bugs (**choice C**) transfer American trypanosomiasis (Chagas' disease); and tsetse flies (**choice G**) transmit African trypanosomiasis (sleeping sickness).
6. **Answer: C.** *Giardia* is common in mountain streams throughout the U.S., and the presentation of prolonged fatty diarrhea and weight loss is pathognomonic. It causes its pathology by its adherence to the mucosa of the upper small intestine with its ventral sucking disc. No toxic metabolites or lytic enzymes are involved in the pathology, which apparently results from blockage of normal digestive absorption. The organism is a flagellate, and thus has flagella, but migration into extraintestinal sites is not a well known problem associated with pathology. And although the organism does encyst as it passes along the intestine, this is not known to produce symptoms.

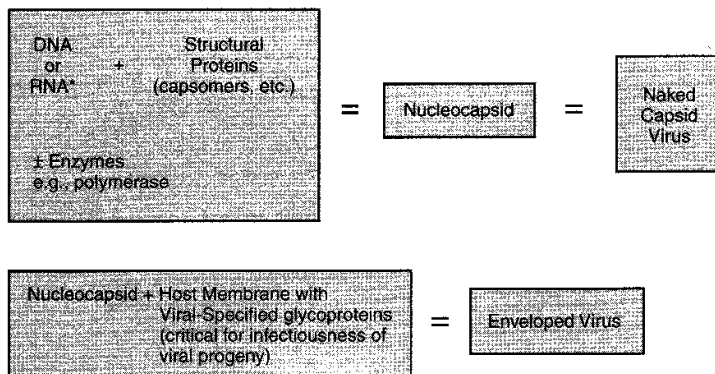
Medically Important Viruses

5

What the USMLE Requires You to Know

- Major concepts of host and tissue specificity
- Major concepts of viral replication
- How viruses cause disease
- Basics of viral diseases (as for bacterial diseases)
- Plus for each virus
 - Nucleic acid (and generalities about how it replicates)
 - Nucleocapsid shape
 - Whether or not it is enveloped

STRUCTURE AND MORPHOLOGY



*Positive sense RNA = (+) RNA
(can be used itself as mRNA)

ss = Single stranded
ds = Double stranded

*Negative sense RNA = (–) RNA

- Complementary to mRNA
- Cannot be used as mRNA
- Requires virion-associated RNA-dependent RNA polymerase (as part of the mature virus)

Figure I-5-1. The Basic Virion

Note

SITV*1 and 2

(Simplified imaginary teaching virus):

- The codon for phenylalanine is UUU.
- Phenylalanine is represented in Figure V-2 by the Greek letter Φ (phi).
- The SIT viruses (SITV + RNA and SITV - RNA) both have a single gene that codes for their capsids, which is made up entirely of phenylalanine.
- Look at the genome in Figure V-2. Which is the positive RNA virus?

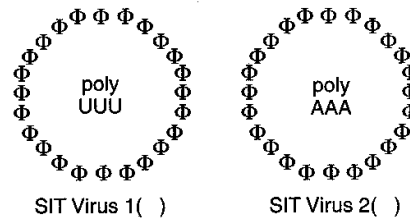


Figure I-5-2. Simplified Imaginary Teaching Viruses

VIRAL STRUCTURE

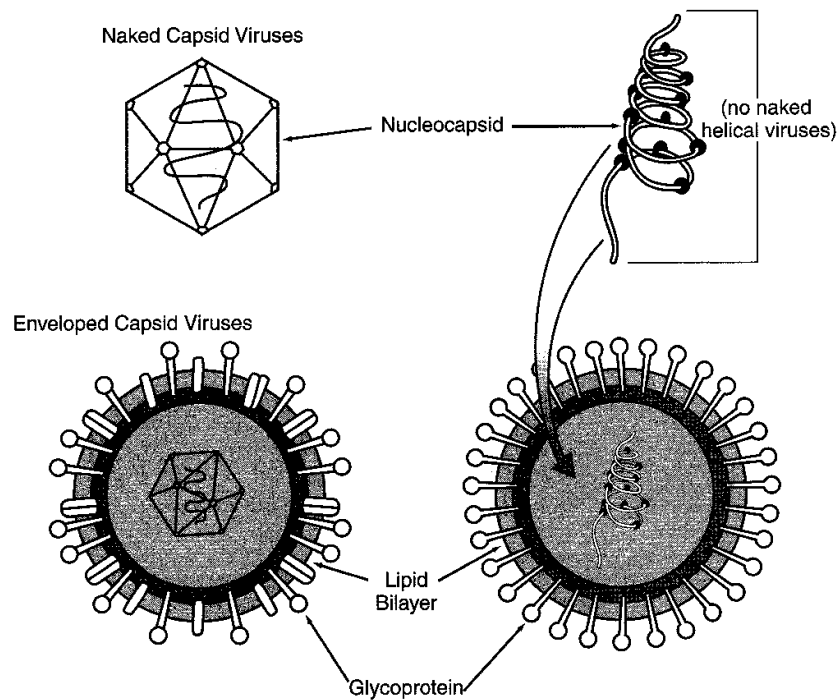


Figure I-5-3. Morphology of Viruses

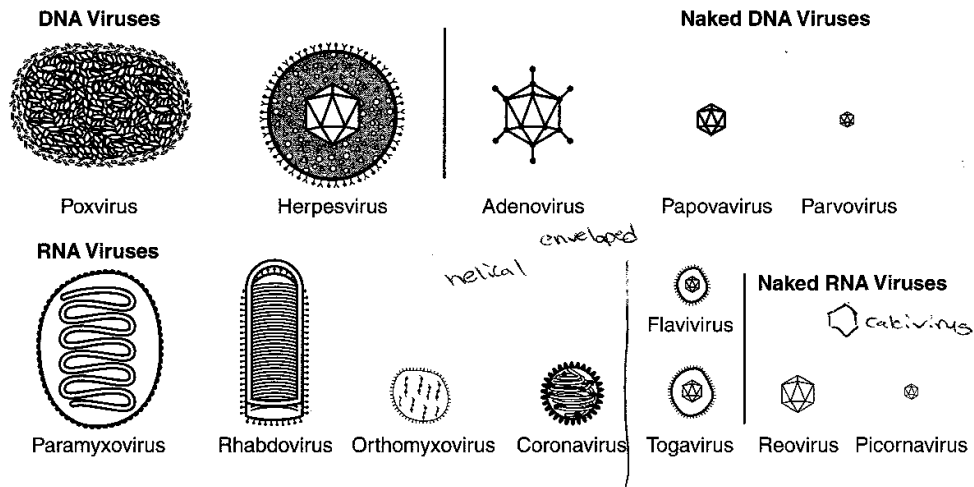


Figure I-5-4. Relative Sizes and Shapes of Different Viruses

VIRAL REPLICATION

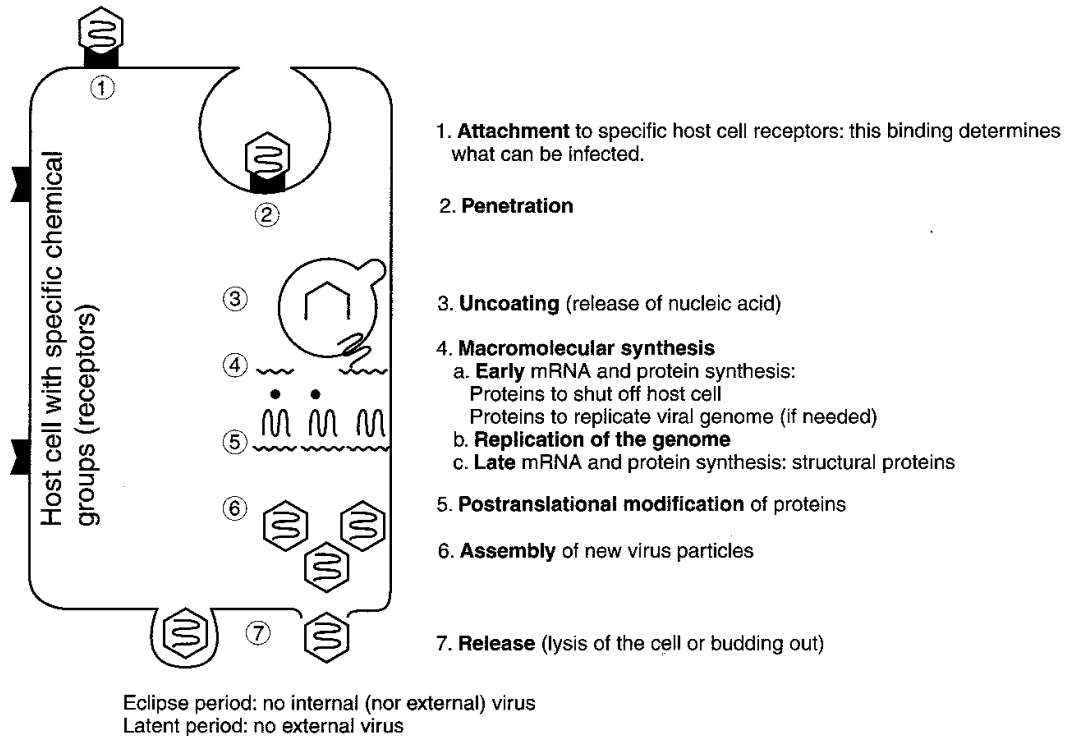
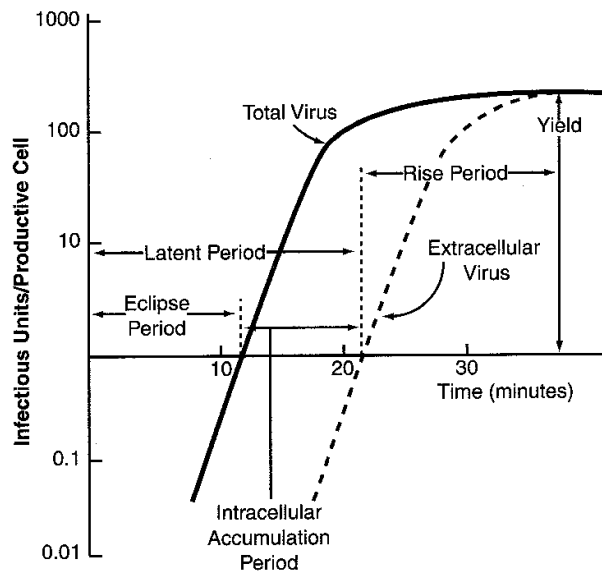


Figure I-5-5. Generalized Viral Replication Scheme



Internal virus is present **after** the end of the eclipse period.
External virus is present **after** the end of the latent period.

Figure I-5-6. One Step Viral Growth Curve

IMPORTANT STEPS IN VIRAL REPLICATION

Spread

Viruses are spread basically by the same mechanisms (e.g., respiratory droplets or sexually) as other pathogens.

Arthropod-borne viruses are referred to as **arboviruses**.

Most belong to three formal taxonomic groups

- Togavirus encephalitis viruses (a.k.a. alphaviruses)
- Flavivirus
- Bunyavirus

Vectors

- Mosquitoes are most common vectors.
- Ticks, biting midges, and sandflies are less common.

Attachment

Viruses bind through specific interaction with the host cell surface components and

- Specific viral surface glycoproteins of enveloped viruses, or
- Specific viral surface proteins of naked viruses.

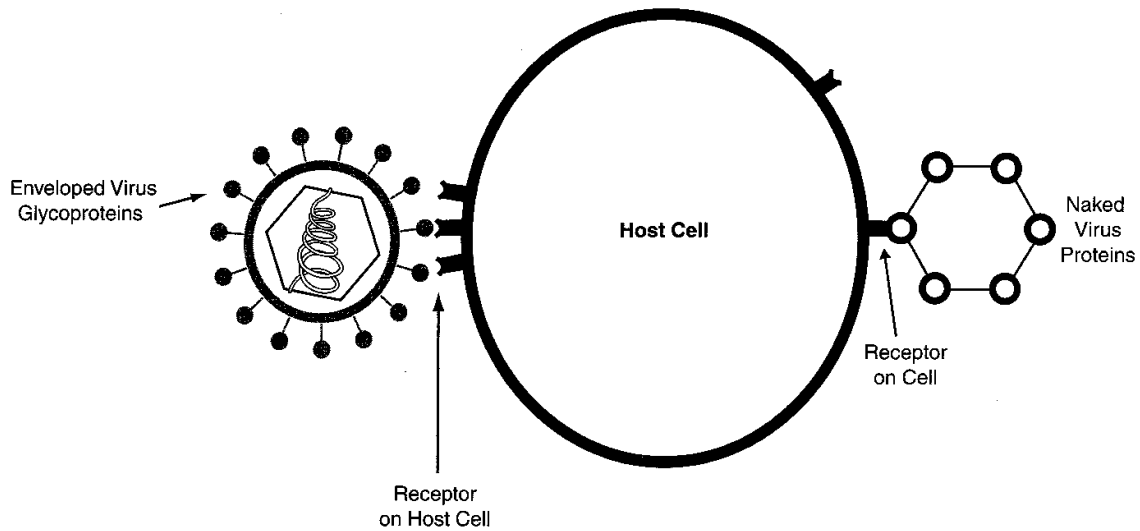


Figure I-5-7. Attachment

These interactions (and the distribution of the receptors) **determine viral**

1. **Host range** (e.g., horses or humans)
2. **Tissue specificity** (e.g., liver versus heart)

Table I-5-1. Specific Viral Receptors to Know

Virus	Target Cell	Receptor on Host Cell
HIV	Th cells, macrophages, microglia	CD4
EBV	B lymphocytes	CD21= CR2
Reovirus	Neurons	β -adrenergic receptor
Rabies	Neurons	Acetylcholine receptor

Table I-5-2. Difference Between Naked and Enveloped Viruses

	Naked	Enveloped
Inactivated by heat, detergents, and organic solvents like ether and alcohols?	No	Yes, since the lipid envelope holds the glycoproteins essential for attachment. Dissolving the envelope inhibits attachment and therefore uptake.
Immune response	Prominently antibody	Antibody and prominent cell-mediated immunity

Viral Entry Into Host Cell

Viral entry is by

- Receptor-mediated endocytosis
- Uptake via coated pits
- Or for those enveloped viruses with fusion proteins via fusion of the cell membrane with the viral envelope

Macromolecular Synthesis

How do the various viruses make their mRNA? mRNA production is diagrammed below.

- The major types of viral genomes are shown on the right.
- The replication intermediates necessary to make mRNA are shown in the gray area.

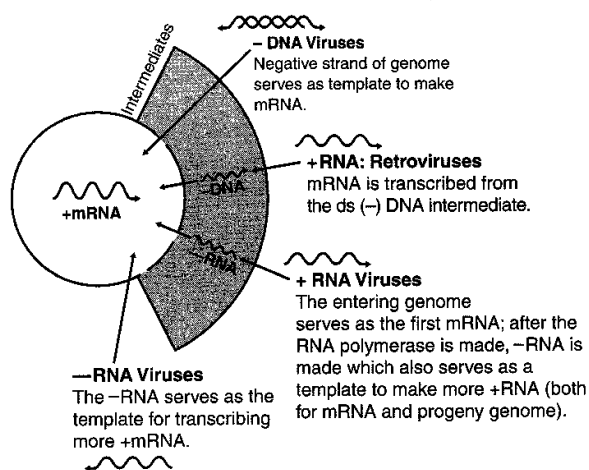


Figure I-5-8. Viral Messenger RNA

Replication of the Genomic Nucleic Acid (NA)

Progeny viruses have a nucleic acid sequence identical to the parent virus.

All single-stranded RNA viruses replicate through a replicative intermediate.

Going back to the Simplified Imaginary Teaching Viruses:

- If the parental genomic sequence is UUUUUUUUU, then the progeny must have the same sequence.
- (Poly AAA would make a polylysine capsid instead.)
- To make more poly UUU, a replicative intermediate of AAAAAAAAAA would be required.
- The replicative intermediate is used to make new poly UUU.

Table I-5-3. Strategy for Viral Genome Replication

Virus Type	Parental Genome	Intermediate Replicative Form	Progeny Genome
Most +ssRNA viruses	+ssRNA	-ssRNA	+ssRNA
Retroviruses	+ssRNA →	dsDNA	+ssRNA
-ssRNA viruses	-ssRNA	+ssRNA	-ssRNA
Most dsDNA viruses	dsDNA		dsDNA
Hepatitis B	dsDNA	ssRNA →	dsDNA

- means an RNA which can serve as mRNA (or for the retroviruses has the same sequence.)

→ = RNA-dependent DNA polymerase

- Called reverse transcriptase for the retroviruses.
- Called the DNA polymerase for hepatitis B.
- Both actually make the first strand of the DNA using the RNA template and then breakdown the RNA and use the single strand of DNA as template to make the second strand.

Table I-5-4. Analogies between Cellular* Nucleic Acid and Viral Nucleic Acid

Cellular DNA Coding Strand Similar/Analogous to:	Cellular DNA Template Strand Similar/Analogous to:
Cellular mRNA (T/U substitution)	No major class of "negative" cellular RNA "Antisense" RNA may occasionally be formed to control gene expression
Positive strand of viral DNA	Negative strand of viral DNA
Positive strand of viral RNA (serves as mRNA for virus)	Negative strand of viral RNA (used as template to produce positive mRNA)

*Cellular DNA includes both eukaryotic and prokaryotic cells.

Release of Viruses

Naked viruses lyse the host cells. Thus, there are **no persistent productive infections** with naked viruses (only cytolytic productive or latent infections).

Release of enveloped viruses: Budding leads to cell senescence (aging), but cells may produce a low level of virus for years as occurs in chronic hepatitis B.

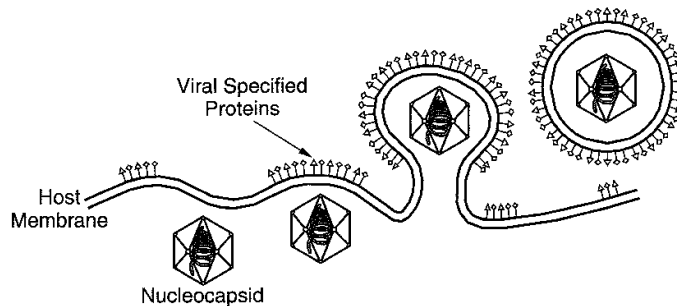


Figure I-5-9. Release of Enveloped Virus

The glycoproteins on the enveloped viral surface are essential for viral infectivity.

PATTERNS OF VIRAL INFECTIONS

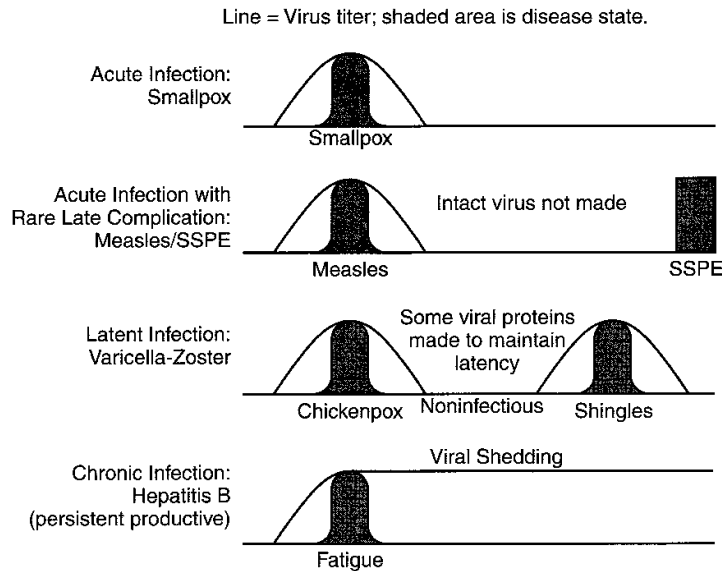


Figure I-5-10. Time Courses: Acute and Persistent Viral Infections

Table I-5-5. Cellular Effects

Infection Type	Virus Production	Fate of Cell
Abortive	—	No effect: No virus is made nor is latency established; Virus is terminated
Cytolytic Naked viruses lyse host cells. Some enveloped viruses also are cytolytic, killing the cell in the process of replication.	+	Lysis of the host cell (death)
Persistent Productive (enveloped viruses)	+	Senescence (premature aging)
Latent	—	No overt damage to host; no production of virus, but viral production may be turned on later.
Transforming	±	Immortalization

In Utero	At Birth	Infants	Children	Adolescents and Young Adults	Adults	Senior Citizens
Cytomegalovirus	→				→	
Rubella →			Rubella →	→		
HSV 2 →	→			HSV 2 →	→	
HIV →	→			HIV →	→	→
	Hepatitis B →			Hepatitis B →	→	→
	HSV 1 →				→	→
		Respiratory Syncytial (bronchiolitis)	→		→	
		Parainfluenza (croup)	→		(colds) →	
		Rotavirus (infant diarrhea)				
		Influenza	→		→	→
			Measles →	→	→	
			Mumps →	→	→	
			Hepatitis A →	→	→	
			Epidemic Gastroenteritis (Norwalk virus)	→	→	→
			Varicella (chickenpox)			Zoster St. Louis Encephalitis

Figure I-5-11. The Most Common Age Groups for Viral Infections

HOST RESISTANCE TO VIRAL INFECTION

Primary Defenses

- Skin barrier (dead keratinized cells impervious to viruses)
- Skin has acids and other inhibitors produced by normal bacterial flora
- Mucociliary elevator

Immune Defense

Innate immune response

- Interferon
- Complement
- Natural killer cells

Adaptive immune response

- Antibody
- Lymphocyte-mediated response

Interferon Production

Interferons (IFNs) are a family of eukaryotic cell proteins classified according to the cell of origin. IFN-alpha and IFN-beta are produced by a variety of virus-infected cells. They:

- Act on target cells to **inhibit viral replication**.
- Do not act directly on the virus.
- Are **not virus-specific**.
- Are **species-specific** (e.g., mouse IFN versus human IFN).

Interferon inhibits viral protein synthesis

- Through activation of an RNA endonuclease, which digests viral RNA.
- By activation (by phosphorylation) of protein kinase that inactivates eIF2 inhibiting viral protein synthesis.

Exogenous human IFN (produced by recombinant DNA technology) may be used in antiviral therapy for chronic, active HBV and HCV infections.

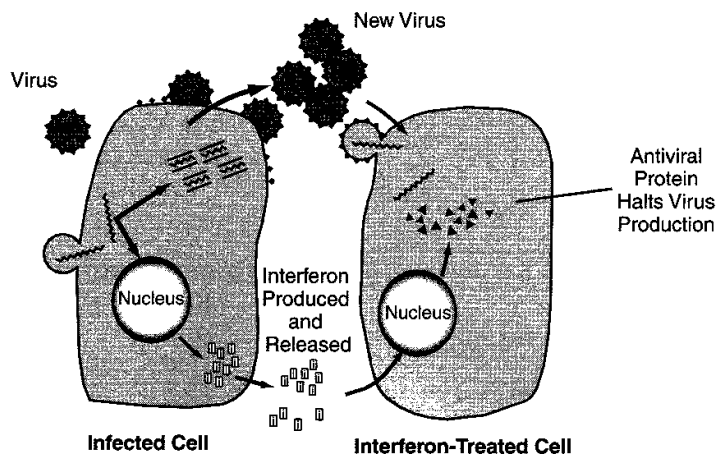


Figure I-5-12. Interferon Production

IMMUNOPROPHYLAXIS

Table I-5-6. Vaccines

	Attenuated (Live)	Killed	Component
Can it revert to a pathogenic form?	Possibly	No	No
Can it cause infections in immunocompromised hosts?	Sometimes	No	No
Immunogenicity?	High	Lower	Middle
Special storage?	Yes; viable organisms	No	No
Potential for contamination with other viruses?	Yes; high	Reduced	No

Active Immunization—Killed Vaccines

RIP-A (Rest In Peace Always—the killed viral vaccines):

Rabies (killed human diploid cell vaccine) and immunoglobulins

Influenza

Polio (Salk)

A Hepatitis

Active Immunization—Live Viral Vaccines

All but adenovirus are attenuated

(mnemonic: Mr. V.Z. Mopsy)

Mumps

Rubella

Varicella - Zoster

Measles

Adenovirus (pathogenic [not attenuated] respiratory strains given in enteric coated capsules)

Polio (Sabin)

Small Pox

Yellow Fever

Active Immunization—Component Vaccines

Hepatitis B

Passive Immunotherapy: Transfer of Immunoglobulins

Hepatitis A
Hepatitis B
Measles
Rabies
Varicella-Zoster

VIRAL HEPATITIS

Symptoms of Hepatitis

Fever, malaise, headache, anorexia, vomiting, dark urine, jaundice.

Table I-5-7. Hepatitis Viruses (Hepatotropic)

Virus	Hepatitis A	Hepatitis B	Hepatitis C	Hepatitis D	Hepatitis E
	"Infectious" (HAV)	"Serum" (HBV)	"Post transfusion Non A Non B" (HCV)	"Delta" (HDV)	"Enteric" (HEV)
Family Features	Picornavirus RNA Naked Capsid	Hepadnavirus DNA Enveloped	Flavivirus RNA Enveloped	Defective Circular RNA Enveloped	Calicivirus RNA Naked capsid
Transmission	Fecal-oral	Parenteral, sexual	Parenteral, sexual	Parenteral, sexual	Fecal-oral
Disease No chronic	Mild acute No chronic No sequelae	Acute is occasionally severe Chronic: 5–10% adults 90% infants 1' hepatocellular carcinoma, cirrhosis	Acute is usually subclinical 80% become chronic Primary hepatocellular carcinoma, cirrhosis	Co-infection with HBV: occa- sionally severe; Superinfection with HBV: often severe Cirrhosis, fulminant hepatitis	Normal patients mild Pregnant patients: severe No chronic
Mortality	<0.5%	1–2%	0.5–1%	High to very high	Normal patients 1–2%; pregnant patients: 20%
Diagnosis acute hepatitis- symptoms &:	IgM to HAV	HBsAg, IgM to HBcAg	Antibody to HCV ELISA	Hepatitis D Ab, HBsAg	Antibody to HEV ELISA

Remember that hepatitis also may occur in other viral diseases (e.g., CMV and EBV infections, congenital rubella, and yellow fever).

Hepatitis B (HBV) Terminology and Markers

Dane particle = infectious HBV

HBsAg

- Surface antigen
- Found during acute disease and persistent infections.
- Presence of HBsAg past 6 months indicates chronic infection.

HBsAb = Antibody to the surface antigen; provides immunity to HBV.

HBs window is the period **between:**

1. **the end of detection of surface antigen and**
2. **the beginning of the detection of surface antibody.**

The absence of these two markers is why the diagnostic test for core antibody is so important.

HBcAg = HBV core antigen

HBcAb = antibody to HbcAg:

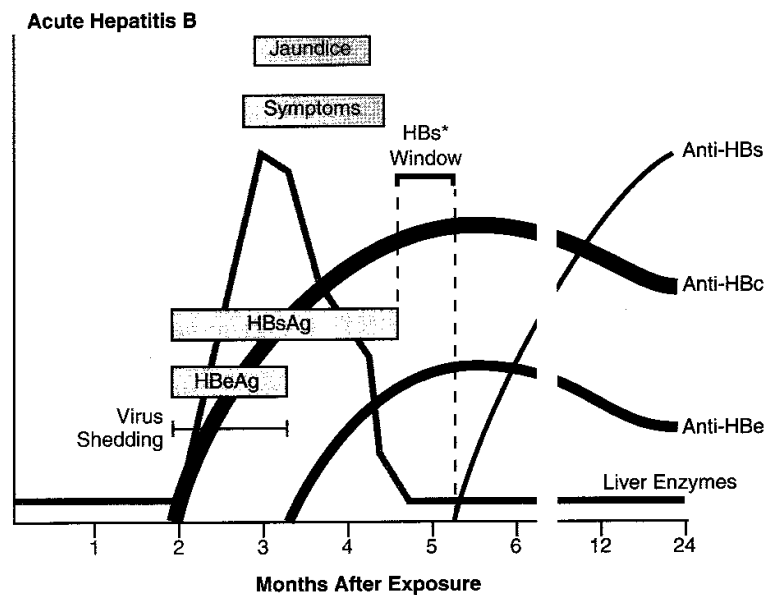
- First antibody to appear
- IgM antibody to core is important test in screening for recent infection.

HBeAg = a second antigenic determinant found in the core of the virus. Its presence correlates with:

- Active viral production
- Infectivity

HBeAb = antibody to HbeAg:

- Generally is detectable after virus is no longer detectable.
- Used to suggest lower risk of transmission.



*The window is the time between the disappearance of the HB surface antigen and before antibody to the surface antigen is detected.

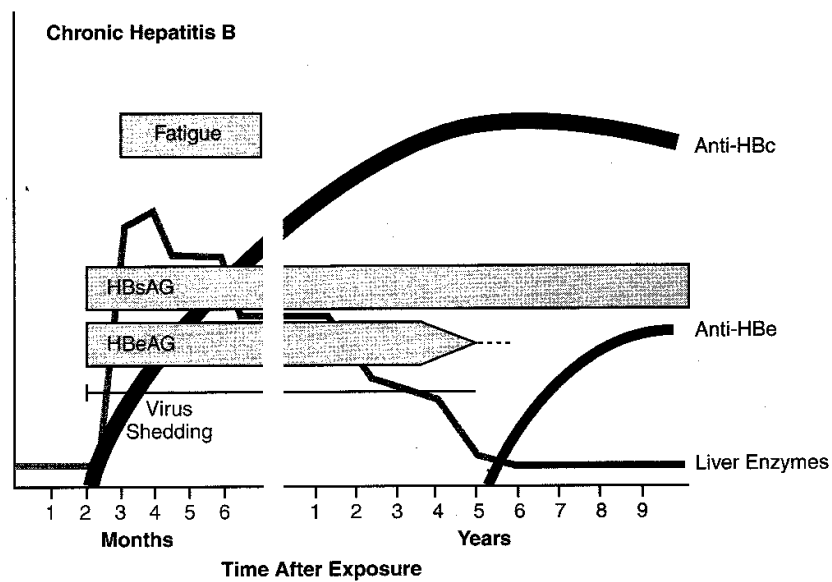


Figure I-5-13. Hepatitis B

DNA VIRUSES: CHARACTERISTICS

All DNA viruses:

- Are double-stranded, except parvovirus
- Are icosahedral, except poxviruses, which are brick-shaped “complex”
- Replicate their DNA in the nucleus, except poxvirus

Table I-5-8. DNA Viruses*

Virus Family	DNA type	Virion-Associated Polymerase	Envelope	DNA Replicates in:	Major Viruses
Parvovirus	ssDNA	No	Naked	Nucleus	B-19
Papovavirus	dsDNA circular	No	Naked	Nucleus	Papilloma Polyoma
Adenovirus	dsDNA linear	No	Naked	Nucleus	Adenoviruses
Herpes virus	dsDNA linear	No	Enveloped (nuclear)	Nucleus; virus assembled in nucleus	HSV Varicella-Zoster Epstein-Barr Cytomegalovirus
Poxvirus	dsDNA linear	Yes**	Enveloped	Cytoplasm	Variola Vaccinia Molluscum contagiosum
Hepadnavirus	partially dsDNA circular	Yes***	Enveloped	Nucleus, RNA intermediate	Hepatitis B

* Mnemonic: Parva's Papa Adds Her Poxes to Hepa's** Poxviruses have a **virion-associated transcriptase** (DNA dependent RNA polymerase) so it can transcribe its own DNA in the cytoplasm and make all of the enzymes and factors necessary for replication of the poxvirus DNA in the cytoplasm.

*** Hepadnaviruses: DNA viruses that carry a DNA polymerase with reverse transcriptase activity to synthesize an RNA intermediate that is then used to make the genomic DNA. Hepatitis B is partially double-stranded with one complete strand.

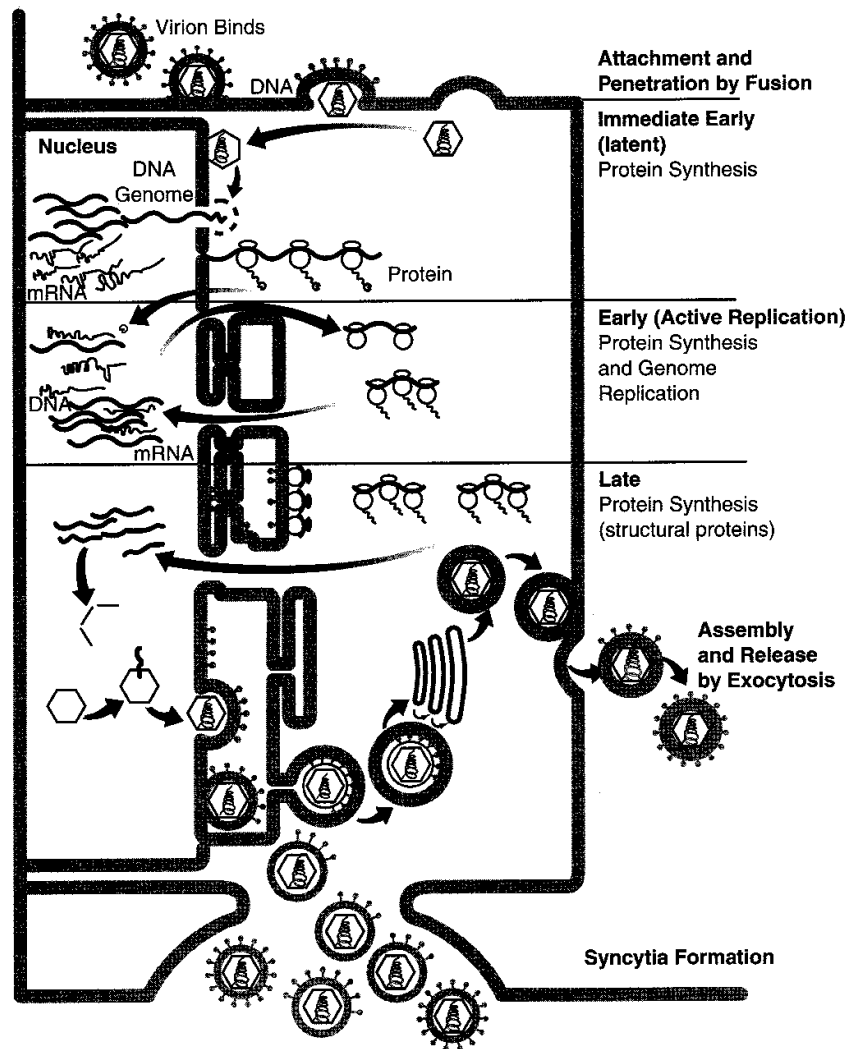


Figure I-5-14. DNA Virus: Life Cycle of Herpes

DNA VIRAL DISEASES

Parvoviridae

- ssDNA (linear)
- Naked icosahedral



Figure I-5-15. Parvovirus

B-19 virus

- Fifth disease = erythema infectiosum = slapped cheek fever:
A mild, febrile disease with facial rash followed by lacy body rash
- Linked to aplastic crises in Sickle Cell Anemia patients.
B-19 infects only immature red cells causing lysis, hence the anemia, which is only clinically significant in Sickle Cell Disease.
- Potential cause of hydrops fetalis

Papovaviridae

- Circular dsDNA
- Naked icosahedral



Figure I-5-16. Papovavirus

Human papilloma virus

- Contact
- Plantar warts: HPV 1 & 4
- Anogenital (condyloma acuminatum) and laryngeal papillomas; HPV types 6 and 11 are most common (considered benign).
- Cervical intraepithelial neoplasia (HPV 98%). Types HPV 16 and 18 are the most common causes of neoplasia.
U.S.A. deaths 4,000–5,000/year
- Pathological findings: koilocytic cells on Pap smear (perinuclear cytoplasmic vacuolization and nuclear enlargement)
- Some strains of HPV associated with cancer; these have two genes which inactivate tumor suppressor genes.

Polyoma viruses

- BK virus: in immunocompromised hosts, causes renal disease
- JC virus: causes progressive multifocal leukoencephalopathy (slow conventional virus).

Adenoviruses

- dsDNA, naked
- Hexons, penton bases, and fibers
- Fibers bind host cell receptors (will bind to most human cells and many animal cells) and also act as hemagglutinins. Fibers, when purified, are toxic to human cells.

Adenovirus Diseases Include:

- Upper respiratory disease in kids.
- Pharyngoconjunctivitis ("pink eye" or "swimming pool" conjunctivitis), non-purulent (unlike *Haemophilus aegyptius*).